

The High-Frequency Factor Zoo - Online Appendix

Saketh Aleti*

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In this appendix, I include additional information about my dataset, empirical results, and theoretical results. First, I begin by giving more detail on how I construct my high-frequency portfolios. I also motivate my portfolio construction methodology and compare my high-frequency portfolios with their low-frequency analogues. Second, I give additional information about my jump detection procedure and my regression methodology. Third, I provide robustness checks for my main results. Fourth, I state my theoretical results and the associated proofs. And, last, I report a table of all the factors in my zoo and their original sources.

*Department of Economics, Duke University, Durham, NC 27708; email: saketh.aleti@duke.edu.

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1 High-Frequency Factor Construction

In this section, I provide additional details regarding the construction of the high-frequency factors. I begin by discussing the construction of the underlying high-frequency stock return data, then discuss my portfolio methodology, and finish with a comparative analysis of the high-frequency factors relative to their low-frequency analogues.

1.1 High-Frequency Prices

As stated in the main text, the dataset consists of common stocks traded on NYSE, NASDAQ, and NYSEMKT. More specifically, common stocks refer to stocks with share codes 10 and 11 on CRSP, while the main exchanges are given by exchange codes 1, 2, and 3. The sample period is January 1996 to December 2020. The initial dataset of high-frequency prices for the stocks takes four steps to construct: (i) obtain high-frequency prices from NYSE TAQ, (ii) obtain open/close prices and intradaily adjusted returns from CRSP, (iii) merge the TAQ and CRSP prices and construct returns, and (iv) construct high-frequency value-weights. I now discuss each of these steps in greater detail.

To start, the high-frequency prices from TAQ are obtained from the “TAQ Monthly” and “TAQ Daily” datasets. The former is used for data before 2015 while the latter is used for 2015 onwards. I begin by collecting trades for each stock, while cleaning the data using rules from Patton and Sheppard (2015) which are similar to Barndorff-Nielsen et al. (2009). Specifically, for each stock, I only consider trades (i) between 09:30:00 and 16:00:00, (ii) with positive prices and size, (iii) belonging to the exchange with the highest volume for that stock, (iv) with condition codes E, F, or blank, (v) with transaction prices between the CRSP ask-high and bid-low for the day, and (vi) with fewer than 5 immediate reversals within a 50-sample moving window. This last rule is enforced using the same approach as Q4 in Barndorff-Nielsen et al. (2009). In addition to the Patton and Sheppard (2015) rules, I take a few more steps to further clean the data. Firstly, when two trades have the same timestamp, a common occurrence in the TAQ Monthly dataset, I take the trade with the median price – this is similar to T3 in Barndorff-Nielsen et al. (2009). Secondly, I only keep trades with correction numbers 0 or 1. And thirdly, I drop all trades on exchange “D” from consideration – this is FINRA’s ADF, not a real exchange. Finally, with the cleaned trades, I construct 5-minute prices for a total of 79 prices per day. For the 9:30:00 price, I use the first trade in the 09:30:00-09:35:00 bucket. Otherwise, each price uses the last trade within each bucket.

I then collect low-frequency prices from CRSP to merge with the high-frequency prices from TAQ. The purpose of merging TAQ data with CRSP data is to handle the overnight return appropriately; this means accounting for dividends, share splits, and delisting. I begin this procedure by first collecting daily CRSP data for all stocks in the universe described above. The CRSP data contains open prices, close prices, adjusted close-to-close-returns, delisting returns, and shares outstanding. Using the adjusted close-to-close return along with the prices for each day, it is simple to infer the adjusted overnight (close-to-open) return. I can then assign this adjusted return to the 09:30:00 observation for each day. Additionally, I adjust the 16:00:00 return using the delisting return; I use the same procedure as in Hou, Xue, and Zhang (2020). This gives me a series of adjusted returns for the open and close of each day, which I then merge with the TAQ data.

However, note that the adjusted returns are a function of the *CRSP* open/close prices, so I must use the *CRSP* prices as well in the final dataset for consistency.

Consequently, I merge the low-frequency *CRSP* prices with the high-frequency 5-minute *TAQ* prices as in Aït-Sahalia, Kalnina, and Xiu (2020). This merge is done by simply replacing the 09:30:00 and 16:00:00 *TAQ*-trades-inferred prices with the *CRSP* open and close prices. In order to ensure *TAQ* and *CRSP* stocks are matched accurately, I use the *TAQ-CRSP Link* table provided by WRDS; matches are based on *CRSP*’s permanent issue identifier known as *PERMNO*.¹ After merging the data, I obtain a dataset with 79 prices from 09:30:00 to 16:00:00 with *CRSP*-adjusted 09:30:00 and 16:00:00 returns for each day; intradaily 5-minute returns are then straightforward to compute. All in all, one can think of this dataset as the *CRSP Daily* dataset with “in-filled” *TAQ* 5-minute prices.

When constructing the factor-sorted portfolios later on, I will be using value-weighted returns which requires specifying the value weights. The value weights are constructed in the same way as in Aït-Sahalia, Kalnina, and Xiu (2020). For each stock, I compute the market capitalization using the close price along with the shares outstanding and associated price/shares-adjustment factors from *CRSP*; this accounts events like stock splits. Then, with the previous day’s closing market capitalization, I use the high-frequency return data to produce a sequence of high-frequency market caps. Finally, I produce the value weights by lagging the market caps by one unit of time. So, for instance, the value-weight at 09:30:00 for a particular stock would be that stock’s market cap on the previous day at 16:00:00; next, at 09:35:00, we would use the 09:30:00 market cap which is simply the previous day’s market cap multiplied by the 09:30:00 return. This procedure is repeated to get high-frequency value weights for the full day for each stock. These high-frequency value-weights make it possible to compute high-frequency returns on value-weighted factor portfolios.

1.2 Factors and Signals

In this subsection, I give some additional information about my choice of factors. To start, I report the 268 factors mentioned in the main text in Table 28. I produce this table by combining the factors previously compiled by Chen and Zimmermann (2020) and Jensen, Kelly, and Pedersen (2021). In addition, I include two columns that report the factor classifications given by these two papers. The *CZ Classification* column reports 34 factor classifications that Chen and Zimmermann construct by hand; the *JKP Classification* column reports the 13 factor classifications that Jensen, Kelly, and Pedersen construct using hierarchical agglomerative clustering (Murtagh & Legendre, 2014) on the factor

¹ This table does not provide a link for every single stock, but it is able to link $\approx 98\%$ of the *CRSP* stock universe to *TAQ*. For the remaining fraction of links, I use the *TAQ Master Table* for each day to obtain CUSIPs for each *TAQ* entry and match those CUSIPs with *CRSP* entries. Additionally, I merge the data such that every *CRSP* entry, for each *PERMNO* and day, is assigned a set of high-frequency prices from *TAQ*, but not vice versa; this means that all *CRSP* data will be in the final dataset but, on some days and for some stocks, there will be no intradaily high-frequency prices. In order to handle missing intradaily data, I do a simple log-linear interpolation on the open and close prices – this is equivalent to assuming constant log returns. Note that the fraction of stocks missing data is usually $< 1\%$ and such stocks tend to be nanocaps. Hence, when computing value-weighted returns for the factor portfolios later, the missing data and interpolation has little effect on the final portfolio returns.

CAPM-residuals. In addition to these classifications, I also include a column containing a simple description of each factor. This description defaults to the description given by JKP, but uses the CZ description for factors that only show up in the latter paper.

Next, I filter these 268 factors further to obtain 218 unique signals. To start, I handle factors with duplicate signal data – that is, factors that appear in both the JKP and CZ datasets. In my 268 factors (153 JKP and 204 CZ), there are 89 that appear in both datasets. I prioritize signal data from JKP, so I take 153 signals from JKP and 115 signals from CZ.² Note that the total number of factors thus far is unchanged, since, at this point, I’ve only specified which paper I use for each signal. Next, I filter out factors that have missing signals; this filter affects the total factor count. Additionally, since the signal data is available for all of the JKP factors, this filter only affects the factors that only appear in the CZ dataset. The total number of unique CZ factors missing signal data is 49. Thus, after dropping these factors, I am left with 65 signals based on CZ and 153 signals based on JKP for a total of 218 signals.

1.3 High-Frequency Portfolio Adjustment

The cumulative returns on high-frequency “high-minus-low” portfolios can differ from their low-frequency equivalents. As a specific example, consider the operating profitability factor proposed by Fama and French (1993), also known as RMW. This factor portfolio is produced by subtracting the return on a “high” portfolio from that on a “low” portfolio. Specifically, the formula is:

$$\begin{aligned} RMW &= (1/2)(SmallRobust + BigRobust) - (1/2)(SmallWeak + BigWeak) \\ &= Robust - Weak \end{aligned}$$

where robust refers to assets high in operating profitability. Now consider a scenario where we have three time periods, $\{t, t+1, t+2\}$. The return from time t to time $t+2$ on such a portfolio is given by

$$\begin{aligned} \text{Low-Freq:} & \quad 1 + r_{t,t+2}^{Robust} - r_{t,t+2}^{Weak} \\ \text{High-Freq:} & \quad (1 + r_{t,t+1}^{Robust} - r_{t,t+1}^{Weak}) \times (1 + r_{t+1,t+2}^{Robust} - r_{t+1,t+2}^{Weak}) \end{aligned}$$

where r represents the percent return on a portfolio. A first-order Taylor series expansion tells us that $r_{t,t+2} \approx r_{t,t+1} + r_{t+1,t+2}$ and that $(1 + r_a)(1 + r_b) \approx 1 + r_a + r_b$. Consequently, the low-frequency and high-frequency cumulative returns will be *approximately* the same. But, they will not be *exactly* the same. This problem arises because net-zero investment portfolios, like RMW, technically have an undefined return – they cost \$0 to enter but

² Note that sometimes a CZ factor may appear to be the same as one of the JKP factors. The reason this occurs is because I treat factors from different papers as different factors. So, for example, the Fama and French (1992) book-to-market factor using December ME appears in the CZ dataset as “BMdec.” A book-to-market factor also appears in the JKP dataset but it is based on Rosenberg, Reid, and Lanstein (1985). So, it may appear that there is a duplication; however, I include both factors because they are based on different papers and are thus constructed differently. Additionally, even if one views this duplication as a problem, it will not present an issue later on when I cluster similar factors together (discussed in Section 2.1).

have non-zero payoffs.³ Since all of the FF+JKP+CZ factors are net-zero, high-minus-low-style portfolios, this issue matters for my dataset.

As a result, the cumulative return on high-frequency factors will deviate from their low-frequency equivalents. The discrepancies for returns on a daily basis are generally very small because they are all second-order. However, when we have high sampling frequencies and long time-spans, the errors can add up. In order to handle this, I inject a small “sampling frequency adjustment” into intradaily returns to align them with daily returns. Specifically, suppose that intradaily portfolio returns indexed by i for an arbitrary day t are given by $r_{t,i}$. Let $R_H \equiv \prod_i (1 + r_{t,i})$ be the gross portfolio return for the day using this high-frequency data; also, let R_L be the gross portfolio return when using daily data instead. The adjusted high-frequency return is defined as

$$\tilde{r}_{t,i} = r_{t,i} + \left(\frac{R_L}{R_H} - 1 \right) / [1/\Delta_n]$$

where $[1/\Delta_n]$ is the number of return observations per day. Then, a simple Taylor series argument can be used to show that the cumulative gross adjusted return $\prod_i (1 + \tilde{r}_{t,i})$ approximately equals R_L , the gross return based on daily data. Essentially, this procedure simply nudges the intradaily returns by a very small amount to ensure that they properly line up with the daily returns; the adjusted returns are not perfect but the adjustment reduces the discrepancy between high-frequency versus low-frequency portfolio returns by an order of magnitude. So, with my high-frequency dataset, I compute both low-frequency and high-frequency portfolio returns and then adjust the high-frequency returns to ensure that cumulative returns for both frequencies are close. Again, this “sampling frequency adjustment” is only needed/used for the net-zero portfolios; it has no effect on portfolios that are not net-zero.⁴

Additionally, some portfolios lack a sufficient number of stocks in their high and low portfolios. In months where a high/low portfolio has fewer than 100 stocks, I instead use the median signal value as the breakpoint for the factor portfolio itself rather than doing a 33/33/33 split. This procedure has very little effect on the final JKP+CZ factors used in the analysis. It primarily improves the construction of the “BrandInvest” factor in the Chen and Zimmermann (2020) dataset (originally from Belo, Lin, and Bazdresch (2014)), which would otherwise lack a sufficient number of stocks in its “high” portfolio.

1.4 Portfolio Methodology Motivation

Having detailed my methodologies for the factor zoo portfolios in main text, it remains to be addressed whether my approach is the best or most appropriate way to reproduce the literatures’ factors. Hence, in this subsection, I motivate my portfolio construction methodology.

I use two portfolio construction methodologies – one for the FF6 factors and another for the remaining factors. This highlights two competing concerns: *uniformity* with respect to the methodologies and *consistency* with respect to the original work. These two

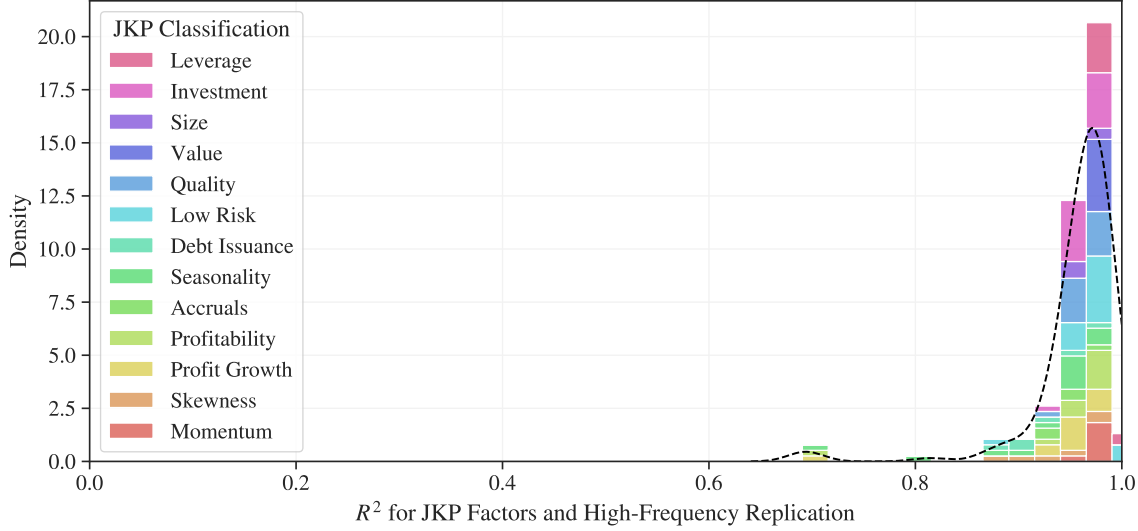
³ On the other hand, for portfolios that have strictly positive or negative total weight, the sampling frequency is totally irrelevant.

⁴ Notice that, for portfolios that are not net-zero portfolios, $R_H = R_L$. This immediately causes the adjustment factor to be 0, resulting in $\tilde{r}_{t,i} = r_{t,i}$.

concerns also show up in Jensen, Kelly, and Pedersen (2021) and Chen and Zimmermann (2020). For instance, Jensen, Kelly, and Pedersen use a single, uniform methodology to construct their factor portfolios. The authors argue that their approach is motivated by what Hamermesh (2007) calls a “scientific” replication – essentially, a replication that slightly tweaks the methodology or sample but tests the same hypothesis. The value of such a replication is that it presents a good test of external validity. In contrast, Chen and Zimmermann (2020) aim to do a “pure” replication; this kind of replication primarily tests internal validity. Hence, Chen and Zimmermann construct each factor using a methodology consistent with the original, associated paper. Since methodologies differ across papers, the CZ factor portfolios are constructed in more than 200 different ways. For instance, some researchers decide to equally weight their factor portfolios while others use value weights. Similarly, researchers may differ in what quantiles they deem to be “high” or “low” for a signal, how long they hold each factor portfolio before rebalancing, what month/day they rebalance on, what exchanges they consider, and so on. Having so many degrees of freedom makes it possible for a researcher to try every approach until they find a methodology that “passes” the standard statistical tests. Hence, doing a pure replication of the original factors may just confirm a result that was hacked through multiple testing; moreover, having so many different methodologies makes it difficult to sensibly compare factors. This creates a case for uniformity. On the other hand, we do not want to misspecify the factor portfolios, which might cause them to lose their explanatory power and become inconsistent with the literature. So, consistency matters as well.

By using two separate methodologies for the 6 Fama-French factors and the 194 JKP+CZ factors, I believe that I strike an appropriate balance between consistency and uniformity. Firstly, this approach is fairly consistent with the original work. The Fama-French 5+1 factors are consistent, because I manually construct them while following the authors’ exact methodology. The 194 factor portfolios from CZ+JKP are still reasonably consistent for two reasons. One, the underlying signals for the factors are faithful replications of the original signals; documentation on these signals is also provided by Jensen, Kelly, and Pedersen (2021) and Chen and Zimmermann (2020). And two, although the 194 JKP+CZ portfolios are not reproduced in the same way as the original papers, many of them appear in Jensen, Kelly, and Pedersen (2021) and my portfolio construction methodology is fairly consistent with theirs. Furthermore, because their paper finds that the majority of factors replicate, it seems reasonable to believe that following their approach will produce high-frequency factor portfolios that capture economically-significant factor risk. So, in some sense, my high-frequency factor portfolios will be faithful to the original portfolios. Secondly, my approach is quite uniform. I only use two methodologies for all 200 factors in my dataset and a single methodology for 194 of them; also, all portfolios are limited to the Fama and French (1993) universe of stocks. This seems to be as uniform as I can be without giving up consistency with either Fama and French (2015) or Jensen, Kelly, and Pedersen (2021). And, recall that my decision to use a consistent methodology for the Fama-French factors is motivated by the fact that their factors are well-established in the asset pricing literature and it would be problematic to deviate from their portfolios. So overall, I believe that my methodology strikes the balance between being as uniform and as consistent as possible.

Figure 1: R^2 S FOR JENSEN, KELLY, AND PEDERSEN 153 PORTFOLIOS



Note: For each of the 153 factors in the Jensen, Kelly, and Pedersen (2021) dataset, I regress my aggregated high-frequency portfolio against the associated, original low-frequency portfolio. All regressions are done using monthly returns. Then, I plot a histogram of the R^2 s associated with each factor. Colors are used to distinguish different types of factors by their classification given in Jensen, Kelly, and Pedersen (2021).

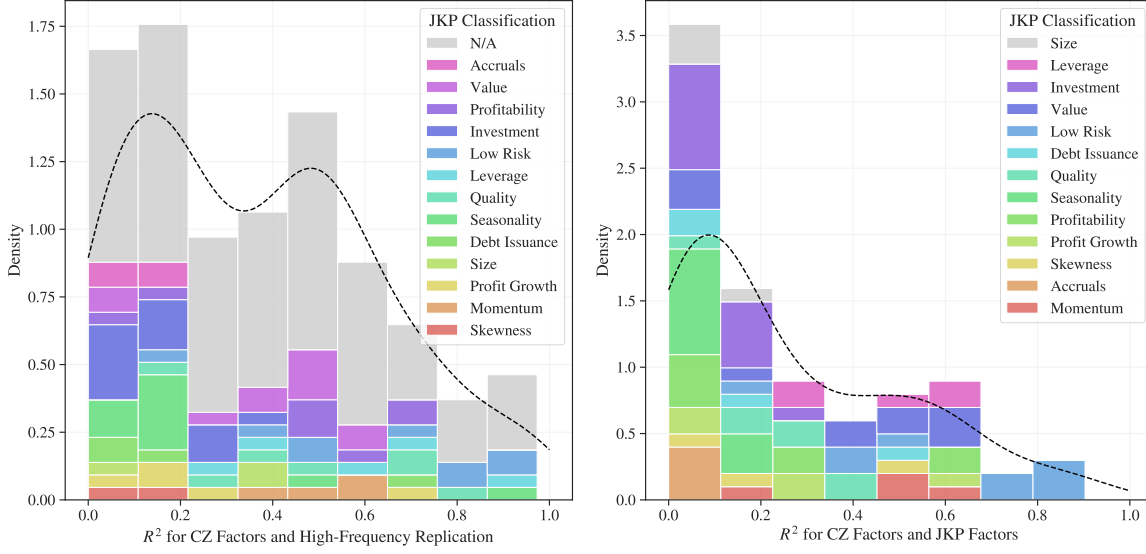
1.5 Accuracy

Because the high-frequency factors have been constructed from scratch, it is important to ensure that they appear sensible. For the Fama-French 5+1 factors, the high-frequency analogues should appear identical since I use the same methodology as the original factors. This was shown to be true in the main text. On the other hand, the 194 JKP+CZ high-freq portfolios may not necessarily match the original portfolios due to methodological differences with the original papers. The differences between my portfolios and the original portfolios are greatest with the CZ factors; this is because Chen and Zimmermann (2020) follow the original papers and end up using more than a hundred portfolio construction approaches to ensure a pure replication. The differences are far weaker between my portfolios and the JKP factors; this is to be expected since I use nearly the same methodology as Jensen, Kelly, and Pedersen (2021).

To start, I compare my high-frequency factors with the lower-frequency analogues from Jensen, Kelly, and Pedersen (2021) in Figure 1.⁵ To do so, I first aggregate my high-frequency factors to get monthly returns. Then, for each factor, I compute the R^2 between my aggregated portfolio and the low-frequency analogue from Jensen, Kelly, and Pedersen (2021). I plot a histogram of the R^2 s with colors indicating the factor's classification given by Jensen, Kelly, and Pedersen. The average R^2 is 95.7% with a standard deviation of 4.5%; only six percent of factors have an R^2 lower than 90% and less than a quarter have an R^2 lower than 95%. There appear to be three relatively weak matches which all have an R^2 of about 69%; the associated factors are “Change in short-

⁵ The authors provide code and data to replicate their results [here](#). I compare my high-freq portfolios with their low-freq portfolios that they make available; specifically, I compare with the data file `Country Factors/USA.csv`.

Figure 2: R^2 s FOR CHEN AND ZIMMERMANN PORTFOLIOS



Note: For each of the factors from the Chen and Zimmermann (2020), I regress my aggregated high-frequency portfolio against the associated, original low-frequency portfolio. All regressions are done using monthly returns. Then, I plot a histogram of the R^2 s associated with each factor. Colors are used to distinguish different types of factors by their classification given in Jensen, Kelly, and Pedersen (2021); factors missing classifications are given the label N/A. The results are shown in the left subplot. The right subplot is constructed similarly but compares the original CZ portfolio returns with the original JKP portfolio returns.

term investments” from Richardson et al. (2005), “Net debt issuance” from Bradshaw, Richardson, and Sloan (2006), and “Pitroski F-score” from Piotroski (2000). So, overall, it appears that my high-frequency factor portfolios match the original low-frequency JKP portfolios quite closely.

On the other hand, the CZ portfolios are much less similar. In Figure 2, I repeat the previous exercise for the CZ factors.⁶ The left histogram shows the R^2 values comparing my factors with the original CZ factors. The stacked densities are colored using classifications from Jensen, Kelly, and Pedersen (2021); for factors not found in the latter dataset, the classification is set to N/A. The right histogram is constructed similarly except it compares the JKP portfolio returns with the CZ portfolio returns. Overall, we can see that both my portfolios and the original JKP portfolios cannot achieve moderate or high R^2 with the CZ factors. Again, this is expected, because my portfolio construction methodology follows that of JKP; hence, both my portfolios and the original JKP portfolios will likely look different from the original CZ portfolios.

2 Descriptive Statistics

In this section, I give some additional descriptive statistics for the high-frequency factor zoo. Again, I focus on the 218 JKP+CZ portfolios.

⁶ The original portfolio returns come from the “PredictorLSretWide.csv” found on the author’s “Open Source Asset Pricing” project website.

2.1 Cluster Assignments

I cluster my portfolios by reusing the cluster assignments from Jensen, Kelly, and Peder-
sen (2021) for the JKP portfolios while manually assigning clusters for the factors from
Chen and Zimmermann (2020). There are two main reasons I reuse their classifications.
Firstly, many existing papers in the factor zoo literature offer their own classifications
for different types of existing factors. Hence, there is now a “zoo” of factor classifica-
tions. So, for simplicity, I opt to reuse a previous set of factor classifications rather than
unnecessarily contribute to this zoo; this decision also avoids any further data mining.
Secondly, the clusters from JKP (2021) are apt for my dataset. Recall that 153 factors
in my zoo also appear in JKP and are based on their signal data. Consequently, it is
natural to reuse their classifications at least for their factors.

As stated in the main text, assignments for the CZ factors are done using the following
formula:

$$\text{Classification}(k_{CZ}) = \arg \max_{C \in \mathcal{C}} \left(\frac{1}{\#C} \sum_{k_{JKP} \in C} \rho(k_{CZ}, k_{JKP}) \right). \quad (1)$$

Again, k refers to an arbitrary factor from a particular dataset, $\rho(\cdot, \cdot)$ is the correlation
between the CAPM residuals of two portfolios, and $\mathcal{C}$ is the set of sets of JKP factors
grouped by classification.

The cluster assignments are given in Table 1. The second-to-last column gives the
cluster assignment for each of the CZ factors. Additionally, the last column reports the
average correlation between a given CZ factor and the set of JKP factors within its cluster
assignment; in other words, this is the term $\frac{1}{\#C} \sum_{k_{JKP} \in C} \rho(k_{CZ}, k_{JKP})$ where C is the
cluster assignment for the corresponding factor k_{CZ} . Some factors are given their natural
classifications; for instance, the Customer Momentum factor from Cohen and Frazzini
(2008) is classified as a momentum factor. But, this does not hold for all the factors;
for example, the debt issuance factor from Spiess and Affleck-Graves (1999) is classified
into the Value cluster rather than the Debt Issuance cluster. In fact, this factor has a
negative average correlation with the Debt Issuance cluster.⁷ Furthermore, based the
high average correlations with the Value cluster, this factor does actually seem to be
capturing value-related risk. Hence, even though factors may not always fall into their
natural clusters, this procedure does still appear to be effective.

I have also considered clustering based on maximum correlation rather than average
correlation. Such a procedure can still be represented using Equation 1 but the parenthet-
ical term would be replaced with $\max_{k_{JKP} \in C} \rho(\dots, \dots)$. The results from this procedure
are almost exactly the same as the results from the procedure described above. There
are only 11 CZ factors that switch clusters and these switches are not necessarily im-
provements. Furthermore, using maximum correlation may be an ineffective approach in
theory, because estimated correlations may be noisy. In addition, sorted portfolios are
not perfect proxies of the systematic risk they are trying to isolate. Hence, using average
correlation may be a better approach in theory, although it seems to make little difference

⁷ This result is not related to the direction of the factor portfolio; for each factor, I go long in the
direction proposed by the original paper. Furthermore, computing average absolute correlation still gives
a result of only 13.8% with respect to the debt issuance cluster. The equivalent statistic for the Value
cluster is 63.7%. Additionally, the result does not substantially change when using the geometric average
or maximum ρ statistic.

in practice.

Table 1: CZ FACTORS – CLUSTER ASSIGNMENTS

Citation	Factor	CZ Description	Cluster Assignment	$Avg(\rho)$
Abarbanell and Bushee, 1998	<i>cz_chinvia</i>	Change in capital inv (ind adj)	Leverage	0.1086
Abarbanell and Bushee, 1998	<i>cz_grsaletogroverhead</i>	Change sales minus change SGA	Value	0.1192
Alwathainani, 2009	<i>cz_earningsconsistency</i>	Earnings consistency	Quality	0.1915
Ang et al., 2006	<i>cz_idiorisk</i>	Idiosyncratic risk	Leverage	0.3647
Ang et al., 2006	<i>cz_betavix</i>	Systematic volatility	Leverage	0.1359
Barry and Brown, 1984	<i>cz_firmage</i>	Firm age based on CRSP	Low Risk	0.3882
Belo, Lin, and Vitorino, 2014	<i>cz_brandinvest</i>	Brand capital investment	Profitability	0.2702
Baik and Ahn, 2007	<i>cz_orderbacklogchg</i>	Change in order backlog	Leverage	0.1002
Bradshaw, Richardson, and Sloan, 2006	<i>cz_xfin</i>	Net external financing	Leverage	0.3076
Cohen and Frazzini, 2008	<i>cz_customermomentum</i>	Customer momentum	Momentum	0.0984
Chan, Jegadeesh, and Lakonishok, 1996	<i>cz_announcementreturn</i>	Earnings announcement return	Momentum	0.2924
Chan, Lakonishok, and Sougiannis, 2001	<i>cz_adexp</i>	Advertising Expense	Value	0.4982
Chandrashekar and Rao, 2011	<i>cz_cashprod</i>	Cash Productivity	Quality	0.6298
Cohen, Diether, and Malloy, 2013	<i>cz_rdability</i>	R&D ability	Value	0.0357
Daniel and Titman, 2006	<i>cz_intanbm</i>	Intangible return using BM	Quality	0.5813
Daniel and Titman, 2006	<i>cz_intansp</i>	Intangible return using Sale2P	Quality	0.5108
Daniel and Titman, 2006	<i>cz_intanep</i>	Intangible return using EP	Quality	0.5248
Daniel and Titman, 2006	<i>cz_intancfp</i>	Intangible return using CFtoP	Quality	0.5226
Daniel and Titman, 2006	<i>cz_compequiss</i>	Composite equity issuance	Value	0.4560
Daniel and Titman, 2006	<i>cz_shareiss5y</i>	Share issuance (5 year)	Leverage	0.3661
De Bondt and Thaler, 1985	<i>cz_mrreversal</i>	Medium-run reversal	Quality	0.2983
Dechow et al., 2001	<i>cz_shortinterest</i>	Short Interest	Size	0.3574
Dharan and Ikenberry, 1995	<i>cz_exchswitch</i>	Exchange Switch	Size	0.2308
Eisfeldt and Papanikolaou, 2013	<i>cz_orgcap</i>	Organizational capital	Low Risk	0.3181
Elgers, Lo, and Pfeiffer, 2001	<i>cz_sfe</i>	Earnings Forecast to price	Value	0.4450
Fama and French, 1992	<i>cz_bmdec</i>	Book to market using December ME	Value	0.5894
Frankel and Lee, 1998	<i>cz_aop</i>	Analyst Optimism	Leverage	0.4395
Franzoni and Marin, 2006	<i>cz_fr</i>	Pension Funding Status	Profitability	0.0784
Guo, Lev, and Shi, 2006	<i>cz_rdipo</i>	IPO and no R&D spending	Size	0.2296
Haugen and Baker, 1996	<i>cz_volumetrend</i>	Volume Trend	Leverage	0.0985
Haugen and Baker, 1996	<i>cz_varcf</i>	Cash-flow to price variance	Size	0.5070

continued. . .

CZ Factors – Cluster Assignments (continued)

Citation	Factor	CZ Description	Cluster Assignment	$Avg(\rho)$
Haugen and Baker, 1996	cz_volmkt	Volume to market equity	Leverage	0.4594
Hou and Robinson, 2006	cz_herf	Industry concentration (sales)	Low Risk	0.4530
Hou and Robinson, 2006	cz_herfasset	Industry concentration (assets)	Low Risk	0.4312
Hou and Robinson, 2006	cz_herfbe	Industry concentration (equity)	Low Risk	0.3444
Hou, 2007	cz_earnsupbig	Earnings surprise of big firms	Profitability	0.1261
Hou, 2007	cz_indretbig	Industry return of big firms	Momentum	0.0715
Jegadeesh and Titman, 1993	cz_mom12m	Momentum (12 month)	Momentum	0.6980
Kelly and Jiang, 2014	cz_betatailrisk	Tail risk beta	Leverage	0.3007
Lakonishok, Shleifer, and Vishny, 1994	cz_meanrankrevgrowth	Revenue Growth Rank	Value	0.5224
Landsman et al., 2011	cz_rds	Real dirty surplus	Investment	0.2965
Lee and Swaminathan, 2000	cz_momvol	Momentum in high volume stocks	Momentum	0.7099
Liu, 2006	cz_zerotradealt12	Days with zero trades (6 months)	Low Risk	0.4299
Liu, 2006	cz_zerotradealt1	Days with zero trades (6 months)	Low Risk	0.5117
Lockwood and Prombutr, 2010	cz_cheq	Growth in book equity	Leverage	0.4365
Loh and Warachka, 2012	cz_earningsstreak	Earnings surprise streak	Quality	0.2696
Lou, 2014	cz_gradexp	Growth in advertising expenses	Leverage	0.3963
Loughran and Wellman, 2011	cz_entmult	Enterprise Multiple	Leverage	0.4293
Lyandres, Sun, and Zhang, 2008	cz_compositedebtissuance	Composite debt issuance	Leverage	0.1229
Mohanram, 2005	cz_ms	Mohanram G-score	Leverage	0.2716
Moskowitz and Grinblatt, 1999	cz_indmom	Industry Momentum	Momentum	0.6299
Nagel, 2005	cz_rio_mb	Inst Own and Market to Book	Size	0.4389
Nagel, 2005	cz_rio_turnover	Inst Own and Turnover	Size	0.3293
Nagel, 2005	cz_rio_volatility	Inst Own and Idio Vol	Size	0.2766
Nguyen and Swanson, 2009	cz_frontier	Efficient frontier index	Value	0.4408
Penman, Richardson, and Tuna, 2007	cz_bpebm	Leverage component of BM	Low Risk	0.3344
Ritter, 1991	cz_ageipo	IPO and age	Value	0.3583
Ritter, 1991	cz_indipo	Initial Public Offerings	Leverage	0.4290
Rajgopal, Shevlin, and Venkatachalam, 2003	cz_orderbacklog	Order backlog	Value	0.2208
Soliman, 2008	cz_chnwc	Change in Net Working Capital	Low Risk	0.0932
Spies and Affleck-Graves, 1999	cz_debtissuance	Debt Issuance	Value	0.6376
Tuzel, 2010	cz_realestate	Real estate holdings	Leverage	0.2607
Valta, 2016	cz_convdebt	Convertible debt indicator	Leverage	0.4216

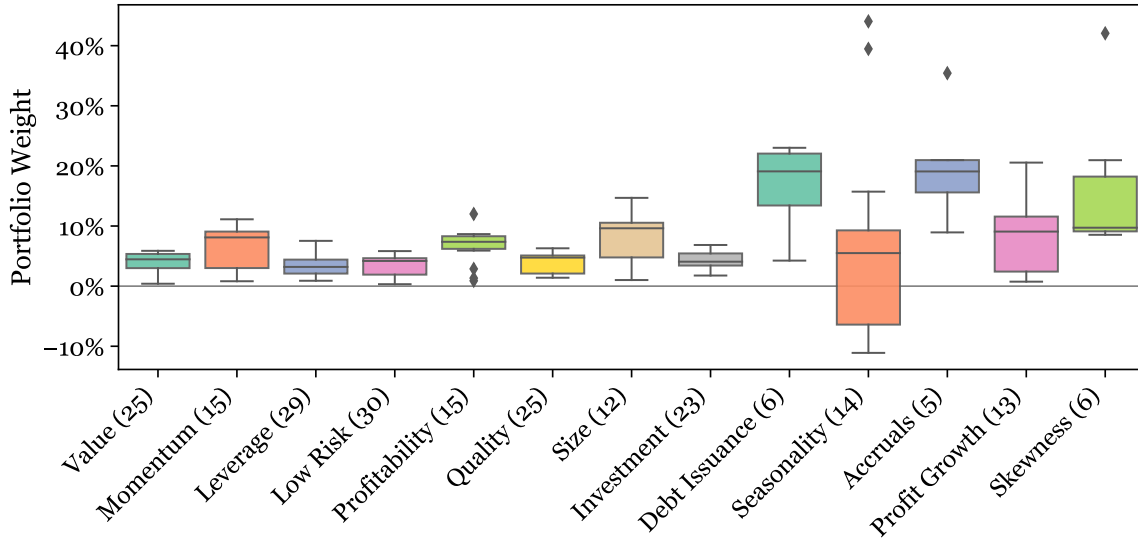
continued. . .

CZ Factors – Cluster Assignments (continued)

Citation	Factor	CZ Description	Cluster Assignment	$Avg(\rho)$
Xie, 2001	cz_abnormalaccruals	Abnormal Accruals	Value	0.4055
Zhang, 2006	cz_firmagemom	Firm Age - Momentum	Momentum	0.4483

This table reports the cluster assignments for the CZ factors. The description column is based on Chen and Zimmermann ([2020](#)). The average correlation between a given CZ factor and the JKP factors within that cluster is given by the column labelled $Avg(\rho)$; note that correlations are realized correlations computed on CAPM residuals using high-frequency data.

Figure 3: DISTRIBUTION OF CLUSTER PORTFOLIO WEIGHTS

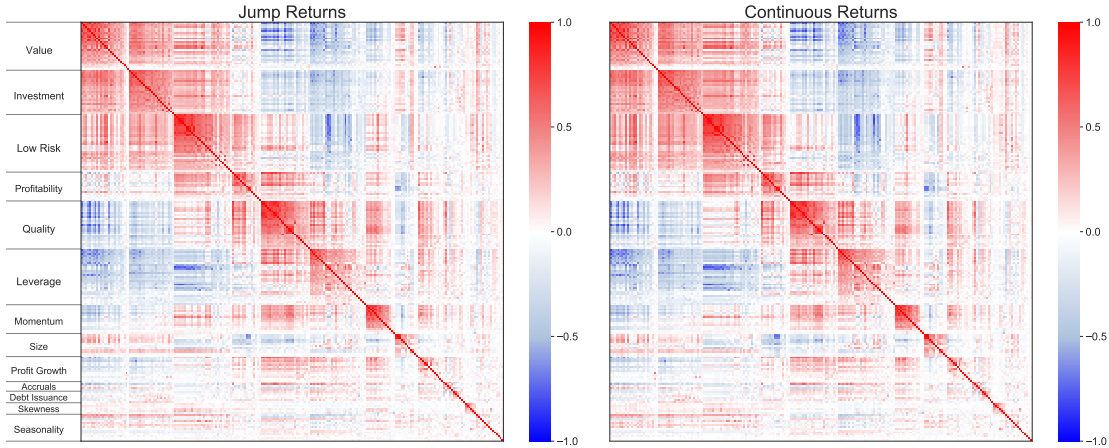


Note: Each boxplot shows the distribution of the portfolio weights for the representative portfolio of each cluster. Each observation represents the weight on particular factor portfolio within that cluster. The x-axis labels also report the number of factors within each cluster.

2.2 Cluster Portfolio Weights

I use the first principle component of each cluster's factor returns to construct a cluster representative portfolio. To ensure the resulting portfolios are reasonable, I normalize the sum of the principal component weights to add up to 100%. This normalization has no effect on the final portfolios besides a superficial rescaling which itself is immaterial since the portfolios are net-zero investment. I report the distribution of the portfolio weights for each cluster in [Figure 3](#). Across all clusters, the minimum portfolio weight is -11%, the median is 8.3%, and the maximum is 44%. Overall, the weights seem fairly sensible.

Figure 4: CORRELATIONS BETWEEN THE JKP+CZ FACTORS – CONTINUOUS RETURNS VERSUS JUMP RETURNS



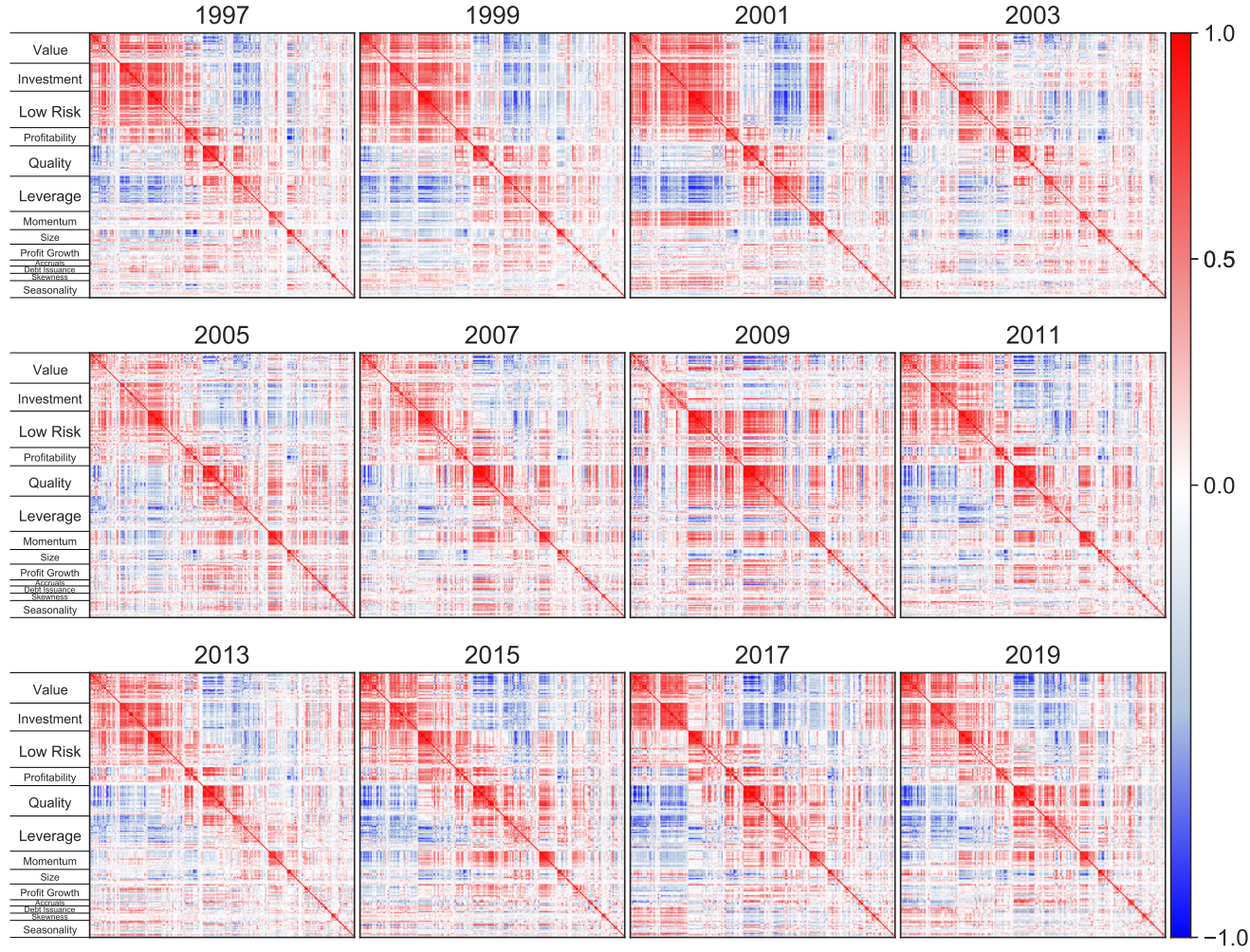
Note: I plot the correlations between the CAPM residuals of the 218 JKP+CZ factors. The residuals are computed separately for the jump/continuous returns and are obtained using simple regressions on the high-frequency Fama-French market factor. The left-hand-side of the first graph shows the cluster assignments for each of the factors; these clusters originally come from Jensen, Kelly, and Pedersen (2021). The sample period is the full sample, 1996-2020.

2.3 Factor Correlations & Cluster Structure

In the main text, I discuss the correlations between the CAPM residuals of my portfolios; these residuals are computed for all returns. Here, I produce figures for the continuous and jump CAPM residuals separately. These two types of residuals are constructed by running separate time-series regressions on continuous and jump returns. The results are shown in Figure 4. The first heatmap shows the correlations for the jump components while the second heatmap shows the correlations for the continuous components. In both heatmaps, the patterns look almost identical. In fact, to quantify the similarity, I flatten the lower triangular of the correlation matrices into two arrays for the two heatmaps. I regress the jump correlations against the continuous correlations and find an R^2 of 96.9%. The associated OLS coefficient, with continuous correlations on the left-hand-side, is 1.03 (95% CI, 1.027 – 1.032); this coefficient suggests that the continuous correlations are only marginally higher. So, for practical purposes, the correlations are about the same.

Next, in Figure 5, I show correlations between the factors over time. Specifically, I produce heatmaps based on the realized correlations between the CAPM residuals of the JKP+CZ factors; this methodology is the same as before and in the main text. But, unlike the previous figures, the correlations are computed for specific years; I sample every second year starting from 1997 and ending with 2019. The results show that the block structure between the factors appears to change over time. But, the clusters themselves remain fairly stable; that is, most of the factors within each of the 13 JKP clusters appear to maintain their within-cluster correlations across the sample. At the same time, there are clear instances – such as the heatmap for 2001 – where multiple clusters of factors merge into one. So, overall, between-cluster correlations appear to strongly vary over time, while within-cluster correlations are relatively stable.

Figure 5: CORRELATIONS BETWEEN THE JKP+CZ FACTORS BY YEAR



Note: I plot the correlations between the CAPM residuals of the 218 JKP+CZ factors for 12 individual years in my 25-year dataset. The residuals are obtained using simple regressions on the high-frequency Fama-French market factor. The left-hand-side of the first column of graphs shows the cluster assignments for each of the factors; these clusters originally come from Jensen, Kelly, and Pedersen (2021).

2.4 Cluster Portfolio Cojumps

For a factor to draw a jump risk premia, it must co-jump with other assets. In the main text, I test for cojumps using the procedure from Jacod and Todorov (2009). The authors propose the following test statistic

$$S^{(NoCoJ)}(\Delta_n) \equiv \sum_i (\Delta_i^n X^{(1)} \times \Delta_i^n X^{(2)})^2, \quad (2)$$

which is simply the sum of the squared product of two asset returns denoted $X^{(1)}$ and $X^{(2)}$. This statistic is used to test the null hypothesis of disjoint jumps between assets against the alternative of at least one cojump. Intuitively, the statistic works by forcing the returns in one asset to interact with the other, resulting in “explosions” whenever both assets jump together. If only one asset jumps or neither does, the statistic necessarily converges since Brownian motion scales in magnitude with the sampling frequency Δ_n .

To implement this test, I compute $S^{(NoCoJ)}(\Delta_n)$ using 15-minute returns over each calendar month in my sample period of 1996-2020 (i.e., 300 months in total). I perform the tests on a monthly basis to avoid an economically uninteresting rejection of the null.⁸ As for the choice of assets, I consider two groups: (1) my thirteen cluster portfolios plus the FF6 and (2) the FF6 plus a sample of large-cap stocks; this gives me 456 pairs in total. By focusing on the cluster portfolios, I vastly reduce the dimension of my analysis, ensuring interpretability and computational feasibility while still capturing the majority of the systematic risk (and thus systematic jumps) spanned by the zoo.

After computing the cojump test statistics, I compute p-values to perform statistical significance tests. However, computing these p-values is non-trivial since the asymptotic distribution of this test statistic under the null is highly non-Gaussian. This behavior arises because the test statistic is the sum of the squared product of returns, which will naturally produce a positive, right-skewed distribution. Hence, to compute the p-values, a Monte Carlo is necessary. The exact details are given in Jacod and Todorov (2009) and also summarized in Aït-Sahalia and Jacod (2014) – the reader should refer to the latter source for a concise description (see section 14.1.3).

To restate shortly here, the simulated distribution depends on two terms: one term that captures the interaction between the continuous components of the returns of the two assets and another that captures the interaction between the continuous and jump components. The first term ($\hat{H}_n$ in Aït-Sahalia and Jacod, 2014) merely “shifts” the asymptotic distribution and requires estimating spot covariance matrices. Here, it is necessary to define a window size and a truncation threshold. For the window size, I use 10 observations to balance the bias-variance trade-off. For the truncation threshold, I use the same bipower-variation-based approach described in the main text. Next is simulating the second term (S_n in Aït-Sahalia and Jacod, 2014), which does not converge in the limit and thus gives rise to a non-trivial asymptotic distribution. For this term, I reuse the estimated spot covariance matrices from earlier and draw 10,000 copies for the simulation.

⁸ Running the cojump tests on the entire sample at once will necessarily reject the null since it is highly likely that two assets will cojump at least once over a 25-year sample period. In contrast, it is much less likely that two assets cojump over an arbitrary month, particularly if they are not exposed to the same systematic risk. Hence, it is far more interesting to check if assets “regularly” cojump using the monthly tests.

This latter choice is sufficiently large while also computationally feasible. Finally, armed with the simulated asymptotic distribution, I compute the p-values. Although not directly defined in either text, these are easily computed by taking one minus the cumulative density of the simulated distribution evaluated at the test statistic estimate. The resulting p-values naturally correspond to one-sided tests, since the decision to reject is simply based on whether the estimated test statistic is “too large.”

3 Methodology

3.1 Jump Detection Notes

In the main text, I state that I identify jumps using the BV truncation method. For reference, this method uses the following equation:

$$u_n^s(i, m) \equiv \alpha \sqrt{\tau_{\ell(i),m}^s BV_{d(i),m}^s} \Delta_n^\varpi \quad (3)$$

$$u_n^f(i, k) \equiv \alpha \sqrt{\tau_{\ell(i),k}^f BV_{d(i),k}^f} \Delta_n^\varpi, \quad (4)$$

where $\ell(i)$ is the time-of-day for return i , $d(i)$ is the day for return i , $u_n^s(i, m)$ is the threshold for asset m and return i , and $u_n^f(i, k)$ is the threshold for factor k and return i . The intradaily bipower variation, BV , is defined in the usual way. Here, I define how the time-of-day correction, τ , is computed.

I compute time-of-day corrections on a yearly basis and use similar corrections for the intradaily and the overnight returns.⁹ The time-of-day correction is given by

$$I_i^y = \{j \in \{1, \dots, n\} : year(j) = year(i)\}$$

$$I_i^\ell = \{j \in \{1, \dots, n\} : \ell(j) = \ell(i)\}$$

$$b_{\ell(i),m}^s = \begin{cases} b_{\ell(i)+1,m}^s & \text{if } \ell(i) = 2 \\ \frac{1}{\#(I_i^\ell \cap I_i^y)} \sum_{j \in (I_i^\ell \cap I_i^y)} |\Delta_j^n P^m| |\Delta_{j-1}^n P^m| & \text{otherwise} \end{cases} \quad (5)$$

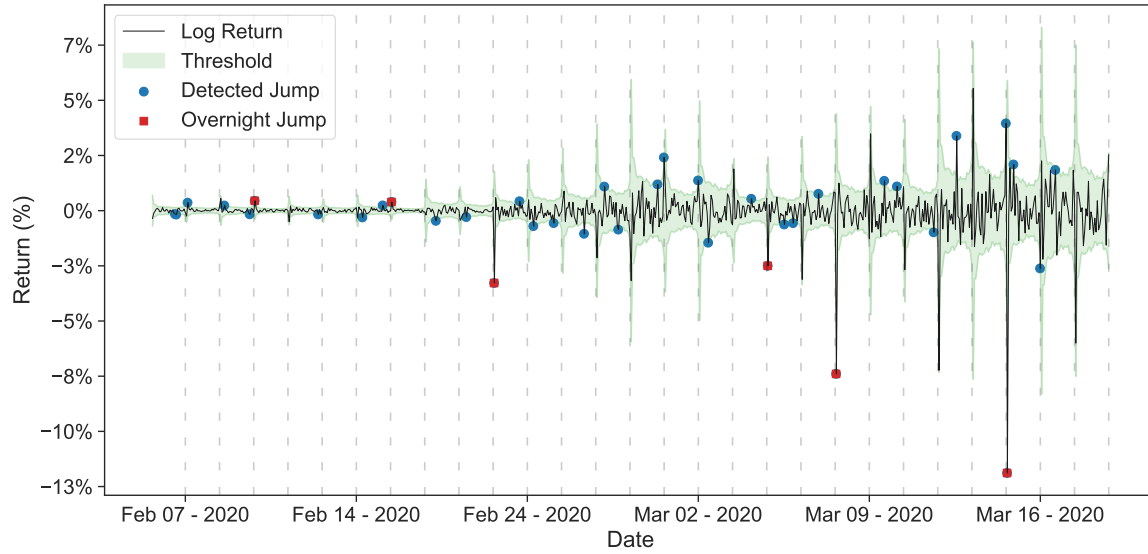
$$\tau_{\ell(i),m}^s = \begin{cases} \frac{b_{\ell(i),m}^s}{\frac{1}{\#I_i^\ell - 1} \sum_{j \in (I_i^\ell \setminus \{1\})} b_{\ell(j),m}^s} & \text{(intradaily)} \\ \frac{b_{\ell(i),m}^s}{\frac{1}{\#I_i^\ell - 1} \sum_{j \in (I_i^\ell \setminus \{1\})} b_{\ell(j),m}^s} \cdot \frac{\frac{1}{\#(I_i^\ell \cap I_i^y)} \sum_{j \in (I_i^\ell \cap I_i^y)} |\Delta_j^n P^m|^2}{\frac{1}{\#((I_n \setminus I_i^\ell) \cap I_i^y)} \sum_{j \in ((I_n \setminus I_i^\ell) \cap I_i^y)} |\Delta_j^n P^m|^2} & \text{(overnight)} \end{cases} \quad (6)$$

This correction is defined for all intervals i , their times of day $\ell(i)$, and assets m . The set I_i^y contains all the indices that fall within the same year as index i , while the set I_i^ℓ contains all the indices that fall within the same time-of-day as index i . Next, the variable $b_{\ell(i)}^m$ is the average of the product of all adjacent absolute returns for indices in the same year and time-of-day as index i . To avoid involving the overnight return when constructing the intradaily version of this variable, I set the first intradaily value, b_2^m , equal to the second intradaily value, b_3^m .

Lastly, the time-of-day correction given by Equation 6 and takes on two forms. For the intradaily returns, I define this correction in the usual way – $b_{\ell(i),m}^s$ is divided by its average across all intradaily times-of-day. The overnight term, $b_{1,m}^s$, is excluded from the summation to prevent the denominator from becoming excessively large. In the data, this will create a u-shape pattern in the correction, since volatility tends to be higher near the beginning and end of the day than the middle of the day. For the overnight returns, I take a slightly different approach. Recall that the truncation equation (Equation 4) involves the sequence Δ_n^ϖ , which would be extremely large if Δ_n is set to the length of the overnight period. Rather than adjusting Δ_n , which would result in nonsensical thresholds, I introduce another term to the correction to indirectly adjust for the length of

⁹One could also treat all the overnight returns as jumps, but this will not substantially change the results in the main text.

Figure 6: JUMP DETECTION EXAMPLE – FF MARKET FACTOR, FEB-MAR 2020



Note: This figure shows the log returns for the high-frequency Fama-French Market factor from February 7, 2020 to March 18, 2020. The green area is defined by the jump truncation threshold given by Equation 4. The blue markers indicate detected intradaily jumps, while the red markers indicate detected overnight jumps.

the overnight period. In the definition, this adjustment term is the second fraction, which is simply the ratio of the average overnight realized variance to the average intradaily realized variance. One could interpret this as implicitly transforming the Δ_n variable for the overnight return into a business-time equivalent.¹⁰

The time-of-day indicators for the factors are defined analogously. Overall, the approach for the intradaily returns is equivalent to the approach generally used in the literature except that I recompute the correction for each year. The modification for the overnight returns is basically a heuristic solution. The results are also generally robust to simply treating all the overnight returns as jumps.

Is this procedure reasonable for classifying jumps? In Figure 6, I plot the jump detection results for the Fama-French market factor. The data starts near the beginning of February and goes up to the middle of March; this particular period captures the beginning of the COVID panic.¹¹ The vertical dashed lines separate trading days, the green area shows the jump detection threshold defined by Equation 4, and the marked points are detected jump returns.

The truncation threshold shows some clear patterns. Firstly, volatility drives the threshold wider as it should. Secondly, there is a u-shaped time-of-day pattern in the

¹⁰ Note that the overnight period is 17.5 hours long, while my sample frequency is 15-minutes. Consequently, based on naive Brownian scaling, adjusting the Δ_n^ϖ term in the standard truncation equation for the length of the overnight period would make it $\sqrt{66}$ times larger. This is clearly far too large for the overnight return truncation, since the overnight volatility is not simply $\sqrt{66}$ times higher than the intradaily volatility. Hence, a natural way to scale the Δ_n term would be to instead scale $\Delta_n = 15 \text{ minutes}$ by the ratio between the average overnight and intradaily realized vol. Since the exponent on Δ_n in the truncation equation (ϖ) is close to $1/2$, this natural solution can be heuristically implemented by adjusting the time-of-day indicator.

¹¹ The overnight jump for March 15, 2020 is related to the following FOMC statement: <https://www.federalreserve.gov/newsevents/pressreleases/monetary20200315a.htm>.

threshold – this pattern comes from the u-shaped pattern in volatility, which is well known by the literature. Thirdly, the detected jumps generally appear to be jumps. Each day, there is about one intraday jump. Since I use $\alpha = 4.0$ in Equation 4, it is generally unlikely that these returns are generated by Brownian motion; 4σ Gaussian returns occur with a $p < 0.00007$. Additionally, based on Figure 6, my overnight return classification procedure appears to be reasonable, if not slightly conservative. There are 6 overnight returns classified as jumps in the plotted 28-day sample. But, there are multiple large overnight returns in March that do not appear to be classified as jumps. So, there may be a few more overnight jumps than I have actually classified. Still, my main results would be qualitatively the same if I classified 100% of overnight returns as jumps. This robustness check strongly suggests that a more liberal threshold for overnight returns would not substantially change my results.

3.2 Regression Notes

To simplify the exposition for the regression methodology, I omitted the tuning parameters $\rho, \rho' > 0$ which are used to define the jump partitions $\bar{B}^h(\rho, \rho')$ from Assumption B.V. To understand this notation, consider a specific example. Imagine we split jumps into two buckets: jumps larger than 1% and jumps not larger than 1%. A jump that is exactly 1% in size would be categorized in the first bucket. However, a small amount of contamination from the Brownian motion component of a factor may cause a particular jump to become misclassified in practice; this is particularly problematic for jumps that lie exactly on the boundary. In order to avoid this misclassification, we would instead define the jump risk factors by doing

$$\hat{R}_{j,k,h} = \Delta_j^n F^{k(h)} \cdot \mathbb{1}_{[\Delta_j^n F^{k(h)} \in \bar{B}^h(\rho, \rho')]}$$

with $\rho > 0$ small and $\rho' > 0$ large. So, in the previous example, we might assign all detected jumps that are larger than 1.01% to the first bucket and all detected jumps that are smaller than 0.99% to the second bucket. This would prevent any misclassification, because the contamination from the Brownian motion component would approach zero as $\Delta_n \rightarrow 0$. However, I implement my regressions using positive and negative jumps, which are much less likely to be misclassified. In fact, as an artifact of the bipower truncation approach, jumps close to zero are naturally omitted in a finite-sample setting. In fact, the issue described above is really only a concern in an asymptotic setting. In finite sample and in practice, one could pick ρ to be extremely small and ρ' to be extremely large to the point where using $\bar{B}^h(\rho, \rho')$ has no effect on the definition of the regressors. All in all, the tuning parameters not mentioned in the main text have no effect in the practical implementation of the regression.

Additionally, I simplified the exposition for the central limit theorem. The CLT requires defining the variable V_t , a process that determines the variance of the risk premia estimates. This process is defined element-wise as:

$$V_t^{j,j'} = \sum_{m,m'=1}^M \int_0^t \eta_s^{j,m} \eta_s^{j',m'} d[P^m, P^{m'}]_s. \quad (7)$$

In order to obtain a CLT, I need to assume that $(1/t)V_t \xrightarrow{p} V_\infty$ and that there exists a sequence of subsets $\Omega_n \subset \Omega$ with $P(\Omega_n) \rightarrow 1$ where $\hat{V}$ is invertible on Ω_n . This allows us to treat $\hat{V}$ as an estimator for V_∞ . Finally, under the assumptions given in the main text, Theorem 1 from Aït-Sahalia, Jacod, and Xiu (2021) gives,

$$T_n \hat{V}^{-1/2} (\hat{\Lambda} - \Lambda_{T_n}) \xrightarrow{d} \mathcal{N}(0, I_{K+H}). \quad (8)$$

on the sequence Ω_n .

There are a few things to note about this result. Firstly, unlike the standard Fama-MacBeth estimator, we use the realized covariance matrix rather than the standard covariance matrix to compute the variance of $\hat{\Lambda}$. Recall that the spot risk premia converge to zero as $\Delta_n \rightarrow 0$, so a 'realized' variance is necessary in this context.¹² Secondly, the rate of convergence, after normalizing the variance $\hat{V}$, is $\sqrt{T_n}$. This is obtained through a martingale CLT and an asymptotic negligibility argument stated explicitly in Assumption B.VI. In other words, $\Delta_n \rightarrow 0$ is still necessary for consistent estimation, but error from T_n dominates by assumption. Lastly, there is a sequence of sets Ω_n on which convergence is obtained. This is necessary in a theoretical setting to ensure that the variance matrix is invertible but is inconsequential in practice.

¹² Note that my estimator of this realized variance (given in the main text) is constructed slightly differently than Aït-Sahalia, Jacod, and Xiu (2021). That is, instead of using the monthly realized covariance matrix of the spot risk premia estimates (intervals of $q_n \Delta_n$), I use the higher frequency realized covariance matrix (intervals of Δ_n). This decision does not affect the consistency of $\hat{V}$ but will improve its precision in finite sample. Both estimators are also fairly intuitive – they simply estimate the realized volatility of the risk premia estimates which are just portfolio returns defined by the cross-sectional regressions.

4 Robustness Checks

I now discuss the robustness of my main results with respect to various tuning parameters. In particular, I focus on the risk premia estimates and the variance decomposition. For the risk premia estimates, I consider just the workhorse factors and cluster portfolios, since the latter portfolios essentially summarize the factor zoo. I check robustness with respect to three tuning parameters: the jump truncation threshold α (used in the sequence u_n), the jump beta estimation window, and the continuous beta estimation window.

I provide a map of my results in Table 2. Each entry provides a link to a robustness check table. These tables represent my main results and are produced by modifying one of the several tuning parameters. Note that my default tuning parameters are the same as those used in the main text.¹³ As shown by the table, my robustness checks are fairly extensive, so I will briefly summarize my findings.

4.1 Results

There are three key sets of results to perform robustness checks on: the risk premia estimates for the workhorse factors, the risk premia estimates for the cluster portfolios, and the variance decomposition. To start, with respect to the workhorse factor risk premia, my results are robust to altering all three tuning parameters. That is, regardless of the tuning parameters used, I find statistically and economically significant risk premia associated with the jumps and negative semijumps in FF-MKT along with the continuous returns in FF-RMW. Second, with respect to the cluster portfolio risk premia, I find exactly four risk factors that survive all the robustness checks. These are negative jumps in Accruals, positive jumps in Skewness, positive jumps in Investment, and continuous returns in Profitability; all four obtain a q-value less than 10% in every robustness check. The remaining factors that were significant in main text fail to pass at least one of the robustness checks.¹⁴ Thirdly, the variance decomposition results are broadly the same. That is, in all of the robustness checks, the lower bound of the variance explained by the jump component exceeds the upper bound explained by the continuous component. So, in general, my results appear to be robust with respect to the tuning parameters.

¹³ For the bipower truncation method used to identify jumps, I use $\alpha = 4.0$ and $\varpi = 0.49$. Next, for the continuous beta estimation, I use a monthly rolling window; for the jump beta estimation, I use a yearly rolling window. And, in order to cut down on microstructure noise, I use a 15-minute sampling frequency. Lastly, as test assets, I use the top 1000 stocks by market cap and the factor zoo itself.

¹⁴ It should be noted that the q-value computation here relies on estimate(s) of π_0 ; to be consistent with the main text, I reused $\pi_0 = 40\%$ when identifying these four statistically significant risk factors. However, if I instead use all of the cluster risk premia estimates from the robustness checks to recompute an estimate of π_0 , I obtain $\hat{\pi}_0 = 41\%$. In other words, a more “accurate” estimate of π_0 would not change these findings.

4.2 Tables

Table 2: ROBUSTNESS CHECKS – GUIDE

	Tuning Parameter Value	Risk Premia Estimates		Cluster Portfolios	Variance Decomposition
		Workhorse Factors			
		Cts/Jmp	SemiJump		
Jump Truncation (α)	2.5	Table 3	Table 4	Table 5	Table 6
	3.5	Table 7	Table 8	Table 9	Table 10
Continuous Beta Window	14 days	Table 11	Table 12	Table 13	Table 14
	60 days	Table 15	Table 16	Table 17	Table 18
Jump Beta Window	180 days	Table 19	Table 20	Table 21	Table 22
	540 days	Table 23	Table 24	Table 25	Table 26

Note: For each robustness check and result, I report the associated table. For the variance decomposition results, I only report results for the Cts/Jmp specification. The tuning parameter α refers to the standard deviation threshold used to define the truncation parameter u_n . The default values for each of the left-hand-side tuning parameters are 4.0, 365 days, and 30 days.

4.2.1 Jump Truncation Threshold: 2.5

Table 3: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND JUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	1.52	(0.40)	−0.05	(−0.01)	1.62	(0.48)	1.30	(0.39)
	Jump	2.65	(0.90)	6.22**	(2.31)	6.93***	(2.66)	7.51***	(2.90)
FF SMB	Continuous			−0.40	(−0.20)	−0.10	(−0.05)	0.08	(0.04)
	Jump			0.76	(0.39)	2.53	(1.36)	3.16*	(1.73)
FF HML	Continuous			−1.32	(−0.71)	−2.30	(−1.30)	−2.39	(−1.37)
	Jump			0.54	(0.34)	0.84	(0.55)	1.18	(0.81)
FF RMW	Continuous					3.58***	(3.01)	3.55***	(3.06)
	Jump					3.62***	(3.31)	3.39***	(3.17)
FF CMA	Continuous					0.44	(0.38)	0.99	(0.87)
	Jump					−0.45	(−0.43)	−0.43	(−0.41)
FF UMD	Continuous							1.68	(0.76)
	Jump							−2.46	(−1.20)
R^2		31.9%		36.2%		38.0%		39.0%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 4: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND SEMIJUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	0.91	(0.24)	−0.04	(−0.01)	1.78	(0.53)	1.47	(0.45)
	Neg Jump	3.88	(1.61)	4.73**	(2.24)	4.74**	(2.29)	4.92**	(2.41)
	Pos Jump	−0.33	(−0.14)	1.99	(0.97)	2.37	(1.19)	2.60	(1.33)
FF SMB	Continuous			−0.55	(−0.28)	−0.34	(−0.18)	−0.20	(−0.11)
	Neg Jump			2.86*	(1.93)	3.12**	(2.17)	3.49**	(2.44)
	Pos Jump			−1.72	(−1.14)	−0.61	(−0.42)	−0.54	(−0.37)
FF HML	Continuous			−1.24	(−0.66)	−2.07	(−1.17)	−2.31	(−1.34)
	Neg Jump			0.60	(0.48)	0.47	(0.39)	0.67	(0.58)
	Pos Jump			−0.69	(−0.61)	−0.09	(−0.08)	0.04	(0.03)
FF RMW	Continuous					3.38***	(2.87)	3.31***	(2.88)
	Neg Jump					2.13**	(2.48)	2.07**	(2.42)
	Pos Jump					1.25	(1.56)	0.94	(1.21)
FF CMA	Continuous					0.64	(0.55)	1.10	(0.97)
	Neg Jump					0.96	(1.20)	0.88	(1.10)
	Pos Jump					−1.75**	(−2.23)	−1.33*	(−1.70)
FF UMD	Continuous							1.73	(0.80)
	Neg Jump							−1.17	(−0.77)
	Pos Jump							−1.43	(−0.94)
R^2		32.4%		36.8%		39.0%		40.1%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 5: CLUSTER PORTFOLIOS – CONTINUOUS AND SEMIJUMP RISK PREMIA

Cluster Portfolio	Component	Risk Premia	t -stat	q-value		
				$\pi_0 = 30\%$	$\pi_0 = 40\%$	$\pi_0 = 50\%$
Accruals	Neg. Jump	2.13%	2.99	0.032	0.043	0.054
Skewness	Pos. Jump	−2.56%	−2.54	0.056	0.074	0.093
Investment	Pos. Jump	−2.08%	−2.42	0.056	0.074	0.093
Profitability	Continuous	2.98%	2.28	0.056	0.074	0.093
Seasonality	Pos. Jump	−2.98%	−2.26	0.056	0.074	0.093
Profitability	Neg. Jump	2.01%	2.11	0.067	0.090	0.112
Size	Neg. Jump	2.30%	1.89	0.077	0.102	0.128
Low Risk	Continuous	4.27%	1.87	0.077	0.102	0.128
Seasonality	Neg. Jump	2.53%	1.84	0.077	0.102	0.128
Debt Issuance	Neg. Jump	0.92%	1.77	0.077	0.102	0.128
Low Risk	Pos. Jump	−2.45%	−1.74	0.077	0.102	0.128
Leverage	Pos. Jump	−1.93%	−1.73	0.077	0.102	0.128
Leverage	Neg. Jump	1.79%	1.72	0.077	0.102	0.128
Investment	Neg. Jump	1.44%	1.55	0.101	0.135	0.169
Quality	Pos. Jump	1.69%	1.43	0.116	0.154	0.193

Note: Each row reports a risk premia estimate for a particular cluster portfolio; clusters are based on the methodology described in [Section 2.1](#) and their representative portfolios are formed using the first principal component of the factor returns within each cluster. I report estimates are obtained from running separate Cts-Time Fama-MacBeth regressions for each factor; for these regressions, the span assets are the FF3 plus a given cluster portfolio. The risk-premia estimates are annualized. The q-values are computed using the procedure given in Storey (2002) under various parameterizations for the true proportion of nulls, π_0 . Only the top fifteen estimates by p-value are included for brevity.

Table 6: VARIANCE DECOMPOSITION RESULTS – CTS-TIME FAMA-MACBETH WITH CONTINUOUS/JUMP RISK FACTORS

<i>Panel A. Using Continuous and Jump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	48.33	6.22	10.56	37.77	42.11
All	C14	57.69	7.39	11.13	46.56	50.30
Stocks	FF6	48.06	5.99	9.56	38.50	42.07
Stocks	C14	57.67	8.06	10.96	46.71	49.62
Portfolios	FF6	50.27	7.82	17.64	32.63	42.45
Portfolios	C14	57.81	2.74	12.31	45.50	55.07
<i>Panel B. Using Continuous and SemiJump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	50.04	5.85	12.94	37.10	44.19
All	C14	59.11	6.93	16.24	42.87	52.17
Stocks	FF6	49.51	5.81	12.10	37.41	43.71
Stocks	C14	59.17	7.48	15.67	43.50	51.69
Portfolios	FF6	53.69	6.14	18.81	34.88	47.55
Portfolios	C14	58.71	3.12	20.21	38.50	55.59

Note: For each specification listed in the second column, I run a Cts-Time Fama-MacBeth regression to estimate betas and spot risk premia for each month. The span assets include the continuous and jump component of each factor in the specification; the test assets are the yearly top 1000 stocks by market cap and all the portfolios in my factor zoo. In the first panel, the regressions involve continuous and jump risk factors, while, in the second panel, the regressions further involve semijump risk factors. For each group of test assets, I then estimate $\tilde{R}_J^2$, $\tilde{R}_C^2$, and $\tilde{R}^2$. The upper and lower bounds for the explanatory power of the jump and continuous components are given by the sets $\mathcal{S}_J^R$ and $\mathcal{S}_C^R$, which are defined in Equation 10 and Equation 11; these sets are also truncated when they exceed 0% or 100%. All values are reported as percentages.

4.2.2 Jump Truncation Threshold: 3.5

Table 7: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND JUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	2.46	(0.67)	0.90	(0.26)	2.69	(0.78)	2.45	(0.71)
	Jump	1.67	(0.73)	4.09**	(2.03)	3.90**	(2.06)	4.49**	(2.41)
FF SMB	Continuous			−0.36	(−0.19)	0.06	(0.03)	0.20	(0.11)
	Jump			1.31	(1.03)	1.54	(1.26)	2.10*	(1.78)
FF HML	Continuous			−1.20	(−0.66)	−2.12	(−1.22)	−1.72	(−0.99)
	Jump			0.41	(0.36)	0.91	(0.84)	0.92	(0.89)
FF RMW	Continuous					4.28***	(3.61)	4.24***	(3.65)
	Jump					1.57**	(2.07)	1.35*	(1.80)
FF CMA	Continuous					0.49	(0.42)	1.08	(0.95)
	Jump					−0.36	(−0.47)	−0.14	(−0.18)
FF UMD	Continuous							2.18	(1.01)
	Jump							−2.20	(−1.56)
R^2		31.7%		36.1%		38.0%		39.2%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 8: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND SEMIJUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	2.16	(0.59)	0.64	(0.18)	2.58	(0.75)	2.42	(0.71)
	Neg Jump	2.71	(1.57)	3.28**	(2.26)	2.21	(1.60)	2.32*	(1.68)
	Pos Jump	−0.47	(−0.27)	1.67	(1.13)	2.45*	(1.73)	2.74*	(1.96)
FF SMB	Continuous			−0.50	(−0.26)	−0.04	(−0.02)	0.12	(0.07)
	Neg Jump			1.30	(1.30)	1.30	(1.32)	1.56	(1.64)
	Pos Jump			0.00	(0.00)	0.38	(0.45)	0.57	(0.68)
FF HML	Continuous			−1.22	(−0.67)	−1.98	(−1.13)	−1.79	(−1.04)
	Neg Jump			0.16	(0.18)	0.03	(0.03)	0.23	(0.28)
	Pos Jump			−0.37	(−0.46)	0.29	(0.39)	0.38	(0.51)
FF RMW	Continuous					4.11***	(3.50)	4.04***	(3.51)
	Neg Jump					1.36**	(2.15)	1.23**	(1.96)
	Pos Jump					0.41	(0.89)	0.11	(0.24)
FF CMA	Continuous					0.55	(0.47)	1.06	(0.93)
	Neg Jump					0.66	(1.07)	0.73	(1.20)
	Pos Jump					−1.18**	(−2.22)	−0.96*	(−1.79)
FF UMD	Continuous							2.13	(0.99)
	Neg Jump							−0.66	(−0.64)
	Pos Jump							−1.11	(−1.08)
R^2		32.2%		36.8%		39.0%		40.2%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 9: CLUSTER PORTFOLIOS – CONTINUOUS AND SEMIJUMP RISK PREMIA

Cluster Portfolio	Component	Risk Premia	t -stat	q-value		
				$\pi_0 = 30\%$	$\pi_0 = 40\%$	$\pi_0 = 50\%$
Skewness	Pos. Jump	−2.29%	−3.22	0.015	0.020	0.025
Accruals	Neg. Jump	1.41%	2.85	0.018	0.024	0.030
Profitability	Continuous	3.61%	2.83	0.018	0.024	0.030
Size	Neg. Jump	2.04%	2.64	0.025	0.033	0.041
Investment	Pos. Jump	−1.59%	−2.54	0.026	0.035	0.043
Quality	Continuous	3.28%	2.14	0.063	0.084	0.105
Leverage	Neg. Jump	1.48%	2.01	0.064	0.085	0.106
Debt Issuance	Pos. Jump	−0.76%	−2.01	0.064	0.085	0.106
Low Risk	Continuous	4.47%	1.97	0.064	0.085	0.106
Debt Issuance	Continuous	1.39%	1.80	0.081	0.109	0.136
Low Risk	Pos. Jump	−1.76%	−1.74	0.081	0.109	0.136
Debt Issuance	Neg. Jump	0.65%	1.73	0.081	0.109	0.136
Profitability	Neg. Jump	1.14%	1.63	0.093	0.124	0.155
Seasonality	Pos. Jump	−1.54%	−1.56	0.100	0.133	0.167
Momentum	Continuous	3.13%	1.50	0.102	0.136	0.171

Note: Each row reports a risk premia estimate for a particular cluster portfolio; clusters are based on the methodology described in [Section 2.1](#) and their representative portfolios are formed using the first principal component of the factor returns within each cluster. I report estimates are obtained from running separate Cts-Time Fama-MacBeth regressions for each factor; for these regressions, the span assets are the FF3 plus a given cluster portfolio. The risk-premia estimates are annualized. The q-values are computed using the procedure given in Storey (2002) under various parameterizations for the true proportion of nulls, π_0 . Only the top fifteen estimates by p-value are included for brevity.

Table 10: VARIANCE DECOMPOSITION RESULTS – CTS-TIME FAMA-MACBETH WITH CONTINUOUS/JUMP RISK FACTORS

<i>Panel A. Using Continuous and Jump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	49.92	9.52	21.72	28.19	40.39
All	C14	57.33	12.31	20.63	36.70	45.02
Stocks	FF6	49.32	8.13	20.88	28.44	41.19
Stocks	C14	56.63	11.09	20.93	35.70	45.55
Portfolios	FF6	54.10	19.33	27.69	26.41	34.77
Portfolios	C14	62.23	18.48	20.89	41.34	43.75
<i>Panel B. Using Continuous and SemiJump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	50.49	8.90	22.31	28.17	41.59
All	C14	57.05	10.36	18.22	38.84	46.70
Stocks	FF6	49.72	7.92	21.47	28.25	41.80
Stocks	C14	56.37	9.56	18.20	38.17	46.81
Portfolios	FF6	55.90	15.80	28.29	27.61	40.10
Portfolios	C14	61.85	15.96	18.35	43.50	45.89

Note: For each specification listed in the second column, I run a Cts-Time Fama-MacBeth regression to estimate betas and spot risk premia for each month. The span assets include the continuous and jump component of each factor in the specification; the test assets are the yearly top 1000 stocks by market cap and all the portfolios in my factor zoo. In the first panel, the regressions involve continuous and jump risk factors, while, in the second panel, the regressions further involve semijump risk factors. For each group of test assets, I then estimate $\tilde{R}_J^2$, $\tilde{R}_C^2$, and $\tilde{R}^2$. The upper and lower bounds for the explanatory power of the jump and continuous components are given by the sets $\mathcal{S}_J^R$ and $\mathcal{S}_C^R$, which are defined in Equation 10 and Equation 11; these sets are also truncated when they exceed 0% or 100%. All values are reported as percentages.

4.2.3 Continuous Beta Window: 14 Days

Table 11: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND JUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	1.97	(0.57)	0.52	(0.16)	0.57	(0.18)	0.40	(0.13)
	Jump	1.45	(0.57)	4.10*	(1.76)	5.68**	(2.51)	6.45***	(2.86)
FF SMB	Continuous			1.51	(0.91)	1.49	(0.94)	1.70	(1.09)
	Jump			−0.22	(−0.14)	0.61	(0.40)	1.22	(0.82)
FF HML	Continuous			−0.56	(−0.35)	−1.31	(−0.86)	−1.14	(−0.76)
	Jump			0.07	(0.05)	0.03	(0.02)	−0.07	(−0.05)
FF RMW	Continuous					3.38***	(3.45)	3.33***	(3.46)
	Jump					2.76***	(3.00)	2.68***	(2.97)
FF CMA	Continuous					0.80	(0.82)	1.01	(1.06)
	Jump					−0.62	(−0.70)	−0.50	(−0.57)
FF UMD	Continuous							1.86	(0.98)
	Jump							−1.06	(−0.62)
R^2		30.8%		35.0%		36.9%		37.8%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 12: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND SEMIJUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	1.74	(0.51)	0.33	(0.10)	0.52	(0.17)	0.53	(0.17)
	Neg Jump	1.60	(0.84)	2.93*	(1.68)	3.25*	(1.91)	3.15*	(1.84)
	Pos Jump	0.32	(0.16)	1.80	(1.04)	2.41	(1.43)	2.86*	(1.72)
FF SMB	Continuous			1.47	(0.90)	1.32	(0.84)	1.45	(0.94)
	Neg Jump			0.95	(0.78)	1.10	(0.91)	1.24	(1.06)
	Pos Jump			−1.27	(−1.10)	−0.61	(−0.55)	−0.25	(−0.22)
FF HML	Continuous			−0.52	(−0.33)	−1.05	(−0.69)	−1.06	(−0.71)
	Neg Jump			−0.48	(−0.46)	−0.63	(−0.62)	−0.67	(−0.68)
	Pos Jump			−0.38	(−0.41)	0.18	(0.20)	0.06	(0.07)
FF RMW	Continuous					3.25***	(3.34)	3.16***	(3.30)
	Neg Jump					1.54**	(2.07)	1.44*	(1.95)
	Pos Jump					1.12*	(1.86)	1.04*	(1.79)
FF CMA	Continuous					0.87	(0.90)	1.01	(1.07)
	Neg Jump					0.62	(0.88)	0.57	(0.82)
	Pos Jump					−1.42**	(−2.22)	−1.05*	(−1.65)
FF UMD	Continuous							1.78	(0.94)
	Neg Jump							−0.80	(−0.62)
	Pos Jump							−0.01	(−0.01)
R^2		31.2%		35.7%		37.9%		38.9%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 13: CLUSTER PORTFOLIOS – CONTINUOUS AND SEMIJUMP RISK PREMIA

Cluster Portfolio	Component	Risk Premia	t -stat	q-value		
				$\pi_0 = 30\%$	$\pi_0 = 40\%$	$\pi_0 = 50\%$
Size	Neg. Jump	2.46%	2.57	0.042	0.056	0.070
Debt Issuance	Continuous	1.54%	2.48	0.042	0.056	0.070
Skewness	Pos. Jump	−2.03%	−2.39	0.042	0.056	0.070
Accruals	Neg. Jump	1.44%	2.37	0.042	0.056	0.070
Investment	Pos. Jump	−1.76%	−2.36	0.042	0.056	0.070
Leverage	Neg. Jump	1.81%	2.11	0.059	0.078	0.098
Profitability	Continuous	2.30%	2.11	0.059	0.078	0.098
Quality	Continuous	2.68%	2.01	0.064	0.086	0.107
Debt Issuance	Neg. Jump	0.86%	1.96	0.064	0.086	0.107
Seasonality	Pos. Jump	−2.14%	−1.88	0.070	0.093	0.116
Low Risk	Continuous	3.68%	1.77	0.081	0.108	0.135
Low Risk	Pos. Jump	−2.01%	−1.69	0.089	0.118	0.148
Leverage	Pos. Jump	−1.56%	−1.54	0.101	0.135	0.169
Quality	Pos. Jump	1.42%	1.53	0.101	0.135	0.169
Profitability	Pos. Jump	1.16%	1.51	0.101	0.135	0.169

Note: Each row reports a risk premia estimate for a particular cluster portfolio; clusters are based on the methodology described in [Section 2.1](#) and their representative portfolios are formed using the first principal component of the factor returns within each cluster. I report estimates are obtained from running separate Cts-Time Fama-MacBeth regressions for each factor; for these regressions, the span assets are the FF3 plus a given cluster portfolio. The risk-premia estimates are annualized. The q-values are computed using the procedure given in Storey (2002) under various parameterizations for the true proportion of nulls, π_0 . Only the top fifteen estimates by p-value are included for brevity.

Table 14: VARIANCE DECOMPOSITION RESULTS – CTS-TIME FAMA-MACBETH WITH CONTINUOUS/JUMP RISK FACTORS

<i>Panel A. Using Continuous and Jump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	46.36	1.61	9.44	36.92	44.74
All	C14	54.49	7.32	12.71	41.78	47.17
Stocks	FF6	45.28	0.90	8.25	37.03	44.38
Stocks	C14	53.25	6.40	12.42	40.84	46.85
Portfolios	FF6	53.82	6.53	17.64	36.18	47.28
Portfolios	C14	63.00	13.65	14.73	48.27	49.35
<i>Panel B. Using Continuous and SemiJump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	47.02	1.68	13.33	33.69	45.34
All	C14	53.98	6.32	14.03	39.95	47.66
Stocks	FF6	45.76	1.08	12.33	33.43	44.68
Stocks	C14	52.86	5.59	13.49	39.37	47.27
Portfolios	FF6	55.82	5.85	20.31	35.50	49.96
Portfolios	C14	61.68	11.32	17.75	43.93	50.35

Note: For each specification listed in the second column, I run a Cts-Time Fama-MacBeth regression to estimate betas and spot risk premia for each month. The span assets include the continuous and jump component of each factor in the specification; the test assets are the yearly top 1000 stocks by market cap and all the portfolios in my factor zoo. In the first panel, the regressions involve continuous and jump risk factors, while, in the second panel, the regressions further involve semijump risk factors. For each group of test assets, I then estimate $\tilde{R}_J^2$, $\tilde{R}_C^2$, and $\tilde{R}^2$. The upper and lower bounds for the explanatory power of the jump and continuous components are given by the sets $\mathcal{S}_J^R$ and $\mathcal{S}_C^R$, which are defined in Equation 10 and Equation 11; these sets are also truncated when they exceed 0% or 100%. All values are reported as percentages.

4.2.4 Continuous Beta Window: 60 Days

Table 15: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND JUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	2.68	(0.68)	0.36	(0.10)	2.16	(0.58)	1.89	(0.51)
	Jump	0.87	(0.31)	5.10**	(2.27)	5.18**	(2.46)	6.12***	(2.95)
FF SMB	Continuous			1.39	(0.64)	1.78	(0.85)	1.40	(0.67)
	Jump			−0.29	(−0.19)	1.22	(0.81)	2.10	(1.46)
FF HML	Continuous			−0.69	(−0.34)	−2.51	(−1.30)	−2.36	(−1.23)
	Jump			0.30	(0.23)	0.44	(0.36)	0.39	(0.34)
FF RMW	Continuous					6.13***	(4.38)	5.98***	(4.34)
	Jump					0.89	(1.00)	0.61	(0.70)
FF CMA	Continuous					2.22*	(1.66)	2.23*	(1.69)
	Jump					−0.72	(−0.85)	−0.47	(−0.56)
FF UMD	Continuous							1.47	(0.63)
	Jump							−1.95	(−1.21)
R^2		31.1%		36.0%		38.3%		39.4%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 16: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND SEMIJUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	1.55	(0.40)	−0.21	(−0.06)	1.70	(0.46)	1.61	(0.44)
	Neg Jump	2.71	(1.36)	3.83**	(2.27)	3.28**	(2.04)	3.71**	(2.33)
	Pos Jump	−0.24	(−0.12)	2.37	(1.42)	2.67*	(1.66)	2.96*	(1.87)
FF SMB	Continuous			1.35	(0.62)	1.71	(0.81)	1.51	(0.73)
	Neg Jump			0.62	(0.51)	0.35	(0.30)	0.80	(0.70)
	Pos Jump			−1.14	(−1.02)	0.44	(0.40)	0.62	(0.58)
FF HML	Continuous			−0.77	(−0.38)	−2.37	(−1.23)	−2.33	(−1.22)
	Neg Jump			0.39	(0.40)	−0.12	(−0.13)	0.17	(0.18)
	Pos Jump			−0.31	(−0.34)	0.58	(0.68)	0.36	(0.43)
FF RMW	Continuous					5.83***	(4.19)	5.56***	(4.06)
	Neg Jump					0.58	(0.80)	0.46	(0.65)
	Pos Jump					0.09	(0.15)	−0.08	(−0.14)
FF CMA	Continuous					2.24*	(1.69)	2.06	(1.59)
	Neg Jump					0.32	(0.48)	0.34	(0.50)
	Pos Jump					−1.29**	(−2.09)	−0.90	(−1.47)
FF UMD	Continuous							1.55	(0.67)
	Neg Jump							−0.73	(−0.62)
	Pos Jump							−0.97	(−0.82)
R^2		31.5%		36.6%		39.2%		40.4%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 17: CLUSTER PORTFOLIOS – CONTINUOUS AND SEMIJUMP RISK PREMIA

Cluster Portfolio	Component	Risk Premia	t -stat	q-value		
				$\pi_0 = 30\%$	$\pi_0 = 40\%$	$\pi_0 = 50\%$
Investment	Pos. Jump	−2.19%	−3.00	0.015	0.020	0.025
Accruals	Neg. Jump	1.71%	2.94	0.015	0.020	0.025
Low Risk	Continuous	7.24%	2.89	0.015	0.020	0.025
Size	Neg. Jump	2.60%	2.77	0.016	0.022	0.027
Leverage	Neg. Jump	2.21%	2.63	0.020	0.027	0.034
Skewness	Pos. Jump	−2.05%	−2.46	0.023	0.031	0.039
Leverage	Continuous	−4.43%	−2.45	0.023	0.031	0.039
Profitability	Continuous	3.49%	2.41	0.023	0.031	0.039
Investment	Continuous	3.39%	2.20	0.033	0.044	0.054
Seasonality	Pos. Jump	−2.39%	−2.20	0.033	0.044	0.054
Debt Issuance	Neg. Jump	0.92%	2.13	0.033	0.044	0.055
Low Risk	Pos. Jump	−2.48%	−2.12	0.033	0.044	0.055
Seasonality	Continuous	4.47%	2.02	0.039	0.052	0.065
Skewness	Continuous	−2.57%	−1.62	0.088	0.117	0.146
Quality	Continuous	2.61%	1.54	0.093	0.124	0.155

Note: Each row reports a risk premia estimate for a particular cluster portfolio; clusters are based on the methodology described in [Section 2.1](#) and their representative portfolios are formed using the first principal component of the factor returns within each cluster. I report estimates are obtained from running separate Cts-Time Fama-MacBeth regressions for each factor; for these regressions, the span assets are the FF3 plus a given cluster portfolio. The risk-premia estimates are annualized. The q-values are computed using the procedure given in Storey (2002) under various parameterizations for the true proportion of nulls, π_0 . Only the top fifteen estimates by p-value are included for brevity.

Table 18: VARIANCE DECOMPOSITION RESULTS – CTS-TIME FAMA-MACBETH WITH CONTINUOUS/JUMP RISK FACTORS

<i>Panel A. Using Continuous and Jump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	48.67	5.08	5.09	43.58	43.59
All	C14	58.75	8.61	14.67	44.08	50.14
Stocks	FF6	48.57	4.41	4.85	43.72	44.15
Stocks	C14	58.11	9.82	13.19	44.92	48.29
Portfolios	FF6	49.39	6.83	9.77	39.62	42.55
Portfolios	C14	63.25	0.07	25.13	38.12	63.19
<i>Panel B. Using Continuous and SemiJump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	49.71	3.94	4.35	45.36	45.76
All	C14	58.43	10.59	12.67	45.75	47.83
Stocks	FF6	49.30	3.43	3.85	45.45	45.87
Stocks	C14	57.65	11.16	11.73	45.92	46.49
Portfolios	FF6	52.58	7.57	7.88	44.71	45.01
Portfolios	C14	63.88	6.58	19.33	44.55	57.30

Note: For each specification listed in the second column, I run a Cts-Time Fama-MacBeth regression to estimate betas and spot risk premia for each month. The span assets include the continuous and jump component of each factor in the specification; the test assets are the yearly top 1000 stocks by market cap and all the portfolios in my factor zoo. In the first panel, the regressions involve continuous and jump risk factors, while, in the second panel, the regressions further involve semijump risk factors. For each group of test assets, I then estimate $\tilde{R}_J^2$, $\tilde{R}_C^2$, and $\tilde{R}^2$. The upper and lower bounds for the explanatory power of the jump and continuous components are given by the sets $\mathcal{S}_J^R$ and $\mathcal{S}_C^R$, which are defined in Equation 10 and Equation 11; these sets are also truncated when they exceed 0% or 100%. All values are reported as percentages.

4.2.5 Jump Beta Window: 180 Days

Table 19: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND JUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	1.42	(0.37)	0.08	(0.02)	1.89	(0.53)	1.84	(0.52)
	Jump	3.67*	(1.68)	4.76**	(2.52)	5.01***	(2.81)	5.44***	(3.07)
FF SMB	Continuous			1.89	(0.97)	2.36	(1.27)	2.70	(1.47)
	Jump			0.42	(0.36)	1.16	(1.02)	1.28	(1.17)
FF HML	Continuous			−1.04	(−0.56)	−2.27	(−1.28)	−1.84	(−1.05)
	Jump			−0.16	(−0.15)	0.38	(0.38)	0.35	(0.36)
FF RMW	Continuous					4.53***	(3.75)	4.41***	(3.73)
	Jump					0.85	(1.19)	0.87	(1.25)
FF CMA	Continuous					0.81	(0.69)	1.39	(1.21)
	Jump					−0.15	(−0.22)	0.12	(0.18)
FF UMD	Continuous							1.92	(0.89)
	Jump							−1.25	(−0.90)
R^2		30.9%		35.2%		37.1%		38.1%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 20: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND SEMIJUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	1.83	(0.48)	0.00	(0.00)	1.79	(0.51)	1.76	(0.50)
	Neg Jump	2.60	(1.64)	3.38**	(2.42)	3.38**	(2.52)	3.53***	(2.66)
	Pos Jump	0.54	(0.35)	1.52	(1.12)	1.40	(1.07)	1.45	(1.12)
FF SMB	Continuous			2.16	(1.12)	2.56	(1.38)	2.70	(1.47)
	Neg Jump			0.71	(0.78)	0.83	(0.94)	0.98	(1.12)
	Pos Jump			−0.53	(−0.61)	−0.27	(−0.33)	−0.24	(−0.30)
FF HML	Continuous			−1.01	(−0.55)	−2.02	(−1.15)	−1.65	(−0.95)
	Neg Jump			0.54	(0.66)	0.75	(0.95)	0.70	(0.89)
	Pos Jump			−1.12	(−1.53)	−0.72	(−1.03)	−0.74	(−1.07)
FF RMW	Continuous					4.26***	(3.56)	4.15***	(3.54)
	Neg Jump					0.78	(1.37)	0.84	(1.48)
	Pos Jump					0.10	(0.21)	0.07	(0.15)
FF CMA	Continuous					0.90	(0.77)	1.37	(1.21)
	Neg Jump					0.70	(1.31)	0.71	(1.34)
	Pos Jump					−0.86*	(−1.74)	−0.63	(−1.29)
FF UMD	Continuous							1.97	(0.92)
	Neg Jump							0.60	(0.60)
	Pos Jump							−1.77*	(−1.85)
R^2		31.3%		35.9%		38.1%		39.1%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 21: CLUSTER PORTFOLIOS – CONTINUOUS AND SEMIJUMP RISK PREMIA

Cluster Portfolio	Component	Risk Premia	t -stat	q-value		
				$\pi_0 = 30\%$	$\pi_0 = 40\%$	$\pi_0 = 50\%$
Investment	Pos. Jump	−1.87%	−3.24	0.014	0.019	0.024
Accruals	Neg. Jump	1.40%	2.96	0.018	0.024	0.030
Size	Neg. Jump	1.99%	2.76	0.022	0.030	0.037
Skewness	Pos. Jump	−1.59%	−2.46	0.034	0.045	0.056
Low Risk	Pos. Jump	−2.31%	−2.40	0.034	0.045	0.056
Profitability	Continuous	3.03%	2.33	0.034	0.045	0.056
Low Risk	Continuous	5.38%	2.32	0.034	0.045	0.056
Leverage	Neg. Jump	1.46%	2.20	0.041	0.054	0.068
Quality	Continuous	3.22%	2.11	0.045	0.060	0.075
Value	Neg. Jump	1.39%	2.07	0.045	0.060	0.075
Momentum	Neg. Jump	2.03%	2.02	0.046	0.062	0.077
Debt Issuance	Pos. Jump	−0.62%	−1.85	0.056	0.075	0.093
Profitability	Neg. Jump	1.13%	1.83	0.056	0.075	0.093
Leverage	Continuous	−2.96%	−1.82	0.056	0.075	0.093
Debt Issuance	Neg. Jump	0.57%	1.79	0.056	0.075	0.093

Note: Each row reports a risk premia estimate for a particular cluster portfolio; clusters are based on the methodology described in [Section 2.1](#) and their representative portfolios are formed using the first principal component of the factor returns within each cluster. I report estimates are obtained from running separate Cts-Time Fama-MacBeth regressions for each factor; for these regressions, the span assets are the FF3 plus a given cluster portfolio. The risk-premia estimates are annualized. The q-values are computed using the procedure given in Storey (2002) under various parameterizations for the true proportion of nulls, π_0 . Only the top fifteen estimates by p-value are included for brevity.

Table 22: VARIANCE DECOMPOSITION RESULTS – CTS-TIME FAMA-MACBETH WITH CONTINUOUS/JUMP RISK FACTORS

<i>Panel A. Using Continuous and Jump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	49.23	6.70	15.69	33.54	42.53
All	C14	55.37	11.73	23.91	31.46	43.64
Stocks	FF6	48.82	6.24	15.26	33.56	42.58
Stocks	C14	55.13	11.63	24.52	30.61	43.50
Portfolios	FF6	52.37	10.24	19.00	33.38	42.13
Portfolios	C14	57.18	12.45	19.16	38.02	44.73
<i>Panel B. Using Continuous and SemiJump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	49.03	5.93	16.80	32.23	43.09
All	C14	55.79	11.37	27.30	28.49	44.42
Stocks	FF6	48.47	5.46	16.43	32.04	43.01
Stocks	C14	55.47	11.16	27.49	27.98	44.31
Portfolios	FF6	53.32	9.59	19.68	33.64	43.73
Portfolios	C14	58.26	13.01	25.83	32.43	45.25

Note: For each specification listed in the second column, I run a Cts-Time Fama-MacBeth regression to estimate betas and spot risk premia for each month. The span assets include the continuous and jump component of each factor in the specification; the test assets are the yearly top 1000 stocks by market cap and all the portfolios in my factor zoo. In the first panel, the regressions involve continuous and jump risk factors, while, in the second panel, the regressions further involve semijump risk factors. For each group of test assets, I then estimate $\tilde{R}_J^2$, $\tilde{R}_C^2$, and $\tilde{R}^2$. The upper and lower bounds for the explanatory power of the jump and continuous components are given by the sets $\mathcal{S}_J^R$ and $\mathcal{S}_C^R$, which are defined in Equation 10 and Equation 11; these sets are also truncated when they exceed 0% or 100%. All values are reported as percentages.

4.2.6 Jump Beta Window: 540 Days

Table 23: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND JUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	2.15	(0.57)	0.50	(0.15)	2.15	(0.64)	1.94	(0.58)
	Jump	1.94	(0.72)	4.76**	(1.98)	4.53**	(1.96)	5.80**	(2.52)
FF SMB	Continuous			0.18	(0.09)	0.20	(0.11)	0.53	(0.29)
	Jump			−0.09	(−0.05)	1.28	(0.75)	1.29	(0.78)
FF HML	Continuous			−1.22	(−0.66)	−1.99	(−1.13)	−1.71	(−0.98)
	Jump			0.05	(0.04)	0.71	(0.53)	0.58	(0.44)
FF RMW	Continuous					3.78***	(3.17)	3.65***	(3.13)
	Jump					2.63***	(2.63)	2.29**	(2.31)
FF CMA	Continuous					0.12	(0.10)	0.65	(0.58)
	Jump					0.27	(0.29)	0.34	(0.37)
FF UMD	Continuous							1.65	(0.76)
	Jump							−1.93	(−1.10)
R^2		31.9%		36.3%		38.2%		39.3%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 24: CTS-TIME FAMA-MACBETH REGRESSIONS – CONTINUOUS AND SEMIJUMP RISK PREMIA

		<i>Specification</i>							
		CAPM		FF3		FF5		FF6	
FF MKT	Continuous	1.19	(0.31)	−0.07	(−0.02)	1.87	(0.56)	1.65	(0.50)
	Neg Jump	3.67*	(1.66)	4.03**	(2.12)	2.88	(1.55)	3.96**	(2.15)
	Pos Jump	−0.47	(−0.22)	1.64	(0.87)	2.10	(1.14)	2.18	(1.18)
FF SMB	Continuous			0.14	(0.07)	0.19	(0.10)	0.39	(0.21)
	Neg Jump			1.82	(1.27)	1.77	(1.27)	1.97	(1.48)
	Pos Jump			−1.97	(−1.48)	−0.87	(−0.68)	−0.95	(−0.76)
FF HML	Continuous			−1.18	(−0.64)	−1.93	(−1.10)	−1.93	(−1.12)
	Neg Jump			0.74	(0.66)	0.48	(0.43)	0.43	(0.41)
	Pos Jump			−1.17	(−1.15)	0.15	(0.16)	0.17	(0.17)
FF RMW	Continuous					3.71***	(3.15)	3.52***	(3.06)
	Neg Jump					2.05**	(2.49)	1.88**	(2.27)
	Pos Jump					0.36	(0.53)	0.19	(0.28)
FF CMA	Continuous					0.25	(0.22)	0.63	(0.56)
	Neg Jump					1.44*	(1.80)	1.57**	(1.99)
	Pos Jump					−1.37*	(−1.90)	−1.28*	(−1.78)
FF UMD	Continuous							1.86	(0.86)
	Neg Jump							−0.37	(−0.28)
	Pos Jump							−1.45	(−1.07)
R^2		32.4%		36.9%		39.1%		40.4%	

Note: I report Cts-Time Fama-MacBeth estimates of the annualized risk premia (%) for each factor along with t-statistics in parentheses. The test assets include every portfolio in the factor zoo along with the top 1000 stocks by market cap in each year. For each specification, I use yearly jump betas and monthly continuous betas. Both betas are estimated using a backwards-looking rolling window on 15-minute returns. Cross-sectional regressions are done on a monthly basis. The R^2 values report the time-series average of the R^2 estimates for each cross-sectional regression. The regressions include 3394 test assets and the risk premia are averaged over a time span of 24.9 years. The notation *, **, and *** refers to 90%, 95%, and 99% levels of significance respectively.

Table 25: CLUSTER PORTFOLIOS – CONTINUOUS AND SEMIJUMP RISK PREMIA

Cluster Portfolio	Component	Risk Premia	t -stat	q-value		
				$\pi_0 = 30\%$	$\pi_0 = 40\%$	$\pi_0 = 50\%$
Size	Neg. Jump	2.87%	2.64	0.036	0.048	0.060
Leverage	Pos. Jump	−2.79%	−2.52	0.036	0.048	0.060
Debt Issuance	Neg. Jump	1.26%	2.51	0.036	0.048	0.060
Investment	Pos. Jump	−2.06%	−2.46	0.036	0.048	0.060
Skewness	Pos. Jump	−2.30%	−2.42	0.036	0.048	0.060
Leverage	Neg. Jump	2.21%	2.33	0.038	0.051	0.063
Accruals	Neg. Jump	1.52%	2.28	0.038	0.051	0.063
Profitability	Continuous	2.76%	2.14	0.048	0.063	0.079
Low Risk	Pos. Jump	−2.70%	−2.03	0.055	0.073	0.091
Investment	Neg. Jump	1.71%	1.84	0.071	0.095	0.119
Profitability	Neg. Jump	1.68%	1.83	0.071	0.095	0.119
Low Risk	Continuous	4.05%	1.79	0.072	0.096	0.121
Value	Pos. Jump	−1.78%	−1.73	0.076	0.101	0.126
Low Risk	Neg. Jump	2.45%	1.64	0.082	0.109	0.137
Debt Issuance	Pos. Jump	−0.79%	−1.62	0.082	0.109	0.137

Note: Each row reports a risk premia estimate for a particular cluster portfolio; clusters are based on the methodology described in [Section 2.1](#) and their representative portfolios are formed using the first principal component of the factor returns within each cluster. I report estimates are obtained from running separate Cts-Time Fama-MacBeth regressions for each factor; for these regressions, the span assets are the FF3 plus a given cluster portfolio. The risk-premia estimates are annualized. The q-values are computed using the procedure given in Storey (2002) under various parameterizations for the true proportion of nulls, π_0 . Only the top fifteen estimates by p-value are included for brevity.

Table 26: VARIANCE DECOMPOSITION RESULTS – CTS-TIME FAMA-MACBETH WITH CONTINUOUS/JUMP RISK FACTORS

<i>Panel A. Using Continuous and Jump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	51.72	7.23	17.40	34.33	44.50
All	C14	58.24	6.99	16.41	41.84	51.25
Stocks	FF6	51.58	6.77	16.46	35.12	44.81
Stocks	C14	57.85	7.09	16.20	41.65	50.75
Portfolios	FF6	52.68	10.31	23.78	28.89	42.36
Portfolios	C14	60.94	6.30	17.84	43.10	54.64
<i>Panel B. Using Continuous and SemiJump Risk Factors</i>						
Assets	Specification	$\tilde{R}^2$	Cts Component		Jump Component	
			Lower	Upper	Lower	Upper
All	FF6	52.92	5.84	14.20	38.71	47.08
All	C14	59.77	5.06	11.04	48.73	54.71
Stocks	FF6	52.81	5.39	13.18	39.63	47.42
Stocks	C14	59.64	5.23	10.35	49.29	54.41
Portfolios	FF6	53.67	8.90	21.23	32.44	44.77
Portfolios	C14	60.63	3.86	15.71	44.91	56.77

Note: For each specification listed in the second column, I run a Cts-Time Fama-MacBeth regression to estimate betas and spot risk premia for each month. The span assets include the continuous and jump component of each factor in the specification; the test assets are the yearly top 1000 stocks by market cap and all the portfolios in my factor zoo. In the first panel, the regressions involve continuous and jump risk factors, while, in the second panel, the regressions further involve semijump risk factors. For each group of test assets, I then estimate $\tilde{R}_J^2$, $\tilde{R}_C^2$, and $\tilde{R}^2$. The upper and lower bounds for the explanatory power of the jump and continuous components are given by the sets $\mathcal{S}_J^R$ and $\mathcal{S}_C^R$, which are defined in Equation 10 and Equation 11; these sets are also truncated when they exceed 0% or 100%. All values are reported as percentages.

5 Continuous and Jump Risk Premia

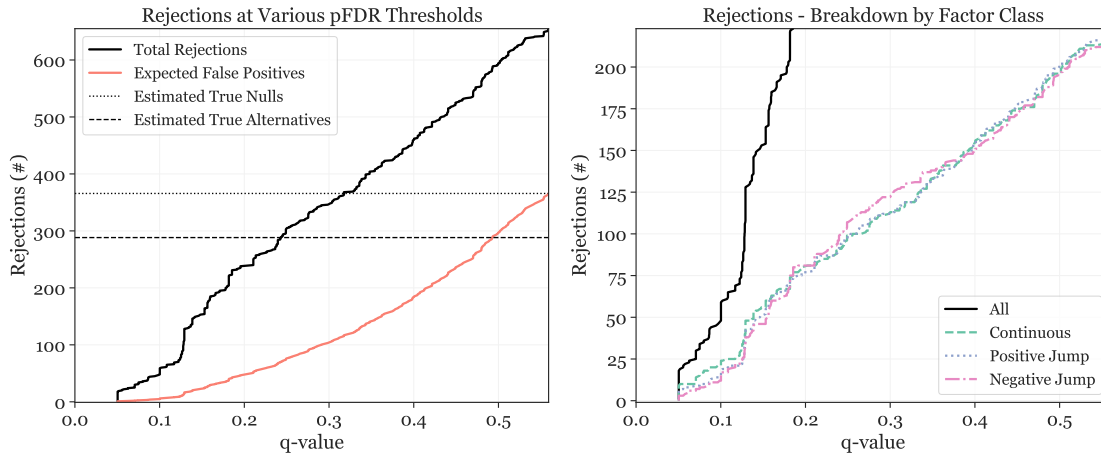
In Table 27, I report the risk premia estimates for all the factors in my zoo. These estimates correspond to those referenced in the main text under “Individual Analysis.” The entries are sorted by the absolute value of their t-statistic and therefore by their q-value as well. For reference, the q-value for hypothesis $m \in \{1, \dots, M\}$ is calculated as:

$$q(m; \pi_0) = \begin{cases} p(m) \cdot \pi_0 & \text{if } m = M \\ \min(q(m+1), p(m) \cdot \pi_0 \cdot M/m) & \text{otherwise} \end{cases}$$

where $p(m)$ is the m 'th p-value corresponding to a two-sided test of the null. The π_0 term is estimated using the procedure described in the main text.

To help summarize the hypothesis tests associated with the q-values, I plot the rejection counts at each q-value threshold in Figure 7. The key takeaway from the first subplot is that there are considerably greater rejections (and thus greater statistical power) when moving from conservative thresholds of 5% and 10% to more liberal thresholds of 15% and 20%. At the same time, transitioning between these thresholds increases the expected number of false positives ($q \times \text{Rejections}$) by only a small amount. For instance, at a threshold of 5%, there are zero rejections and zero false positives; but, at a threshold of 20%, there are 238 rejections with an expected number of 46.5 false positives. The main takeaway from the second subplot, which shows the number of rejections by factor class, is that there is little heterogeneity in terms the factor class of the nulls being rejected at each threshold. That is, whether we use a q-value threshold of 10% or 30%, there are an approximately equal number of rejections of the null for risk factors associated with each of the three types of risk premia (continuous, positive jump, and negative jump).

Figure 7: ESTIMATING THE PROPORTION OF TRUE NULLS



Note: In the first subplot, I plot the number of null hypothesis rejections across various q-value cutoffs for the risk premia estimates from the individual analysis for all of the JKP+CZ portfolios. The orange line reports the product between the given q-value threshold and the number of rejections. The estimated true nulls and alternatives are computed as $M \times \hat{\pi}_0$ and $M \times (1 - \hat{\pi}_0)$ where M represents the total number of hypotheses. In the second subplot, I break down the total rejections, given as “All,” into three groups for each of the three types of risk factors. The y-axis of this latter subplot is truncated for clarity.

Table 27: FACTOR ZOO – ALL CONTINUOUS AND SEMIJUMP RISK PREMIA ESTIMATES

Factor Name	Component	Description	Risk Premia	t-stat	q-value
RESFF3_12_1	Pos. Jump	Residual momentum t-12 to t-1	-3.33%	-3.78	0.051
OPE_BE	Continuous	Operating profits-to-book equity	6.01%	3.57	0.051
SEAS_6_10NA	Pos. Jump	Years 6-10 lagged returns, nonannual	-4.68%	-3.43	0.051
INDIPO	Pos. Jump	Initial Public Offerings	-5.22%	-3.43	0.051
EBIT_BEV	Continuous	Return on net operating assets	5.54%	3.19	0.051
FCF_ME	Continuous	Free cash flow-to-price	5.18%	3.16	0.051
FCF_ME	Neg. Jump	Free cash flow-to-price	2.97%	3.14	0.051
AT_TURNOVER	Neg. Jump	Capital turnover	2.76%	3.10	0.051
XFIN	Continuous	Net external financing	-5.69%	-3.09	0.051
EQNETIS_AT	Continuous	Net equity issuance	6.27%	3.09	0.051
MISPRICING_MGMT	Neg. Jump	Mispricing factor: Management	3.03%	3.08	0.051
LTI_GR1A	Pos. Jump	Change in long-term investments	-1.92%	-3.05	0.051
NLBE	Continuous	Return on equity	5.22%	3.04	0.051
OPE_BEL1	Continuous	Operating profits-to-lagged book equity	4.58%	3.04	0.051
FR	Pos. Jump	Pension Funding Status	-2.70%	-3.03	0.051
QMJ	Continuous	Quality minus Junk: Composite	5.85%	3.03	0.051
SALE_BEV	Continuous	Assets turnover	4.54%	3.03	0.051
MEANRANKREVGROWTH	Pos. Jump	Revenue Growth Rank	-2.83%	-3.03	0.051
COMPEQUISS	Continuous	Composite equity issuance	-5.39%	-3.00	0.052
ABNORMALACCRUALS	Pos. Jump	Abnormal Accruals	-2.40%	-2.96	0.056
EMP_GR1	Pos. Jump	Hiring rate	-2.92%	-2.95	0.056
OCF_ME	Neg. Jump	Operating cash flow-to-market	3.77%	2.92	0.058
F_SCORE	Neg. Jump	Pitroski F-score	2.18%	2.87	0.063
SALE_BEV	Neg. Jump	Assets turnover	2.44%	2.87	0.063
SALEQ_SU	Pos. Jump	Standardized Revenue surprise	-2.49%	-2.82	0.071
IDIORISK	Continuous	Idiosyncratic risk	-9.46%	-2.78	0.071
OPEX_AT	Continuous	Operating leverage	3.89%	2.77	0.071
NIQ_BE	Continuous	Quarterly return on equity	4.54%	2.76	0.071
CONVDEBT	Neg. Jump	Convertible debt indicator	2.30%	2.76	0.071
AGEIPO	Continuous	IPO and age	11.14%	2.76	0.071
MS	Continuous	Mohanram G-score	5.02%	2.72	0.076

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
GP_AT	Continuous	Gross profits-to-assets	5.16%	2.71	0.076
INDRETBIG	Neg. Jump	Industry return of big firms	3.25%	2.71	0.076
RET_1_0	Pos. Jump	Short-term reversal	-4.10%	-2.70	0.076
ORGCAP	Continuous	Organizational capital	4.65%	2.67	0.080
IVOL_HXZ4_21D	Continuous	Idiosyncratic volatility from the q-factor model	6.95%	2.66	0.080
EBIT_SALE	Neg. Jump	Profit margin	2.57%	2.63	0.086
O_SCORE	Continuous	Ohlson O-score	4.47%	2.61	0.087
ISKEW_HXZ4_21D	Pos. Jump	Idiosyncratic skewness from the q-factor model	-1.68%	-2.61	0.087
NETIS_AT	Neg. Jump	Net total issuance	2.80%	2.58	0.087
CHCSHO_12M	Neg. Jump	Net stock issues	2.78%	2.58	0.087
RSKEW_21D	Pos. Jump	Total skewness	-2.23%	-2.58	0.087
FR	Continuous	Pension Funding Status	-3.83%	-2.57	0.087
NFNA_GR1A	Pos. Jump	Change in net financial assets	-1.60%	-2.55	0.089
BETA_60M	Pos. Jump	Market Beta	-3.95%	-2.52	0.095
QMJ_SAFETY	Continuous	Quality minus Junk: Safety	4.92%	2.51	0.097
GRADEXP	Neg. Jump	Growth in advertising expenses	3.35%	2.49	0.099
NNCOA_GR1A	Pos. Jump	Change in net noncurrent operating assets	-1.92%	-2.47	0.101
DSALE_DSGA	Continuous	Change sales minus change SG\&A	-3.78%	-2.46	0.101
QMJ_PROF	Continuous	Quality minus Junk: Profitability	4.43%	2.45	0.101
COMPEQUISS	Pos. Jump	Composite equity issuance	-2.38%	-2.45	0.101
OACCRUALS_AT	Neg. Jump	Operating accruals	2.41%	2.44	0.101
AT_BE	Neg. Jump	Book leverage	3.16%	2.43	0.101
AT_GR1	Pos. Jump	Asset Growth	-2.63%	-2.43	0.101
BE_GR1A	Pos. Jump	Change in common equity	-2.66%	-2.42	0.101
INTRINSIC_VALUE	Neg. Jump	Intrinsic value-to-market	2.67%	2.41	0.101
SHORTINTEREST	Neg. Jump	Short Interest	2.53%	2.41	0.101
VARCF	Continuous	Cash-flow to price variance	-4.92%	-2.41	0.101
VOLMKT	Neg. Jump	Volume to market equity	3.77%	2.41	0.101
AGE	Pos. Jump	Firm age	-2.99%	-2.39	0.103
AT_TURNOVER	Continuous	Capital turnover	3.53%	2.37	0.108
RIO_MB	Pos. Jump	Inst Own and Market to Book	-3.41%	-2.36	0.109
PL_NIX	Neg. Jump	Taxable income-to-book income	1.76%	2.35	0.109

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
SFE	Neg. Jump	Earnings Forecast to price	3.37%	2.34	0.109
RESFF3_6_1	Pos. Jump	Residual momentum t-6 to t-1	-2.12%	-2.34	0.109
DSALE_DINV	Neg. Jump	Change sales minus change Inventory	1.66%	2.30	0.118
VARCF	Neg. Jump	Cash-flow to price variance	2.50%	2.30	0.118
EBIT_SALE	Continuous	Profit margin	3.74%	2.29	0.118
BRANDINVEST	Neg. Jump	Brand capital investment	5.02%	2.29	0.118
IVOL_CAPM_21D	Continuous	Idiosyncratic volatility from the CAPM (21 days)	6.47%	2.26	0.123
NLME	Neg. Jump	Earnings-to-price	3.14%	2.26	0.123
IVOL_FF3_21D	Continuous	Idiosyncratic volatility from the Fama-French 3...	6.09%	2.25	0.123
REALESTATE	Continuous	Real estate holdings	3.95%	2.25	0.123
ZERO_TRADES_21D	Continuous	Number of zero trades with turnover as tiebreak...	5.69%	2.24	0.125
OCFQ_SALEQ_STD	Continuous	Cash flow volatility	4.06%	2.23	0.125
PRC_HIGHPRC_252D	Continuous	Current price to high price over last year	5.87%	2.22	0.125
CONVDEBT	Pos. Jump	Convertible debt indicator	-1.93%	-2.22	0.125
CHEQ	Neg. Jump	Growth in book equity	1.72%	2.21	0.126
ZERO_TRADES_21D	Pos. Jump	Number of zero trades with turnover as tiebreak...	-2.75%	-2.20	0.126
PL_NIX	Continuous	Taxable income-to-book income	-2.82%	-2.20	0.126
NIQ_AT	Neg. Jump	Quarterly return on assets	2.57%	2.20	0.126
SEAS_11_15AN	Pos. Jump	Years 11-15 lagged returns, annual	-2.03%	-2.19	0.126
RVOL_21D	Continuous	Return volatility	7.26%	2.19	0.126
BETATAILRISK	Neg. Jump	Tail risk beta	2.94%	2.17	0.127
XFIN	Pos. Jump	Net external financing	-2.65%	-2.17	0.127
NETIS_AT	Continuous	Net total issuance	3.94%	2.17	0.127
RIO_TURNOVER	Pos. Jump	Inst Own and Turnover	-3.14%	-2.17	0.127
BETADOWN_252D	Pos. Jump	Downside beta	-3.68%	-2.16	0.127
RMAX1_21D	Continuous	Maximum daily return	6.10%	2.16	0.127
ZERO_TRADES_126D	Continuous	Number of zero trades with turnover as tiebreak...	5.49%	2.14	0.128
EARNSUPBIG	Neg. Jump	Earnings surprise of big firms	1.87%	2.14	0.128
CAPX_GR3	Pos. Jump	CAPEX growth (3 years)	-1.89%	-2.13	0.128
EQNPO_ME	Continuous	Net payout yield	4.80%	2.13	0.128
NOA_GR1A	Neg. Jump	Change in net operating assets	1.84%	2.12	0.128
IVOL_HXZ4_21D	Pos. Jump	Idiosyncratic volatility from the q-factor model	-2.84%	-2.12	0.128

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
RMAX5.RVOL.21D	Pos. Jump	Highest 5 days of return scaled by volatility	-2.20%	-2.12	0.128
ISKEW_FF3.21D	Pos. Jump	Idiosyncratic skewness from the Fama-French 3-f...	-1.45%	-2.12	0.128
EQPO_ME	Pos. Jump	Payout yield	-2.64%	-2.12	0.128
SEAS.16_20AN	Neg. Jump	Years 16-20 lagged returns, annual	2.19%	2.11	0.129
OPEX_AT	Neg. Jump	Operating leverage	1.81%	2.11	0.129
TACCRUALS_AT	Neg. Jump	Total accruals	1.55%	2.10	0.129
ORGCAP	Pos. Jump	Organizational capital	-1.89%	-2.10	0.129
EARNINGSCONSISTENCY	Pos. Jump	Earnings consistency	-2.51%	-2.09	0.129
ZERO_TRADES.252D	Pos. Jump	Number of zero trades with turnover as tiebreak...	-2.63%	-2.09	0.129
ZERO_TRADES.252D	Continuous	Number of zero trades with turnover as tiebreak...	5.28%	2.09	0.129
FIRMAGEMOM	Continuous	Firm Age - Momentum	8.17%	2.08	0.129
SEAS.2_5AN	Neg. Jump	Years 2-5 lagged returns, annual	2.03%	2.08	0.129
COA_GR1A	Pos. Jump	Change in current operating assets	-1.75%	-2.07	0.130
SALE_EMP_GR1	Continuous	Labor force efficiency	-2.69%	-2.07	0.130
TURNOVER.126D	Continuous	Share turnover	5.30%	2.06	0.130
IVOL_FF3.21D	Neg. Jump	Idiosyncratic volatility from the Fama-French 3...	3.44%	2.06	0.130
VOLMKT	Continuous	Volume to market equity	-6.38%	-2.05	0.130
IVOL_CAPM.252D	Continuous	Idiosyncratic volatility from the CAPM (252 days)	6.25%	2.05	0.130
SALEQ_GR1	Neg. Jump	Sales growth (1 quarter)	2.53%	2.05	0.130
NOA_AT	Neg. Jump	Net operating assets	1.64%	2.04	0.130
IVOL_FF3.21D	Pos. Jump	Idiosyncratic volatility from the Fama-French 3...	-2.85%	-2.04	0.130
MISPRICING_PERF	Continuous	Mispricing factor: Performance	4.48%	2.04	0.130
RD_ME	Neg. Jump	R\&D-to-market	2.41%	2.03	0.130
MRREVERSAL	Pos. Jump	Medium-run reversal	-2.62%	-2.02	0.130
COSKEW.21D	Pos. Jump	Coskewness	-2.13%	-2.02	0.130
CONVDEBT	Continuous	Convertible debt indicator	-2.89%	-2.01	0.130
ISKEW_CAPM.21D	Pos. Jump	Idiosyncratic skewness from the CAPM	-1.47%	-2.01	0.130
CHNWC	Pos. Jump	Change in Net Working Capital	-1.42%	-2.01	0.130
RD_SALE	Neg. Jump	R\&D-to-sales	3.11%	2.01	0.130
SALEQ_GR1	Pos. Jump	Sales growth (1 quarter)	-2.23%	-2.01	0.130
OCF_AT_CHG1	Neg. Jump	Change in operating cash flow to assets	1.27%	2.00	0.130
SHORTINTEREST	Continuous	Short Interest	-3.84%	-2.00	0.130

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	<i>t</i> -stat	q-value
BETAVIX	Continuous	Systematic volatility	-4.05%	-2.00	0.130
TACCRUALS_NI	Pos. Jump	Percent total accruals	-1.54%	-1.99	0.133
ALIQ_AT	Neg. Jump	Liquidity of book assets	2.68%	1.98	0.135
NCOA_GR1A	Neg. Jump	Change in noncurrent operating assets	1.62%	1.98	0.135
NLIVOL	Neg. Jump	Earnings volatility	2.15%	1.96	0.137
SALE_GR1	Continuous	Sales Growth (1 year)	3.89%	1.96	0.137
SEAS_2_5NA	Pos. Jump	Years 2-5 lagged returns, nonannual	-2.26%	-1.96	0.137
AGE	Neg. Jump	Firm age	2.00%	1.96	0.137
RIO_VOLATILITY	Pos. Jump	Inst Own and Idio Vol	-3.69%	-1.94	0.139
BETA_60M	Continuous	Market Beta	5.48%	1.94	0.139
ALIQ_AT	Pos. Jump	Liquidity of book assets	-2.39%	-1.94	0.139
RMAX5_21D	Continuous	Highest 5 days of return	6.12%	1.93	0.139
COA_GR1A	Neg. Jump	Change in current operating assets	1.84%	1.93	0.139
EQNETIS_AT	Neg. Jump	Net equity issuance	2.35%	1.92	0.139
ISKEW_HXZ4_21D	Neg. Jump	Idiosyncratic skewness from the q-factor model	1.23%	1.92	0.139
GP_ATL1	Continuous	Gross profits-to-lagged assets	3.85%	1.92	0.139
RIO_VOLATILITY	Continuous	Inst Own and Idio Vol	7.16%	1.92	0.139
REALESTATE	Pos. Jump	Real estate holdings	-1.87%	-1.92	0.139
DIV12M_ME	Pos. Jump	Dividend yield	-2.25%	-1.92	0.139
OCF_AT	Neg. Jump	Operating cash flow to assets	2.29%	1.91	0.141
FNL_GR1A	Pos. Jump	Change in financial liabilities	-1.25%	-1.90	0.142
NIQ_AT	Continuous	Quarterly return on assets	3.62%	1.90	0.142
AOP	Pos. Jump	Analyst Optimism	-1.99%	-1.89	0.144
RESFF3_12_1	Continuous	Residual momentum t-12 to t-1	2.74%	1.88	0.147
ZERO_TRADES_126D	Pos. Jump	Number of zero trades with turnover as tiebreak...	-2.43%	-1.87	0.148
RESFF3_6_1	Continuous	Residual momentum t-6 to t-1	2.89%	1.87	0.149
PL_NIX	Pos. Jump	Taxable income-to-book income	-1.37%	-1.85	0.154
NCOA_GR1A	Pos. Jump	Change in noncurrent operating assets	-1.36%	-1.84	0.154
INV_GR1	Neg. Jump	Inventory growth	1.60%	1.84	0.154
AOP	Continuous	Analyst Optimism	-3.41%	-1.84	0.154
SEAS_6_10NA	Neg. Jump	Years 6-10 lagged returns, nonannual	2.52%	1.83	0.154
SEAS_16_20NA	Pos. Jump	Years 16-20 lagged returns, nonannual	-2.83%	-1.83	0.154

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
NFNA_GR1A	Neg. Jump	Change in net financial assets	1.14%	1.83	0.154
ALIQ_MAT	Continuous	Liquidity of market assets	3.35%	1.83	0.154
LNOA_GR1A	Pos. Jump	Change in long-term net operating assets	-1.42%	-1.82	0.154
NOA_AT	Continuous	Net operating assets	2.53%	1.82	0.154
CAPX_GR1	Continuous	CAPEX growth (1 year)	2.85%	1.82	0.154
AT_GR1	Neg. Jump	Asset Growth	1.97%	1.80	0.157
IVOL_CAPM_252D	Pos. Jump	Idiosyncratic volatility from the CAPM (252 days)	-2.84%	-1.80	0.157
RD5_AT	Neg. Jump	R\&D capital-to-book assets	2.48%	1.80	0.157
CUSTOMERMOMENTUM	Neg. Jump	Customer momentum	5.10%	1.80	0.157
DEBT_ME	Pos. Jump	Debt-to-market	-2.42%	-1.80	0.157
EQNPO_12M	Neg. Jump	Equity net payout	2.10%	1.79	0.157
LNOA_GR1A	Continuous	Change in long-term net operating assets	2.62%	1.79	0.157
VOLMKT	Pos. Jump	Volume to market equity	-3.23%	-1.79	0.157
FRONTIER	Pos. Jump	Efficient frontier index	-2.46%	-1.79	0.157
AGEIPO	Neg. Jump	IPO and age	4.46%	1.78	0.157
IVOL_CAPM_252D	Neg. Jump	Idiosyncratic volatility from the CAPM (252 days)	3.24%	1.78	0.158
GRADEXP	Pos. Jump	Growth in advertising expenses	-2.37%	-1.78	0.158
IVOL_HXZ4_21D	Neg. Jump	Idiosyncratic volatility from the q-factor model	2.95%	1.77	0.158
SEAS_11_15NA	Neg. Jump	Years 11-15 lagged returns, nonannual	2.04%	1.77	0.159
DEBT_GR3	Neg. Jump	Growth in book debt (3 years)	1.25%	1.76	0.160
TAX_GR1A	Neg. Jump	Tax expense surprise	1.40%	1.76	0.160
CHCSHO_12M	Continuous	Net stock issues	3.10%	1.76	0.160
COL_GR1A	Pos. Jump	Change in current operating liabilities	-1.54%	-1.75	0.160
BIDASKHL_21D	Continuous	The high-low bid-ask spread	-5.39%	-1.75	0.160
TANGIBILITY	Neg. Jump	Asset tangibility	1.43%	1.75	0.160
CAPX_GR1	Pos. Jump	CAPEX growth (1 year)	-1.54%	-1.75	0.160
CHEQ	Pos. Jump	Growth in book equity	-1.54%	-1.73	0.164
DSALE_DINV	Pos. Jump	Change sales minus change Inventory	-1.40%	-1.73	0.166
OP_AT	Continuous	Operating profits-to-book assets	3.12%	1.72	0.167
BMDEC	Continuous	Book to market using December ME	-3.64%	-1.72	0.167
EBIT_BEV	Neg. Jump	Return on net operating assets	1.80%	1.71	0.167
OCF_AT_CHG1	Continuous	Change in operating cash flow to assets	-2.00%	-1.71	0.167

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
EQPO_ME	Continuous	Payout yield	4.02%	1.70	0.170
ADEXP	Neg. Jump	Advertising Expense	2.13%	1.69	0.171
ORDERBACKLOG	Pos. Jump	Order backlog	-2.62%	-1.69	0.171
IDIORISK	Pos. Jump	Idiosyncratic risk	-3.70%	-1.69	0.171
PPEINV_GR1A	Neg. Jump	Change PPE and Inventory	1.68%	1.67	0.178
ZEROTRADEALT12	Pos. Jump	Days with zero trades (6 months)	-2.69%	-1.66	0.178
TURNOVER_126D	Pos. Jump	Share turnover	-2.18%	-1.66	0.178
INV_GR1	Pos. Jump	Inventory growth	-1.36%	-1.66	0.178
O_SCORE	Neg. Jump	Ohlson O-score	1.73%	1.66	0.178
BE_GR1A	Neg. Jump	Change in common equity	1.86%	1.65	0.180
COWC_GR1A	Neg. Jump	Change in current operating working capital	1.19%	1.65	0.180
NI_BE	Neg. Jump	Return on equity	1.62%	1.65	0.180
NNCOA_GR1A	Neg. Jump	Change in net noncurrent operating assets	1.41%	1.64	0.182
RDABILITY	Continuous	R\&D ability	-5.51%	-1.63	0.182
OPE_BE	Neg. Jump	Operating profits-to-book equity	1.64%	1.63	0.182
SEAS_1_1NA	Neg. Jump	Year 1-lagged return, nonannual	2.53%	1.63	0.182
BPEBM	Neg. Jump	Leverage component of BM	1.58%	1.62	0.182
OP_AT	Neg. Jump	Operating profits-to-book assets	1.80%	1.62	0.182
BETABAB_1260D	Continuous	Frazzini-Pedersen market beta	5.05%	1.62	0.182
INTANBM	Neg. Jump	Intangible return using BM	2.44%	1.61	0.182
PRC	Continuous	Price per share	-3.36%	-1.61	0.182
BETATAILRISK	Continuous	Tail risk beta	-3.81%	-1.61	0.182
EARNSUPBIG	Continuous	Earnings surprise of big firms	2.48%	1.61	0.182
OCF_AT	Continuous	Operating cash flow to assets	3.00%	1.61	0.182
EBIT_SALE	Pos. Jump	Profit margin	-1.31%	-1.61	0.182
FNL_GR1A	Neg. Jump	Change in financial liabilities	1.03%	1.61	0.182
EQNPO_12M	Pos. Jump	Equity net payout	-1.57%	-1.60	0.182
CASH_AT	Pos. Jump	Cash-to-assets	-2.15%	-1.60	0.182
OP_ATL1	Neg. Jump	Operating profits-to-lagged book assets	1.79%	1.60	0.182
ADEXP	Continuous	Advertising Expense	3.76%	1.60	0.182
ZEROTRADEALT1	Pos. Jump	Days with zero trades (6 months)	-2.64%	-1.60	0.183
BETADOWN_252D	Continuous	Downside beta	4.98%	1.58	0.186

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
EQPO_ME	Neg. Jump	Payout yield	2.24%	1.58	0.186
IDIORISK	Neg. Jump	Idiosyncratic risk	2.73%	1.58	0.186
RIO_MB	Neg. Jump	Inst Own and Market to Book	2.09%	1.57	0.186
FIRMAGE	Neg. Jump	Firm age based on CRSP	1.45%	1.57	0.186
CHEQ	Continuous	Growth in book equity	-2.45%	-1.57	0.186
CHINVIA	Neg. Jump	Change in capital inv (ind adj)	1.27%	1.57	0.186
EXCHSWITCH	Pos. Jump	Exchange Switch	-3.76%	-1.57	0.186
DSALE_DINV	Continuous	Change sales minus change Inventory	-2.07%	-1.57	0.186
FIRMAGE	Pos. Jump	Firm age based on CRSP	-1.30%	-1.55	0.192
COP_AT	Neg. Jump	Cash-based operating profits-to-book assets	1.69%	1.55	0.192
INTANSP	Continuous	Intangible return using Sale2P	3.88%	1.54	0.194
COP_AT	Continuous	Cash-based operating profits-to-book assets	2.76%	1.53	0.196
EQNPO_12M	Continuous	Equity net payout	3.01%	1.53	0.196
DEBT_GR3	Pos. Jump	Growth in book debt (3 years)	-1.05%	-1.53	0.196
SEAS_16_20AN	Pos. Jump	Years 16-20 lagged returns, annual	-1.58%	-1.53	0.196
EARNINGS_VARIABILITY	Continuous	Earnings variability	2.18%	1.50	0.203
BETABAB_1260D	Pos. Jump	Frazzini-Pedersen market beta	-2.51%	-1.48	0.211
CHNWC	Neg. Jump	Change in Net Working Capital	1.03%	1.48	0.211
EQNPO_ME	Neg. Jump	Net payout yield	2.04%	1.47	0.211
SALE_GR1	Neg. Jump	Sales Growth (1 year)	1.70%	1.47	0.211
RET_12_7	Pos. Jump	Price momentum t-12 to t-7	-1.97%	-1.47	0.211
PRC_HIGHPRC_252D	Neg. Jump	Current price to high price over last year	2.48%	1.47	0.211
NETDEBT_ME	Neg. Jump	Net debt-to-price	1.88%	1.47	0.211
BETAVIX	Pos. Jump	Systematic volatility	-1.73%	-1.47	0.211
QMJ_SAFETY	Pos. Jump	Quality minus Junk: Safety	-1.50%	-1.46	0.211
NCOL_GR1A	Pos. Jump	Change in noncurrent operating liabilities	-0.97%	-1.46	0.211
MRREVERSAL	Neg. Jump	Medium-run reversal	1.71%	1.46	0.211
SALE_GR3	Pos. Jump	Sales Growth (3 years)	-1.44%	-1.46	0.211
RET_9_1	Pos. Jump	Price momentum t-9 to t-1	-2.08%	-1.45	0.212
ZEROTRADEALT1	Continuous	Days with zero trades (6 months)	4.68%	1.45	0.214
RVOL_21D	Pos. Jump	Return volatility	-2.48%	-1.44	0.214
RMAX1_21D	Neg. Jump	Maximum daily return	2.46%	1.44	0.214

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
RD_SALE	Pos. Jump	R\&D-to-sales	-2.40%	-1.44	0.214
AGE	Continuous	Firm age	-2.97%	-1.44	0.214
HERFASSET	Continuous	Industry concentration (assets)	2.80%	1.43	0.218
BIDASKHL_21D	Pos. Jump	The high-low bid-ask spread	-2.59%	-1.42	0.218
AT_BE	Continuous	Book leverage	-3.10%	-1.42	0.221
COP_ATL1	Neg. Jump	Cash-based operating profits-to-lagged book assets	1.59%	1.41	0.224
CASHPROD	Neg. Jump	Cash Productivity	1.44%	1.40	0.227
SALE_ME	Neg. Jump	Sales-to-market	1.59%	1.40	0.227
ALIQ_AT	Continuous	Liquidity of book assets	3.23%	1.39	0.227
INV_GR1A	Neg. Jump	Inventory change	1.01%	1.38	0.231
KZ_INDEX	Neg. Jump	Kaplan-Zingales index	1.25%	1.37	0.233
NFNA_GR1A	Continuous	Change in net financial assets	1.57%	1.37	0.234
BETA_DIMSON_21D	Continuous	Dimson beta	3.48%	1.36	0.237
NIQ_SU	Pos. Jump	Standardized earnings surprise	-0.99%	-1.36	0.237
COSKEW_21D	Continuous	Coskewness	2.30%	1.36	0.237
SALEQ_GR1	Continuous	Sales growth (1 quarter)	2.85%	1.36	0.237
NIQ_BE	Neg. Jump	Quarterly return on equity	1.31%	1.35	0.238
PPEINV_GR1A	Pos. Jump	Change PPE and Inventory	-1.16%	-1.35	0.239
RDIP0	Pos. Jump	IPO and no R\&D spending	-3.21%	-1.35	0.239
MS	Neg. Jump	Mohanram G-score	1.42%	1.34	0.239
RET_3_1	Neg. Jump	Price momentum t-3 to t-1	1.88%	1.34	0.239
INV_GR1A	Pos. Jump	Inventory change	-0.97%	-1.33	0.240
DSALE_DREC	Pos. Jump	Change sales minus change receivables	-0.91%	-1.33	0.240
NIQ_AT_CHG1	Continuous	Change in quarterly return on assets	-1.78%	-1.33	0.240
OCFQ_SALEQ_STD	Neg. Jump	Cash flow volatility	1.54%	1.33	0.240
OCF_AT	Pos. Jump	Operating cash flow to assets	1.46%	1.33	0.240
QMJ_PROF	Neg. Jump	Quality minus Junk: Profitability	1.43%	1.32	0.240
CAPX_GR2	Pos. Jump	CAPEX growth (2 years)	-1.15%	-1.32	0.240
BIDASKHL_21D	Neg. Jump	The high-low bid-ask spread	2.21%	1.32	0.240
HERF	Neg. Jump	Industry concentration (sales)	1.47%	1.31	0.243
ORDERBACKLOG	Neg. Jump	Order backlog	2.10%	1.31	0.243
RMAX5_21D	Pos. Jump	Highest 5 days of return	-2.27%	-1.31	0.243

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
SFE	Continuous	Earnings Forecast to price	3.22%	1.31	0.243
SALE_GR1	Pos. Jump	Sales Growth (1 year)	-1.32%	-1.30	0.244
VOLUMETREND	Continuous	Volume Trend	-2.51%	-1.30	0.244
QMJ_GROWTH	Neg. Jump	Quality minus Junk: Growth	1.05%	1.29	0.246
HERF	Continuous	Industry concentration (sales)	2.53%	1.29	0.246
OP_ATL1	Continuous	Operating profits-to-lagged book assets	2.42%	1.29	0.246
RET_12_7	Neg. Jump	Price momentum t-12 to t-7	1.60%	1.29	0.246
NLIVOL	Continuous	Earnings volatility	-2.43%	-1.28	0.249
COL_GR1A	Neg. Jump	Change in current operating liabilities	1.29%	1.27	0.250
LTL_GR1A	Neg. Jump	Change in long-term investments	0.89%	1.27	0.250
TACCRUALS_AT	Pos. Jump	Total accruals	-0.97%	-1.27	0.250
EXCHSWITCH	Continuous	Exchange Switch	-4.91%	-1.27	0.250
GP_ATL1	Neg. Jump	Gross profits-to-lagged assets	1.45%	1.27	0.250
CAPX_GR3	Neg. Jump	CAPEX growth (3 years)	1.24%	1.26	0.250
SEAS_11_15NA	Continuous	Years 11-15 lagged returns, nonannual	2.61%	1.26	0.250
KZ_INDEX	Continuous	Kaplan-Zingales index	-1.98%	-1.26	0.250
EQNETIS_AT	Pos. Jump	Net equity issuance	-1.39%	-1.26	0.250
BRANDINVEST	Pos. Jump	Brand capital investment	-2.50%	-1.26	0.251
NLAR1	Neg. Jump	Earnings persistence	0.79%	1.25	0.254
IVOL_CAPM_21D	Pos. Jump	Idiosyncratic volatility from the CAPM (21 days)	-1.81%	-1.25	0.254
ALIQ_MAT	Neg. Jump	Liquidity of market assets	1.29%	1.24	0.256
RSKEW_21D	Continuous	Total skewness	1.66%	1.24	0.256
BETA_60M	Neg. Jump	Market Beta	2.06%	1.23	0.258
ORGCAP	Neg. Jump	Organizational capital	1.26%	1.22	0.260
AT_BE	Pos. Jump	Book leverage	-1.65%	-1.22	0.262
ENTMULT	Pos. Jump	Enterprise Multiple	-1.32%	-1.22	0.262
OCF_AT_CHG1	Pos. Jump	Change in operating cash flow to assets	0.82%	1.21	0.264
NOA_GR1A	Pos. Jump	Change in net operating assets	-0.95%	-1.21	0.264
INDMOM	Neg. Jump	Industry Momentum	1.61%	1.20	0.265
IVOL_CAPM_21D	Neg. Jump	Idiosyncratic volatility from the CAPM (21 days)	2.06%	1.20	0.266
RMAX1_21D	Pos. Jump	Maximum daily return	-1.85%	-1.20	0.266
NLME	Continuous	Earnings-to-price	2.87%	1.18	0.272

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
RET_9_1	Neg. Jump	Price momentum t-9 to t-1	1.70%	1.18	0.272
RIO.TURNOVER	Continuous	Inst Own and Turnover	3.53%	1.18	0.273
STL_GR1A	Pos. Jump	Change in short-term investments	-0.80%	-1.17	0.274
GP_AT	Neg. Jump	Gross profits-to-assets	1.30%	1.16	0.276
SHAREISS5Y	Continuous	Share issuance (5 year)	-1.80%	-1.16	0.276
MOMVOL	Neg. Jump	Momentum in high volume stocks	2.11%	1.16	0.276
HERFBE	Continuous	Industry concentration (equity)	2.06%	1.16	0.276
RET_1_0	Continuous	Short-term reversal	-2.74%	-1.16	0.276
EBIT_BEV	Pos. Jump	Return on net operating assets	1.07%	1.16	0.276
NETIS_AT	Pos. Jump	Net total issuance	1.13%	1.16	0.276
TANGIBILITY	Continuous	Asset tangibility	-1.87%	-1.15	0.276
Z_SCORE	Pos. Jump	Altman Z-score	-1.24%	-1.15	0.277
RIO.TURNOVER	Neg. Jump	Inst Own and Turnover	2.04%	1.15	0.277
DEBT_GR3	Continuous	Growth in book debt (3 years)	1.41%	1.14	0.278
INTRINSIC_VALUE	Continuous	Intrinsic value-to-market	2.15%	1.13	0.282
QMJ_GROWTH	Continuous	Quality minus Junk: Growth	1.66%	1.13	0.282
STL_GR1A	Neg. Jump	Change in short-term investments	0.74%	1.13	0.282
NI_ME	Pos. Jump	Earnings-to-price	-1.39%	-1.13	0.283
BETAVIX	Neg. Jump	Systematic volatility	1.30%	1.12	0.285
TACCRUALS_NI	Continuous	Percent total accruals	-1.46%	-1.11	0.286
EQNPO_ME	Pos. Jump	Net payout yield	-1.29%	-1.11	0.286
FR	Neg. Jump	Pension Funding Status	0.99%	1.11	0.286
INDIPO	Continuous	Initial Public Offerings	-2.68%	-1.11	0.287
RDABILITY	Neg. Jump	R\&D ability	2.46%	1.10	0.289
RET_60_12	Neg. Jump	Long-term reversal	1.21%	1.09	0.291
INTANBM	Pos. Jump	Intangible return using BM	1.63%	1.09	0.291
BE_ME	Continuous	Book-to-market equity	-2.40%	-1.08	0.299
DOLVOL_VAR_126D	Continuous	Coefficient of variation for dollar trading volume	1.77%	1.07	0.300
EARNINGS.VARIABILITY	Neg. Jump	Earnings variability	-1.01%	-1.07	0.301
EARNINGSSTREAK	Neg. Jump	Earnings surprise streak	1.12%	1.06	0.303
PRC_HIGHPRC_252D	Pos. Jump	Current price to high price over last year	-1.50%	-1.05	0.304
SEAS.16_20AN	Continuous	Years 16-20 lagged returns, annual	1.86%	1.05	0.304

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
CAPX_GR1	Neg. Jump	CAPEX growth (1 year)	1.01%	1.05	0.304
NIQ_BE.CHG1	Neg. Jump	Change in quarterly return on equity	0.90%	1.05	0.306
DEBTISSUANCE	Pos. Jump	Debt Issuance	-0.77%	-1.05	0.306
CAPEX_ABN	Pos. Jump	Abnormal corporate investment	-0.85%	-1.04	0.307
RET_6.1	Neg. Jump	Price momentum t-6 to t-1	1.43%	1.03	0.310
NETDEBT_ME	Pos. Jump	Net debt-to-price	-1.42%	-1.03	0.311
Z_SCORE	Continuous	Altman Z-score	2.08%	1.03	0.312
GRSALETOGROVERHEAD	Pos. Jump	Change sales minus change SGA	-0.94%	-1.02	0.312
COMPOSITEDEBTISSUANCE	Continuous	Composite debt issuance	-1.50%	-1.02	0.312
DIV12M_ME	Neg. Jump	Dividend yield	1.35%	1.02	0.314
MOMVOL	Pos. Jump	Momentum in high volume stocks	-1.73%	-1.01	0.317
CORR_1260D	Continuous	Market correlation	2.15%	1.00	0.318
TAX_GR1A	Pos. Jump	Tax expense surprise	-0.86%	-1.00	0.318
BE_ME	Neg. Jump	Book-to-market equity	-1.18%	-1.00	0.318
GRADEXP	Continuous	Growth in advertising expenses	-2.33%	-1.00	0.318
EBITDA_MEV	Continuous	Ebitda-to-market enterprise value	2.37%	1.00	0.318
ISKEW_CAPM_21D	Neg. Jump	Idiosyncratic skewness from the CAPM	0.68%	1.00	0.318
SEAS_1_1AN	Neg. Jump	Year 1-lagged return, annual	1.09%	0.97	0.329
EMP_GR1	Continuous	Hiring rate	1.77%	0.97	0.330
EARNSUPBIG	Pos. Jump	Earnings surprise of big firms	-0.89%	-0.96	0.331
RD_ME	Pos. Jump	R\&D-to-market	-1.05%	-0.96	0.331
RIO_MB	Continuous	Inst Own and Market to Book	-2.45%	-0.96	0.331
COA_GR1A	Continuous	Change in current operating assets	1.49%	0.95	0.333
MOMVOL	Continuous	Momentum in high volume stocks	2.81%	0.95	0.333
RMAX5_RVOL_21D	Continuous	Highest 5 days of return scaled by volatility	-1.65%	-0.95	0.333
RET_6.1	Pos. Jump	Price momentum t-6 to t-1	-1.27%	-0.95	0.333
BEV_MEV	Continuous	Book-to-market enterprise value	-2.19%	-0.94	0.334
OCF_ME	Pos. Jump	Operating cash flow-to-market	-1.11%	-0.94	0.336
TURNOVER_VAR_126D	Neg. Jump	Coefficient of variation for share turnover	0.82%	0.94	0.336
MISPRICING_PERF	Neg. Jump	Mispricing factor: Performance	1.25%	0.93	0.336
DIV12M_ME	Continuous	Dividend yield	2.16%	0.93	0.336
EARNINGS_VARIABILITY	Pos. Jump	Earnings variability	-0.80%	-0.93	0.336

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
ZERO_TRADES_21D	Neg. Jump	Number of zero trades with turnover as tiebreak...	1.47%	0.93	0.336
CHINVIA	Pos. Jump	Change in capital inv (ind adj)	-0.73%	-0.93	0.336
CASH_AT	Neg. Jump	Cash-to-assets	1.17%	0.92	0.337
CORR_1260D	Pos. Jump	Market correlation	-1.10%	-0.92	0.337
OACCRUALS_NI	Neg. Jump	Percent operating accruals	0.61%	0.92	0.339
VOLUMETREND	Pos. Jump	Volume Trend	-1.01%	-0.92	0.339
CUSTOMERMOMENTUM	Continuous	Customer momentum	3.56%	0.91	0.339
EQ_DUR	Neg. Jump	Equity duration	-1.06%	-0.91	0.339
NIQ_AT_CHG1	Pos. Jump	Change in quarterly return on assets	-0.74%	-0.90	0.343
NIQ_AT	Pos. Jump	Quarterly return on assets	0.99%	0.89	0.344
COP_ATL1	Continuous	Cash-based operating profits-to-lagged book assets	1.71%	0.89	0.344
COMPOSITEDEBTISSUANCE	Pos. Jump	Composite debt issuance	-0.75%	-0.89	0.344
CAPX_GR2	Continuous	CAPEX growth (2 years)	1.38%	0.89	0.344
OCF_ME	Continuous	Operating cash flow-to-market	2.08%	0.89	0.344
ADEXP	Pos. Jump	Advertising Expense	-1.12%	-0.89	0.344
FIRMAGE	Continuous	Firm age based on CRSP	1.33%	0.89	0.344
SEAS_6_10NA	Continuous	Years 6-10 lagged returns, nonannual	2.09%	0.88	0.348
SEAS_16_20NA	Neg. Jump	Years 16-20 lagged returns, nonannual	1.36%	0.87	0.348
NOA_GR1A	Continuous	Change in net operating assets	1.26%	0.87	0.348
CHCSHO_12M	Pos. Jump	Net stock issues	-0.81%	-0.87	0.348
QMJ_PROF	Pos. Jump	Quality minus Junk: Profitability	0.92%	0.87	0.348
TURNOVER_VAR_126D	Continuous	Coefficient of variation for share turnover	1.40%	0.86	0.355
AT_ME	Pos. Jump	Assets-to-market	-1.11%	-0.85	0.355
BETADOWN_252D	Neg. Jump	Downside beta	1.54%	0.85	0.355
SEAS_2_5AN	Continuous	Years 2-5 lagged returns, annual	-1.37%	-0.85	0.355
MISPRICING_MGMT	Continuous	Mispricing factor: Management	1.42%	0.85	0.356
AML_126D	Pos. Jump	Amihud Measure	-1.04%	-0.84	0.358
PPEINV_GR1A	Continuous	Change PPE and Inventory	1.38%	0.84	0.358
MOM12M	Continuous	Momentum (12 month)	2.46%	0.83	0.359
RVOL_21D	Neg. Jump	Return volatility	1.65%	0.83	0.361
EARNINGSSTREAK	Continuous	Earnings surprise streak	-1.51%	-0.82	0.363
INTANBM	Continuous	Intangible return using BM	2.00%	0.82	0.363

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
INV_GR1	Continuous	Inventory growth	1.19%	0.82	0.363
RD5_AT	Pos. Jump	R\&D capital-to-book assets	-1.11%	-0.82	0.363
SFE	Pos. Jump	Earnings Forecast to price	-0.96%	-0.82	0.363
DEBTISSUANCE	Neg. Jump	Debt Issuance	0.62%	0.81	0.363
ORDERBACKLOGCHG	Pos. Jump	Change in order backlog	-1.47%	-0.81	0.363
RET_60_12	Pos. Jump	Long-term reversal	-0.79%	-0.81	0.365
DSALE_DSGA	Neg. Jump	Change sales minus change SG\&A	-0.69%	-0.80	0.365
RIO_VOLATILITY	Neg. Jump	Inst Own and Idio Vol	1.81%	0.80	0.366
NIQ_SU	Neg. Jump	Standardized earnings surprise	0.60%	0.78	0.375
SALE_EMP_GR1	Pos. Jump	Labor force efficiency	-0.60%	-0.78	0.376
PRC	Pos. Jump	Price per share	-0.95%	-0.78	0.376
HERF	Pos. Jump	Industry concentration (sales)	-0.79%	-0.77	0.378
NOA_AT	Pos. Jump	Net operating assets	0.62%	0.77	0.378
DEBTISSUANCE	Continuous	Debt Issuance	-1.06%	-0.76	0.379
SEAS_1_1AN	Pos. Jump	Year 1-lagged return, annual	-0.90%	-0.76	0.379
CHNWC	Continuous	Change in Net Working Capital	1.00%	0.76	0.382
AMI_126D	Continuous	Amihud Measure	-1.69%	-0.75	0.382
MRREVERSAL	Continuous	Medium-run reversal	-1.58%	-0.75	0.383
SEAS_1_1AN	Continuous	Year 1-lagged return, annual	-1.41%	-0.75	0.383
ORDERBACKLOGCHG	Continuous	Change in order backlog	-2.05%	-0.74	0.384
EBITDA_MEV	Pos. Jump	Ebitda-to-market enterprise value	-0.87%	-0.74	0.387
ISKEW_FF3_21D	Continuous	Idiosyncratic skewness from the Fama-French 3-f...	0.85%	0.74	0.387
CUSTOMERMOMENTUM	Pos. Jump	Customer momentum	-1.84%	-0.73	0.387
MARKET_EQUITY	Continuous	Market Equity	-1.66%	-0.73	0.387
RMAX5_RVOL_21D	Neg. Jump	Highest 5 days of return scaled by volatility	0.72%	0.73	0.387
TACCRUALS_NI	Neg. Jump	Percent total accruals	0.53%	0.73	0.387
AGEIPO	Pos. Jump	IPO and age	1.41%	0.73	0.387
DSALE_DSGA	Pos. Jump	Change sales minus change SG\&A	-0.65%	-0.72	0.387
LNOA_GR1A	Neg. Jump	Change in long-term net operating assets	0.64%	0.72	0.389
EMP_GR1	Neg. Jump	Hiring rate	0.74%	0.71	0.392
RET_60_12	Continuous	Long-term reversal	-1.28%	-0.71	0.392
INTANCFP	Pos. Jump	Intangible return using CFtoP	0.92%	0.70	0.394

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
INDRETBIG	Continuous	Industry return of big firms	1.45%	0.70	0.394
BEV_MEV	Pos. Jump	Book-to-market enterprise value	-0.87%	-0.70	0.394
ZEROTRADEALT12	Continuous	Days with zero trades (6 months)	2.20%	0.69	0.398
DOLVOL_VAR_126D	Neg. Jump	Coefficient of variation for dollar trading volume	0.62%	0.69	0.398
COP_ATL1	Pos. Jump	Cash-based operating profits-to-lagged book assets	0.76%	0.69	0.399
RET_12_1	Pos. Jump	Price momentum t-12 to t-1	-1.00%	-0.69	0.399
OPE_BEL1	Pos. Jump	Operating profits-to-lagged book equity	0.63%	0.68	0.399
VARCF	Pos. Jump	Cash-flow to price variance	0.79%	0.68	0.399
SALE_GR3	Neg. Jump	Sales Growth (3 years)	0.69%	0.68	0.399
OPEX_AT	Pos. Jump	Operating leverage	-0.51%	-0.68	0.399
COL_GR1A	Continuous	Change in current operating liabilities	1.15%	0.68	0.399
SEAS_11_15AN	Continuous	Years 11-15 lagged returns, annual	1.14%	0.67	0.400
NNCOA_GR1A	Continuous	Change in net noncurrent operating assets	0.97%	0.67	0.401
RMAX5_21D	Neg. Jump	Highest 5 days of return	1.25%	0.67	0.401
NLINC8Q	Continuous	Number of consecutive quarters with earnings in...	0.86%	0.67	0.401
CAPEX_ABN	Continuous	Abnormal corporate investment	0.95%	0.66	0.405
ZERO_TRADES_252D	Neg. Jump	Number of zero trades with turnover as tiebreak...	1.00%	0.65	0.405
RDIP0	Continuous	IPO and no R\&D spending	2.67%	0.65	0.405
NIQ_BE	Pos. Jump	Quarterly return on equity	0.58%	0.65	0.405
Z_SCORE	Neg. Jump	Altman Z-score	0.73%	0.65	0.405
BE_GR1A	Continuous	Change in common equity	1.24%	0.64	0.408
O_SCORE	Pos. Jump	Ohlson O-score	-0.62%	-0.64	0.408
NIQ_BE_CHG1	Pos. Jump	Change in quarterly return on equity	-0.51%	-0.64	0.408
EBITDA_MEV	Neg. Jump	Ebitda-to-market enterprise value	0.77%	0.64	0.408
SEAS_11_15NA	Pos. Jump	Years 11-15 lagged returns, nonannual	0.76%	0.63	0.408
BETA_DIMSON_21D	Neg. Jump	Dimson beta	1.02%	0.63	0.411
NLINC8Q	Neg. Jump	Number of consecutive quarters with earnings in...	0.43%	0.62	0.413
RET_3_1	Pos. Jump	Price momentum t-3 to t-1	-0.79%	-0.62	0.413
FCF_ME	Pos. Jump	Free cash flow-to-price	0.51%	0.62	0.413
AT_TURN0VER	Pos. Jump	Capital turnover	-0.50%	-0.62	0.413
SEAS_2_5NA	Continuous	Years 2-5 lagged returns, nonannual	1.33%	0.61	0.413
RET_9_1	Continuous	Price momentum t-9 to t-1	1.51%	0.61	0.413

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
DSALE_DREC	Continuous	Change sales minus change receivables	0.74%	0.61	0.414
ALIQ_MAT	Pos. Jump	Liquidity of market assets	-0.60%	-0.61	0.414
DBNETIS_AT	Neg. Jump	Net debt issuance	0.38%	0.60	0.418
SALE_BEV	Pos. Jump	Assets turnover	-0.45%	-0.59	0.422
DSALE_DREC	Neg. Jump	Change sales minus change receivables	0.41%	0.58	0.422
CAPEX_ABN	Neg. Jump	Abnormal corporate investment	0.48%	0.58	0.422
TANGIBILITY	Pos. Jump	Asset tangibility	-0.53%	-0.58	0.422
NIQ_SU	Continuous	Standardized earnings surprise	-0.76%	-0.58	0.422
COWC_GR1A	Continuous	Change in current operating working capital	0.76%	0.58	0.422
COP_AT	Pos. Jump	Cash-based operating profits-to-book assets	0.60%	0.58	0.422
CAPX_GR3	Continuous	CAPEX growth (3 years)	0.94%	0.57	0.422
NIQ_AT_CHG1	Neg. Jump	Change in quarterly return on assets	0.44%	0.57	0.422
DEBT_ME	Neg. Jump	Debt-to-market	0.77%	0.56	0.426
BETA_DIMSON_21D	Pos. Jump	Dimson beta	-0.83%	-0.56	0.427
MEANRANKREVGROWTH	Neg. Jump	Revenue Growth Rank	0.58%	0.56	0.428
FRONTIER	Continuous	Efficient frontier index	1.46%	0.55	0.431
OPE_BEL1	Neg. Jump	Operating profits-to-lagged book equity	0.50%	0.55	0.431
RDABILITY	Pos. Jump	R\&D ability	-1.20%	-0.55	0.431
OACCRUALS_NI	Continuous	Percent operating accruals	-0.68%	-0.54	0.432
ABNORMALACCRUALS	Neg. Jump	Abnormal Accruals	0.46%	0.54	0.432
RD_SALE	Continuous	R\&D-to-sales	-1.53%	-0.54	0.433
EARNINGSSTREAK	Pos. Jump	Earnings surprise streak	0.61%	0.53	0.436
ZERO_TRADES_126D	Neg. Jump	Number of zero trades with turnover as tiebreak...	0.81%	0.53	0.437
INTRINSIC_VALUE	Pos. Jump	Intrinsic value-to-market	-0.60%	-0.52	0.438
TURNOVER_126D	Neg. Jump	Share turnover	0.79%	0.52	0.440
INTANEP	Pos. Jump	Intangible return using EP	0.68%	0.51	0.440
F_SCORE	Pos. Jump	Pitroski F-score	-0.40%	-0.51	0.440
SEAS_11_15AN	Neg. Jump	Years 11-15 lagged returns, annual	0.51%	0.51	0.440
DBNETIS_AT	Continuous	Net debt issuance	-0.60%	-0.51	0.440
MARKET_EQUITY	Pos. Jump	Market Equity	-0.68%	-0.50	0.441
F_SCORE	Continuous	Pitroski F-score	0.67%	0.50	0.441
RET_3_1	Continuous	Price momentum t-3 to t-1	1.14%	0.50	0.441

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
INDIPO	Neg. Jump	Initial Public Offerings	0.60%	0.50	0.441
LTL_GR1A	Continuous	Change in long-term investments	-0.60%	-0.50	0.441
AT_ME	Neg. Jump	Assets-to-market	0.64%	0.50	0.441
RET_12_1	Neg. Jump	Price momentum t-12 to t-1	0.73%	0.50	0.441
OPE_BE	Pos. Jump	Operating profits-to-book equity	0.43%	0.49	0.444
AT_ME	Continuous	Assets-to-market	-1.19%	-0.49	0.444
SALEQ_SU	Continuous	Standardized Revenue surprise	-0.73%	-0.48	0.445
BPEBM	Pos. Jump	Leverage component of BM	0.45%	0.48	0.447
SEAS_2_5NA	Neg. Jump	Years 2-5 lagged returns, nonannual	0.60%	0.47	0.448
HERFASSET	Pos. Jump	Industry concentration (assets)	-0.49%	-0.47	0.450
OACCRUALS_AT	Pos. Jump	Operating accruals	0.48%	0.46	0.450
MARKET_EQUITY	Neg. Jump	Market Equity	-0.62%	-0.46	0.450
ABNORMALACCRUALS	Continuous	Abnormal Accruals	0.70%	0.46	0.450
CORR_1260D	Neg. Jump	Market correlation	0.57%	0.46	0.450
DBNETIS_AT	Pos. Jump	Net debt issuance	-0.29%	-0.45	0.452
SHAREISS5Y	Neg. Jump	Share issuance (5 year)	0.35%	0.45	0.454
RDS	Pos. Jump	Real dirty surplus	-0.29%	-0.45	0.454
OACCRUALS_AT	Continuous	Operating accruals	-0.76%	-0.44	0.458
PRC	Neg. Jump	Price per share	0.52%	0.44	0.458
QMJ	Neg. Jump	Quality minus Junk: Composite	0.50%	0.43	0.458
VOLUMETREND	Neg. Jump	Volume Trend	0.41%	0.43	0.459
ANNOUNCEMENTRETURN	Pos. Jump	Earnings announcement return	-0.31%	-0.42	0.462
SHORTINTEREST	Pos. Jump	Short Interest	-0.49%	-0.41	0.468
ZEROTRADEALT12	Neg. Jump	Days with zero trades (6 months)	0.71%	0.40	0.469
AOP	Neg. Jump	Analyst Optimism	0.40%	0.40	0.471
BE_ME	Pos. Jump	Book-to-market equity	-0.44%	-0.40	0.471
ISKEW_FF3_21D	Neg. Jump	Idiosyncratic skewness from the Fama-French 3-f...	0.27%	0.39	0.471
RDS	Continuous	Real dirty surplus	-0.47%	-0.39	0.471
EARNINGSCONSISTENCY	Continuous	Earnings consistency	0.78%	0.39	0.471
NCOL_GR1A	Continuous	Change in noncurrent operating liabilities	0.47%	0.39	0.471
MOM12M	Pos. Jump	Momentum (12 month)	-0.65%	-0.39	0.471
INTANEP	Continuous	Intangible return using EP	0.83%	0.39	0.471

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
INV_GR1A	Continuous	Inventory change	0.49%	0.39	0.471
ANNOUNCEMENTRETURN	Neg. Jump	Earnings announcement return	0.30%	0.38	0.471
BETATAILRISK	Pos. Jump	Tail risk beta	0.55%	0.38	0.471
NLBE	Pos. Jump	Return on equity	0.33%	0.38	0.472
INTANCFP	Continuous	Intangible return using CFtoP	0.80%	0.37	0.473
INDMOM	Pos. Jump	Industry Momentum	-0.51%	-0.37	0.473
MISPRICING_MGMT	Pos. Jump	Mispricing factor: Management	-0.32%	-0.37	0.473
TAX_GR1A	Continuous	Tax expense surprise	-0.53%	-0.37	0.473
RESFF3_6_1	Neg. Jump	Residual momentum t-6 to t-1	0.32%	0.37	0.473
RET_12_7	Continuous	Price momentum t-12 to t-7	-0.76%	-0.35	0.479
MISPRICING_PERF	Pos. Jump	Mispricing factor: Performance	0.45%	0.35	0.480
BPEBM	Continuous	Leverage component of BM	0.58%	0.35	0.480
ORDERBACKLOG	Continuous	Order backlog	0.93%	0.34	0.480
SALE_ME	Continuous	Sales-to-market	0.74%	0.34	0.480
DOLVOL_126D	Continuous	Dollar trading volume	0.76%	0.34	0.480
COWC_GR1A	Pos. Jump	Change in current operating working capital	-0.25%	-0.34	0.480
FRONTIER	Neg. Jump	Efficient frontier index	0.48%	0.34	0.480
XFIN	Neg. Jump	Net external financing	-0.32%	-0.33	0.480
AT_GR1	Continuous	Asset Growth	0.62%	0.33	0.480
INTANCFP	Neg. Jump	Intangible return using CFtoP	-0.43%	-0.33	0.480
NIQ_BE_CHG1	Continuous	Change in quarterly return on equity	-0.45%	-0.33	0.481
DOLVOL_VAR_126D	Pos. Jump	Coefficient of variation for dollar trading volume	0.33%	0.33	0.481
BMDEC	Pos. Jump	Book to market using December ME	-0.37%	-0.33	0.481
EXCHSWITCH	Neg. Jump	Exchange Switch	0.78%	0.32	0.482
SALE_ME	Pos. Jump	Sales-to-market	-0.35%	-0.32	0.482
COMPOSITEDEBTISSUANCE	Neg. Jump	Composite debt issuance	0.25%	0.32	0.484
GP_AT	Pos. Jump	Gross profits-to-assets	0.31%	0.31	0.484
SALE_GR3	Continuous	Sales Growth (3 years)	0.56%	0.31	0.485
CASHPROD	Pos. Jump	Cash Productivity	0.33%	0.31	0.485
SEAS_1_1NA	Pos. Jump	Year 1-lagged return, nonannual	-0.47%	-0.31	0.485
ISKEW_CAPM_21D	Continuous	Idiosyncratic skewness from the CAPM	0.37%	0.31	0.485
BMDEC	Neg. Jump	Book to market using December ME	-0.33%	-0.30	0.486

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
NCOA_GR1A	Continuous	Change in noncurrent operating assets	0.42%	0.29	0.488
FNL_GR1A	Continuous	Change in financial liabilities	0.33%	0.29	0.489
NLAR1	Continuous	Earnings persistence	0.34%	0.29	0.489
RSKEW_21D	Neg. Jump	Total skewness	0.23%	0.28	0.491
DOLVOL_126D	Pos. Jump	Dollar trading volume	0.33%	0.28	0.493
OCFQ_SALEQ_STD	Pos. Jump	Cash flow volatility	0.26%	0.28	0.493
SHAREISS5Y	Pos. Jump	Share issuance (5 year)	0.24%	0.27	0.493
EARNINGSCONSISTENCY	Neg. Jump	Earnings consistency	0.29%	0.27	0.493
ZEROTRADEALT1	Neg. Jump	Days with zero trades (6 months)	0.49%	0.27	0.493
RET_1.0	Neg. Jump	Short-term reversal	-0.36%	-0.27	0.493
DGP_DSALE	Continuous	Change gross margin minus change sales	0.37%	0.27	0.493
INDRETBIG	Pos. Jump	Industry return of big firms	-0.32%	-0.26	0.493
DOLVOL_126D	Neg. Jump	Dollar trading volume	0.34%	0.26	0.494
RDIPO	Neg. Jump	IPO and no R\&D spending	0.62%	0.26	0.494
FIRMAGEMOM	Neg. Jump	Firm Age - Momentum	-0.51%	-0.25	0.499
COSKEW_21D	Neg. Jump	Coskewness	-0.28%	-0.25	0.499
DGP_DSALE	Pos. Jump	Change gross margin minus change sales	-0.19%	-0.24	0.500
INTANSP	Pos. Jump	Intangible return using Sale2P	0.36%	0.24	0.500
SEAS_16_20NA	Continuous	Years 16-20 lagged returns, nonannual	-0.59%	-0.24	0.500
BEV_MEV	Neg. Jump	Book-to-market enterprise value	-0.29%	-0.23	0.502
RET_6.1	Continuous	Price momentum t-6 to t-1	0.55%	0.23	0.502
STLGR1A	Continuous	Change in short-term investments	0.28%	0.23	0.502
KZ_INDEX	Pos. Jump	Kaplan-Zingales index	-0.21%	-0.23	0.502
SEAS_6_10AN	Continuous	Years 6-10 lagged returns, annual	0.37%	0.22	0.504
RET_12.1	Continuous	Price momentum t-12 to t-1	0.57%	0.22	0.504
INTANEP	Neg. Jump	Intangible return using EP	-0.29%	-0.22	0.504
ENTMULT	Neg. Jump	Enterprise Multiple	-0.22%	-0.21	0.506
CHINVIA	Continuous	Change in capital inv (ind adj)	0.29%	0.21	0.506
EQ_DUR	Pos. Jump	Equity duration	-0.23%	-0.20	0.509
TURNOVER_VAR_126D	Pos. Jump	Coefficient of variation for share turnover	0.20%	0.20	0.510
CASHPROD	Continuous	Cash Productivity	0.38%	0.20	0.510
ORDERBACKLOGCHG	Neg. Jump	Change in order backlog	0.33%	0.19	0.510

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	t-stat	q-value
QMJ_GROWTH	Pos. Jump	Quality minus Junk: Growth	0.17%	0.18	0.512
MS	Pos. Jump	Mohanram G-score	-0.20%	-0.18	0.512
RD5_AT	Continuous	R\&D capital-to-book assets	0.45%	0.18	0.512
RD_ME	Continuous	R\&D-to-market	0.35%	0.18	0.512
HERFASSET	Neg. Jump	Industry concentration (assets)	0.19%	0.18	0.512
INDMOM	Continuous	Industry Momentum	0.42%	0.18	0.512
SALEQ_SU	Neg. Jump	Standardized Revenue surprise	0.14%	0.18	0.512
QMJ_SAFETY	Neg. Jump	Quality minus Junk: Safety	0.20%	0.17	0.514
COMPEQUISS	Neg. Jump	Composite equity issuance	0.19%	0.17	0.514
ISKEW_HXZ4_21D	Continuous	Idiosyncratic skewness from the q-factor model	0.19%	0.16	0.516
SEAS_6_10AN	Pos. Jump	Years 6-10 lagged returns, annual	0.16%	0.16	0.518
BRANDINVEST	Continuous	Brand capital investment	0.52%	0.16	0.518
GRSALETOGROVERHEAD	Neg. Jump	Change sales minus change SGA	-0.14%	-0.15	0.520
OACCRUALS_NI	Pos. Jump	Percent operating accruals	-0.11%	-0.15	0.520
OP_ATL1	Pos. Jump	Operating profits-to-lagged book assets	-0.16%	-0.15	0.521
ENTMULT	Continuous	Enterprise Multiple	-0.28%	-0.14	0.523
ANNOUNCEMENTRETURN	Continuous	Earnings announcement return	0.19%	0.14	0.523
AMI_126D	Neg. Jump	Amihud Measure	-0.18%	-0.14	0.523
TACCRUALS_AT	Continuous	Total accruals	0.17%	0.13	0.524
BETABAB_1260D	Neg. Jump	Frazzini-Pedersen market beta	0.23%	0.12	0.528
NIIVOL	Pos. Jump	Earnings volatility	-0.13%	-0.11	0.529
OP_AT	Pos. Jump	Operating profits-to-book assets	-0.12%	-0.11	0.529
INTANSP	Neg. Jump	Intangible return using Sale2P	-0.17%	-0.11	0.529
DEBT_ME	Continuous	Debt-to-market	0.27%	0.11	0.530
FIRMAGEMOM	Pos. Jump	Firm Age - Momentum	0.22%	0.10	0.530
NCOL_GR1A	Neg. Jump	Change in noncurrent operating liabilities	-0.08%	-0.10	0.530
RDS	Neg. Jump	Real dirty surplus	0.07%	0.10	0.530
HERFBE	Pos. Jump	Industry concentration (equity)	0.09%	0.10	0.532
MEANRANKREVGROWTH	Continuous	Revenue Growth Rank	-0.17%	-0.10	0.532
NIINC8Q	Pos. Jump	Number of consecutive quarters with earnings in...	-0.07%	-0.09	0.532
SALE_EMP_GR1	Neg. Jump	Labor force efficiency	0.07%	0.09	0.532
SEAS_2_5AN	Pos. Jump	Years 2-5 lagged returns, annual	-0.08%	-0.08	0.537

continued. . .

Factor Zoo – All Continuous and SemiJump Risk Premia Estimates (continued)

Factor Name	Component	Description	Risk Premia	<i>t</i> -stat	q-value
NLAR1	Pos. Jump	Earnings persistence	-0.05%	-0.07	0.542
REALESTATE	Neg. Jump	Real estate holdings	-0.06%	-0.06	0.543
GRSALETOGROVERHEAD	Continuous	Change sales minus change SGA	0.06%	0.04	0.554
CAPX_GR2	Neg. Jump	CAPEX growth (2 years)	0.03%	0.03	0.554
MOM12M	Neg. Jump	Momentum (12 month)	-0.06%	-0.03	0.554
HERFBE	Neg. Jump	Industry concentration (equity)	-0.03%	-0.03	0.554
SEAS_1_1NA	Continuous	Year 1-lagged return, nonannual	-0.07%	-0.03	0.554
CASH_AT	Continuous	Cash-to-assets	0.05%	0.02	0.554
NETDEBT_ME	Continuous	Net debt-to-price	0.05%	0.02	0.554
SEAS_6_10AN	Neg. Jump	Years 6-10 lagged returns, annual	-0.02%	-0.02	0.554
QMJ	Pos. Jump	Quality minus Junk: Composite	0.02%	0.02	0.556
DGP_DSALE	Neg. Jump	Change gross margin minus change sales	-0.01%	-0.01	0.559
EQ_DUR	Continuous	Equity duration	-0.01%	-0.00	0.559
RESFF3_12_1	Neg. Jump	Residual momentum t-12 to t-1	0.00%	0.00	0.559
GP_ATL1	Pos. Jump	Gross profits-to-lagged assets	0.00%	0.00	0.559

Each row reports the risk premia estimate for a particular high-frequency risk factor from the set of Jensen, Kelly, and Pedersen (2021) and Chen and Zimmermann (2020) factors along with the associated *t*-stat and q-value. The estimates are obtained from separate Cts-Time Fama-MacBeth regressions for each factor. The description columns are based on JKP and CZ. The risk premia estimates are annualized. The associated citations are given in Table 28. The q-values are computed using the procedure given in Storey (2002) which is further detailed in the main text.

6 Decomposing the Cross-Section of Returns

For reference, some of the key equations from the main text are relisted here:

$$\begin{aligned}\tilde{R}_J^2 &= 1 - \frac{\sum_{m=1}^M \left(\hat{E}(R_m^e) - \hat{\Gamma}_m^J \right)^2}{\sum_{m=1}^M \left(\hat{E}(R_m^e) \right)^2} \\ \tilde{R}_C^2 &= 1 - \frac{\sum_{m=1}^M \left(\hat{E}(R_m^e) - \hat{\Gamma}_m^C \right)^2}{\sum_{m=1}^M \left(\hat{E}(R_m^e) \right)^2}\end{aligned}\tag{9}$$

$$\begin{aligned}\tilde{R}^2 &= 1 - \frac{\sum_{m=1}^M \left(\hat{E}(R_m^e) - \hat{\Gamma}_m^C - \hat{\Gamma}_m^J \right)^2}{\sum_{m=1}^M \left(\hat{E}(R_m^e) \right)^2}, \\ \mathcal{S}_J^R &= \left[\min \left(\tau \left(\tilde{R}_J^2 \right), \tau \left(\tilde{R}^2 - \tilde{R}_C^2 \right) \right), \max \left(\tau \left(\tilde{R}_J^2 \right), \tau \left(\tilde{R}^2 - \tilde{R}_C^2 \right) \right) \right]\end{aligned}\tag{10}$$

$$\mathcal{S}_C^R = \left[\min \left(\tau \left(\tilde{R}_C^2 \right), \tau \left(\tilde{R}^2 - \tilde{R}_J^2 \right) \right), \max \left(\tau \left(\tilde{R}_C^2 \right), \tau \left(\tilde{R}^2 - \tilde{R}_J^2 \right) \right) \right],\tag{11}$$

where $\tau(x) \equiv \min(\max(x, 0), 1)$ is used to truncate values outside 0% and 100%.

7 Theoretical Results

In this section, I provide proofs for my continuous and jump risk premia estimators, $(\hat{\Gamma}^C, \hat{\Gamma}^J)$. The estimators are fairly intuitive and the proofs follow closely from the existing work done by Aït-Sahalia, Jacod, and Xiu (2021) (henceforth, AJX). Still, I am motivated to write these proofs for two reasons. Firstly, getting my estimators to converge requires assumptions beyond those needed for the Cts-Time Fama-MacBeth procedure. In particular, the rate ϖ , used in the jump truncation sequence u_n , needs to be tightened. Additionally, there are some nuances when defining the variance estimator and giving an asymptotic normality result. In other words, my estimators cannot be justified with merely the existing assumptions from AJX. Secondly, the proofs and convergence result may not be entirely obvious. Unlike the majority of the traditional work in the factor literature, my paper works in a continuous-time setting and uses a relatively new procedure from AJX. So, I put some additional care into showing that my estimators make sense.

The rest of this section is organized as follows. Firstly, in Section 7.1, I begin with redefinitions of the Cts-Time Fama-MacBeth estimators. Then, in Section 7.2, I restate the original assumptions for reference. Thirdly, in Section 7.3 I introduce my new estimators. And finally, in Section 7.4, I provide the necessary theorems and their proofs.

7.1 Definitions

Notation: All stochastic processes are defined on some filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$. The set of $k \times m$ matrices is given by $\mathcal{M}_{k,m}$, while the set of PSD $k \times k$ matrices is given by $\mathcal{M}_k^+$. The smallest eigenvalue of a matrix A is given by $\zeta(A)$. The notation ΔX_t for some arbitrary stochastic process X at time t refers to $X_t - X_{t-}$. Additionally, the i 'th time interval is denoted by $I_i^n = ((i-1)\Delta_n, i\Delta_n]$ and the i 'th return is denoted $\Delta_i^n X = X_{i\Delta_n} - X_{(i-1)\Delta_n}$. Furthermore, C will be an arbitrary constant that may change from line to line.

The proofs later on will require some additional/revised notation. In the main text, certain elements of the estimation procedure were simplified or rewritten for ease of exposition. But, in this section, it is important that my notation and definitions are comprehensive and consistent with AJX.

To start, additional notation needs to be introduced to handle assets with finite lifespans. Define the lifespan of each asset m as $\mathcal{L}_m = (\zeta_m, \theta_m)$, where both times $\zeta_m < \theta_m$ are random but at least 0. Furthermore, assume that ζ_m and θ_m are stopping times and $P_{\zeta_m}^m$ is $\mathcal{F}_{\zeta_m}$ -measurable. As in equation (10) of AJX, spurious prices may be added at times $t \notin \mathcal{L}_m$, but they are irrelevant for estimation. Additionally, after an asset ceases to exist, assume that its betas and the processes that define its idiosyncratic risk are set to zero. This assumption merely simplifies the math later on and has no effect on estimation.

Next, the jump betas need to be redefined. For simplicity, I will assume the jump betas are constant over the entire lifespan of each asset as in AJX; the following proofs can also trivially be generalized to accommodate piecewise constant jump betas. In any case, pick $\rho, \rho' > 0$ arbitrarily so that Assumption B.V holds for some $\varepsilon > 0$. Then, the

jump betas are defined as:

$$\begin{aligned}
I_n &= \{1, \dots, n\} \\
I_n^m &= \{i \in I_n : \zeta_m \leq (i-1)\Delta_n \leq \theta_m - 2\Delta_n\}, \\
\hat{R}_{n,m}^{i,h} &= \Delta_i^n F^{k(h)} \cdot \mathbb{1}_{[\bar{B}^h(\rho, \rho')(\Delta_i^n F^{k(h)})]} \cdot \mathbb{1}_{[I_n^m(i)]} \\
\hat{R}'_{n,m} &= (\hat{R}_{n,m})^\top \hat{R}_{n,m}
\end{aligned} \tag{12}$$

$$\hat{R}''_{n,m} = \begin{cases} (\hat{R}'_{n,m})^{-1} (\hat{R}_{n,m})^\top & \text{if } \zeta(\hat{R}'_{n,m}) \geq v_n \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

$$\hat{\beta}_{n,i}^{J,m,h} = \begin{cases} \sum_{j \in I_n^m} \hat{R}_{n,m}^{h,j} \Delta_j^n P^m & \text{if } iq_n \Delta_n \in \mathcal{L}_m \\ 0 & \text{otherwise.} \end{cases} \tag{14}$$

Thirdly, the continuous beta construction needs to change slightly.

$$\hat{c}_{n,i}^{F,k,k'} = \frac{1}{q_n \Delta_n} \sum_{j=1}^{q_n} \Delta_{iq_n+j}^n F^k \Delta_{iq_n+j}^n F^{k'} \cdot \mathbb{1}_{[|\Delta_{iq_n+j}^n F^k| \leq u_n, |\Delta_{iq_n+j}^n F^{k'}| \leq u_n]} \tag{15}$$

$$\hat{\gamma}_i^{m,k} = \begin{cases} \frac{1}{q_n \Delta_n} \sum_{j=1}^{q_n} \Delta_{iq_n+j}^n P^m \Delta_{iq_n+j}^n F^k \cdot \mathbb{1}_{[|\Delta_{iq_n+j}^n P^m| \leq u_n, |\Delta_{iq_n+j}^n F^k| \leq u_n]} & \text{if } iq_n \Delta_n \in \mathcal{L}_m \\ 0 & \text{otherwise.} \end{cases} \tag{16}$$

Note that the truncation is now simplified to just u_n and is the same for both factors and assets. The spot (co)volatilities are basically the same as before but the indexing has changed. This does not fundamentally affect the definition of β^C ,

$$\hat{\beta}_i^C = \begin{cases} \hat{\gamma}_i (\hat{c}_i^F)^{-1} & \text{if } \zeta(\hat{c}_i^F) > 1/v_n \\ \mathbf{0}_{M \times K} & \text{otherwise.} \end{cases} \tag{17}$$

Additionally, the construction of η is essentially the same.

$$\begin{aligned}
\hat{\beta}_{n,i} &= (\hat{\beta}_{n,i}^C, \hat{\beta}_{n,i}^J) \\
\hat{b}_{n,i} &= (\hat{\beta}_{n,i})^\top \hat{\beta}_{n,i}
\end{aligned} \tag{18}$$

$$\widehat{b_{n,i}^{-1}} = \begin{cases} (\hat{b}_{n,i})^{-1} & \text{if } \zeta(\hat{b}_{n,i}) > 1/v_n \\ 0 & \text{otherwise} \end{cases} \tag{19}$$

$$\hat{\eta}_i = \widehat{b_{n,i}^{-1}} (\hat{\beta}_{n,i})^\top \tag{20}$$

Lastly, the risk premia and variance estimators are given by

$$\hat{U} = \frac{1}{T_n} \sum_{i=0}^{[n/q_n]-2} \hat{\eta}_i \left(P_{(i+2)q_n\Delta_n} + P_{(i+1)q_n\Delta_n} \right) \quad (21)$$

$$\hat{U}' = \frac{q_n\Delta_n}{T_n} \sum_{i=0}^{[n/q_n]-2} \hat{\eta}_{n,i} \bar{r}_{iq_n\Delta_n} \quad (22)$$

$$\hat{\Lambda} = \hat{U} - \hat{U}' \quad (23)$$

$$\hat{V} = \sum_{i=1}^{[n/q_n]-1} \hat{\eta}_i \left(P_{(i+2)q_n\Delta_n} + P_{(i+1)q_n\Delta_n} \right) \left(P_{(i+2)q_n\Delta_n} + P_{(i+1)q_n\Delta_n} \right)^\top (\hat{\eta}_i)^\top. \quad (24)$$

The indexing on $\hat{U}$ and $\hat{U}'$ differ slightly but the estimators are essentially unchanged. The variance estimator now computes the realized covariance using $q_n\Delta_n$ windows. This completes the redefinitions.

7.2 Assumptions

The assumptions needed for the Cts-time FM regression are stated in the main text. However, I restate them again here to simplify cross-references. These assumptions come from Ait-Sahalia, Jacod, and Xiu (2021) along with the following property.

Property 1: Let Y_t be some multidimensional, optional process and $\mathcal{L} = (R, R'] \cap (0, \infty)$ be a random interval with $R < R'$ being two stopping times; there is a constant C such that $\forall s \geq 0$ and all finite stopping times S with $R < S \leq R'$, we have:

$$\left\| \mathbb{E}(Y_{(S+s) \wedge R'} - Y_S \mid \mathcal{F}_S) \right\| \leq Cs, \quad \mathbb{E} \left(\left\| Y_{(S+s) \wedge R'} - Y_S \right\|^2 \mid \mathcal{F}_S \right) \leq Cs \quad (25)$$

This is a higher level property which will be imposed on the beta and risk-free rate processes. The remaining assumptions for the Continuous-Time Fama-MacBeth regression, A.I-A.III and B.I-B.VII, are given below.

A.I. Independence of Idiosyncratic Risk: The Brownian motion and Poisson random measure for the idiosyncratic risk component P_t^I are independent of those for the factors: $(W^I, \underline{p}^I)$ is independent with respect to $(W^F, \underline{p}^F)$.

A.II. Factor Structure: Let $\lambda_t^{C,k}$ be the risk premium process for the risk factor $\tilde{F}^k$, $\lambda_t^{J,h}$ be the risk premium process for risk factor $\bar{F}^h$, and $\lambda_t^{I,m}$ be the risk premium process for the idiosyncratic risk. Define λ_t^C and λ_t^J by stacking the risk premium processes for each component; define λ_t similarly. The drift of the log-price processes is given by

$$\mu_t - \bar{r}_t = \beta_t \lambda_t + \lambda_t^I \quad (26)$$

for all t .

A.III. Weakly Unpriced Idiosyncratic Risk: Define $\Lambda_T^I = \frac{1}{T} \int_0^T \lambda_t^I dt$. For all $m \in \{1, \dots, M\}$, we have $\sqrt{T} \Lambda_T^{I,m} \xrightarrow{p} 0$ as $T \rightarrow \infty$.

- B.I. **Factor Process:** The processes μ_t^F and σ_t^F are optional and bounded, $c_t^F = \sigma_t^F (\sigma_t^F)^\top$ is invertible with a bounded inverse, and the function δ^F on $\Omega \times \mathbb{R}_+ \times E$ is predictable and there are a Borel bounded function $\tilde{\Gamma}$ on E and a number $\alpha \in [0, 1]$ such that $\|\delta^F(\omega, t, z)\| \leq \tilde{\Gamma}(z)$ and $\int_E \tilde{\Gamma}(z)^\alpha \nu(dz) < \infty$. Here, E is an arbitrary Polish space.
- B.II. **Idiosyncratic Risk Process:** The processes μ_t^I and σ_t^I are optional and bounded, and we have $\|\delta^I(\omega, t, z)\| \leq \tilde{\Gamma}(z)$ where $\tilde{\Gamma}(z)$ is the same as in the previous assumption.
- B.III. **Risk-Free Asset Process:** The process r_t is optional, bounded, and satisfies Equation 25 on $\mathbb{R}_+$.
- B.IV. **Factor Loadings:** The process β_t^C is optional and bounded, and the $\mathcal{M}_{K+H}^+$ -valued process $b_t = \beta_t^I \beta_t$ is invertible with a bounded inverse.
- B.V. **Jump Partitions:** For arbitrary reals $\chi, \chi' > 0$ and $\partial \bar{B}^h$ as the boundary of $\bar{B}^h$, define

$$\bar{B}^h(\chi, \chi') = \{x \in \bar{B}^h : \chi \leq |x| \leq \chi', d(x, \partial \bar{B}^h) \leq \chi'\} \quad (27)$$

Then, for each $m \in \{1, \dots, M\}$, $t > 0$, and $\rho' > 4\rho > 0$, define

$$\begin{aligned} A(\rho, \rho')_t^m &= (\zeta_m, \theta_m) \cap \left(\bigcup_{i \leq h \leq H} \{s \in (0, t] : \Delta \bar{F}_s^h \in \bar{B}^h(2\rho, \rho'/2)\} \right) \\ N(\rho, \rho')_t^m &= \#(A(\rho, \rho')_t^m) \\ R(m, \rho, \rho', t) &\text{ is the } N(\rho, \rho')_t^m \times H\text{-matrix with entries} \\ R(m, \rho, \rho', t)^{s,h} &= \Delta \bar{F}_s^h \text{ for } s \in A(\rho, \rho')_t^m, 1 \leq h \leq H \end{aligned} \quad (28)$$

When $H \geq 1$, we have $\Pr(\zeta(R(m, \rho, \rho', t)^\top R(m, \rho, \rho', t)) \geq \varepsilon) \rightarrow 1$ as $t \rightarrow \infty$, for each m and some $\varepsilon, \rho, \rho' > 0$; this also implies $\Pr(N(\rho, \rho')_t^m \geq H) \rightarrow 1$ as $t \rightarrow \infty$.

- B.VI. **Relative Asymptotic Behavior of Sequences:** We have

$$u_n \asymp \Delta_n^\varpi, \quad q_n \asymp \Delta_n^{-\varpi'}, \quad v_n \asymp \log(1/(T_n \Delta_n)) \quad (29)$$

for the BV truncation parameter u_n (in the main text, I directly use Δ_n^ϖ), the spot covariance window parameter q_n , and the inverse truncation parameter v_n . Additionally, the sequences above along with Δ_n and T_n satisfy the following

conditions:

$$\sup_n \left(T_n \Delta_n^\tau + \frac{1}{T_n \Delta_n^\tau} \right) < \infty, \quad \text{where} \quad \begin{cases} 0 < \tau < 1 & \text{if } H = 0 \\ 0 < \tau < \frac{2}{11} & \text{if } H \geq 1 \end{cases} \quad (30)$$

$$\begin{cases} \max\left(\frac{t}{2}, 1 - \tau\right) < \varpi' < 1 - \frac{\tau}{2}, & \text{if } H = 0. \\ \max(5\tau, 1 - \tau) < \varpi' < 1 - \frac{\tau}{2}, & \text{if } H \geq 1. \end{cases} \quad (31)$$

$$\begin{cases} 0 < \frac{1}{2} - \varpi < \frac{1-\alpha}{64-2\alpha}, & \text{if } H = 0. \\ 0 < \frac{1}{2} - \varpi < \min\left(\frac{1-\alpha}{64-2\alpha}, (\varpi' - 5\tau), \frac{1-\varpi'}{6}\right), & \text{if } H \geq 1. \end{cases} \quad (32)$$

where α comes from Assumption B.I.

B.VII. Conditions on the Variance Sequence: Define

$$V_t^{j,j'} = \int_{m,m'=1}^M \int_0^t \eta_s^{j,m} \eta_s^{j',m'} d[P^m, P^{m'}]_s \quad (33)$$

for $j \in \{1, \dots, K+H\}$. Assume that $(1/t)V_t$ converges in probability to a limit V_∞ as $t \rightarrow \infty$.

This completes the assumptions.

7.3 Estimators

7.3.1 Decomposition Terms

I now discuss the estimator for the decomposition terms $\Gamma \equiv (\Gamma^C; \Gamma^J) \in \mathcal{M}_{2M,1}$. The concatenated vector Γ simply stacks the two decomposition terms for each of the M assets. To simplify the notation, it also helps to introduce the variable

$$\varkappa_t = \begin{pmatrix} \beta_t^C & \mathbf{0}_{M,H} \\ \mathbf{0}_{M,K} & \beta_t^J \end{pmatrix} \in \mathcal{M}_{2M,K+H}. \quad (34)$$

The decomposition of the estimation target is now:

$$\Gamma_T = U_{\Gamma,T} - U'_{\Gamma,T} - U''_{\Gamma,T} - \frac{1}{T} \bar{U}_{\Gamma,T} \quad (35)$$

$$\text{where} \quad U_{\Gamma,T} = \frac{1}{T} \int_0^T (\varkappa_s \eta_s) dP_s \quad (36)$$

$$U'_{\Gamma,T} = \frac{1}{T} \int_0^T (\varkappa_s \eta_s) \bar{r}_s ds \quad (37)$$

$$U''_{\Gamma,T} = \frac{1}{T} \int_0^T (\varkappa_s \eta_s) \lambda_s^I ds \quad (38)$$

$$\frac{1}{T} \bar{U}_{\Gamma,T} = \frac{1}{T} \int_0^T (\varkappa_s \eta_s) dP_s^{Mart}. \quad (39)$$

The main idea here is to multiply the spot risk premia estimates, $\eta_s dP_s$, by $(\beta_s^C, \mathbf{0}_H)$ and $(\mathbf{0}_K, \beta_s^J)$ to obtain estimates of $\beta_s^C \lambda_s^C$ and $\beta_s^J \lambda_s^J$. Time-averaging these terms will give the parameter of interest.

Additionally, the quadratic variation of $\bar{U}_{\Gamma,t}$ will be used for deriving the variance of Γ . This quadratic variation is defined similarly to before as

$$V_{\Gamma,t}^{j,j'} = \sum_{m,m'=1}^M \int_0^t (\varkappa_s \eta_s)^{j,m} (\varkappa_s \eta_s)^{j',m'} d[P^m, P^{m'}]_s \quad (40)$$

where $j, j' \in \{1, \dots, 2M\}$. However, note that the matrix form of this process is actually rank deficient. To intuitively see why, recall that the variance process for the risk premia estimator has innovations of the form “ $\eta_s d[P, P]_s \eta_s^\top$.” Based on the earlier assumptions, these innovations are full rank and thus V_t is non-singular. On the other hand, the innovations in $V_{\Gamma,t}$ cannot be full rank, since they take on the form “ $\kappa_s \eta_s d[P, P]_s \eta_s^\top \kappa_s^\top$.” Note that $\kappa_s \in \mathcal{M}_{2M, K+H}$, so the rank of $V_{\Gamma,t}$ will only be $K+H$ because $2M > K+H$. Consequently, this variance process will not be invertible; this prevents Equation 39 from achieving a straightforward asymptotic normality result. To this end, define

$$\varkappa_{m,t} = \begin{pmatrix} \beta_{m,t}^C & \mathbf{0}_H \\ \mathbf{0}_K & \beta_{m,t}^J \end{pmatrix} \in \mathcal{M}_{2, K+H} \quad (41)$$

and define the associated decomposition terms ($U_{\Gamma,m,T}, U'_{\Gamma,m,T}, U''_{\Gamma,m,T}$, and $\bar{U}_{\Gamma,m,T}$) accordingly. These terms essentially just target $\Gamma_m = (\Gamma_m^C, \Gamma_m^J)$, which capture the continuous and jump risk premia for some asset m . The quadratic variation process for $\bar{U}_{\Gamma,m'',T}$ will be

$$V_{\Gamma,m'',t}^{j,j'} = \sum_{m,m'=1}^M \int_0^t (\varkappa_{m'',s} \eta_s)^{j,m} (\varkappa_{m'',s} \eta_s)^{j',m'} d[P^m, P^{m'}]_s, \quad (42)$$

where $j \in \{1, 2\}$. It is easy to see that this process will be of rank 2 and consequently invertible. Hence, it will be possible to whether these components are significant for individual assets.

All in all, the estimation targets will be $U_{\Gamma,T}, U'_{\Gamma,T}$, and $V_{\gamma,T}$. Estimating these processes will automatically result in estimating their asset-specific versions defined above. So, I propose the following estimators:

$$\hat{\varkappa}_t = \begin{pmatrix} \hat{\beta}_t^C & \mathbf{0}_H \\ \mathbf{0}_K & \hat{\beta}_t^J \end{pmatrix} \quad (43)$$

$$\hat{U}_\Gamma = \frac{1}{T_n} \sum_{i=0}^{[n/q_n]-2} (\hat{\varkappa}_i \hat{\eta}_i) (P_{(i+2)q_n \Delta_n} + P_{(i+1)q_n \Delta_n}) \quad (44)$$

$$\hat{U}'_\Gamma = \frac{q_n \Delta_n}{T_n} \sum_{i=0}^{[n/q_n]-2} (\hat{\varkappa}_i \hat{\eta}_i) \bar{r}_{i q_n \Delta_n} \quad (45)$$

$$\hat{\Gamma}_n = \hat{U} - \hat{U}' \quad (46)$$

$$\begin{aligned} \hat{V}_\Gamma &= \sum_{i=1}^{[n/q_n]-1} (\hat{\varkappa}_i \hat{\eta}_i) (P_{(i+2)q_n \Delta_n} + P_{(i+1)q_n \Delta_n}) \\ &\quad \times (P_{(i+2)q_n \Delta_n} + P_{(i+1)q_n \Delta_n})^\top (\hat{\varkappa}_i \hat{\eta}_i)^\top. \end{aligned} \quad (47)$$

These estimators are essentially sample analogues of the previous estimation targets. This

notation also makes it possible to write $\hat{\Gamma}_n \equiv (\hat{\Gamma}_n^C \ \hat{\Gamma}_n^J)$ to match its target. Additionally, I can form estimates of $V_{\Gamma,m}$ by simply extracting the appropriate elements from $\hat{V}_\Gamma$; this will be useful for the asymptotic normality results.

The above estimators will achieve consistency but they also ignore a critical point. Specifically, the Γ terms are first moments, so estimation of these terms is only possible for assets that have infinite lifespans. Of course, the same holds true for consistent estimation of simple, expected returns as well. To handle this, I assume that there are $M' < M$ assets with infinite life-spans.¹⁵ Then, I introduce the matrix $\mathcal{S} \in \mathcal{M}_{M,M'}$, which concatenates the M -dimensional basis vectors $\{e_m : \theta_m = \infty\}$. Hence, $\mathcal{S}^* \hat{\Gamma}_n^C$ selects the rows of $\hat{\Gamma}_n^C$ corresponding to assets that exist forever. Consequently, the time-average of its limit, $\frac{1}{T} \mathcal{S}^* \hat{\Gamma}_n^C$, will be non-degenerate. This contrasts with estimates of $\hat{\Gamma}$ for assets with finite lifespans; such estimates converge to zero in practice and by definition, because β_s is set to zero for nonexistent assets.¹⁶ To make use of $\mathcal{S}$, I assume that it is known at time 0. This assumption is trivial when working with portfolios; on the other hand, it is far stronger when working with stocks. In both cases, the assumption is really only necessary for the theoretical results to hold. It is now possible to state the first theorem.

Theorem 1

Suppose that *A.I-A.III* and *B.I-B.VII* hold true. Additionally, assume that

$$1/2 - \varpi < \min \left(\frac{1 - \alpha}{128 - 2\alpha}, \frac{\varpi' - 5\tau}{48/5}, \frac{1 - \varpi'}{20} \right)$$

And, define the matrix $\mathcal{S}$ by concatenating the M -dimensional basis vectors $\{e_m : \theta_m = \infty\}$; assume that this matrix is identified. Define $\mathcal{S}_2 \equiv (\mathcal{S}; \mathcal{S})$ by stacking $\mathcal{S}$. Then,

- (i) The sequence $\sqrt{T_n} \mathcal{S}_2^* (\hat{\Gamma}_n - \Gamma_{T_n})$ is bounded in probability.
- (ii) If we assume that $(1/t)V_{\Gamma,m,t} \rightarrow V_{\Gamma,m,\infty}$ as $t \rightarrow \infty$ for some m where $\theta_m = \infty$, then we have the following results.
 - There exists a sequence Ω_n of subsets of Ω such that $P(\Omega_n) \rightarrow 1$ and $\hat{V}_{\Gamma,m,n}$ is invertible on Ω_n .
 - The variables

$$Z_{\Gamma,n} = \begin{cases} T_n (\hat{V}_{\Gamma,m,n})^{-1/2} (\hat{\Gamma}_{m,n} - \Gamma_{m,T_n}) & \text{on } \Omega_n \\ 0 & \text{on } \Omega \setminus \Omega_n \end{cases} \quad (48)$$

converge in law to $\mathcal{N}(0, I_2)$. Additionally, the sequences $T_n^{1/2} (\hat{V}_{\Gamma,m,n})^{-1/2} \cdot \mathbb{1}_{[\Omega_n]}$ and $T_n^{-1/2} (\hat{V}_{\Gamma,m,n})^{1/2} \cdot \mathbb{1}_{[\Omega_n]}$ are bounded in probability.

¹⁵ The Cts-Time FM regression from AJX implicitly assumes that $M' > K+H$ to achieve an asymptotic normality result for the risk premia; this is discussed in their paper.

¹⁶ Obviously the risk premia earned on assets that do not exist is not well-defined. Setting their risk premia to zero in this case is more so done for mathematical convenience. This is the same motivation behind setting $\hat{\eta}$ to zero for non-existent assets. Both these decisions cause the estimators for Γ defined above to trivially achieve consistency for assets with finite lifespans; this consistency is inherently meaningless.

Proof. See [Section 7.4.2](#). □

This result is quite similar to Theorem 1 from Aït-Sahalia, Jacod, and Xiu (2021). As in the referenced theorem, the set Ω_n is introduced in certain cases to ensure that the variance is invertible; otherwise, this set is inconsequential in practice. Additionally, my theorem also offers a consistency and asymptotic normality result. However, there are a few differences between my theorem and the original theorem for the risk premia estimation.

Firstly, a new condition is introduced. The tuning parameter $(1/2 - \varpi)$ needs to be slightly smaller to obtain convergence. Note that this new bound is simply a rescaled version of the original bound for $(1/2 - \varpi)$ given in Assumption [B.VI](#); it is easy to see that there exist values for $(\alpha, \varpi, \varpi', \tau)$ such that the new bounds are satisfied. The intuition for why this condition arises comes from the estimator $\hat{U}_{\Gamma, T}$. Specifically, note that this estimator contains a $\hat{\lambda}_i$ term to scale the spot risk premia estimates by the risk exposure for each stock; the point here is to estimate “ $\beta\lambda$ ” in each period before taking the time-average. Since λ is estimated with error, I get additional estimation error in $\hat{U}_{\Gamma, T}$ that would not have shown up in the original term $\hat{U}_T$ for the risk premia estimation. For this error to converge to zero, the new bound on $(1/2 - \varpi)$ is necessary.¹⁷

Secondly, instead of a joint asymptotic normality result for all M' assets with infinite lifespans, the equation [Equation 48](#) gives asymptotic normality for a particular asset m with infinite lifespan. This is due to the fact that $\bar{U}_{\Gamma, T}$, defined earlier in [Equation 39](#), does not have a non-singular quadratic variation process. The reason the quadratic variation is singular is because the “ $\int \beta\lambda$ ” term projects the $K + H$ dimensional risk premia up to the same dimension as the number of assets; since there are more assets than risk factors, the quadratic variation process cannot be full rank.

The singularity does not affect part (i), because it is still possible to show that each element of $\sqrt{T_n} \mathcal{S}^\top (\hat{\Gamma}_n - \Gamma_{T_n})$ is bounded in probability. This boundedness is because the quadratic variation for $\bar{U}_{\Gamma, m, T}$, defined in [Equation 42](#), is non-singular. Consequently, the full term must be bounded in probability as well. At the same time, a joint normality result for every element of $\hat{\Gamma}_n$ at once is not possible. However, it may be possible to put this term into a quadratic form and construct χ^2 statistic to jointly test whether the Γ^C and Γ^J are nonzero across assets. However, these tests are not that interesting from an economic perspective, and, in practice, all the tests results will be extremely significant as well. This is because, when testing multiple assets, its very likely that there is at least one asset such that Γ_m^C and Γ_m^J are nonzero.

Thirdly, note that the asymptotic results involve the matrix $\mathcal{S}$ introduced earlier. As previously mentioned, it is impossible to consistently estimate first moments for assets that have finite lifespans. Hence, it would seem that my entire procedure is limited to infinite lifespan assets. However, this is not entirely case. Specifically, notice that Γ estimation can be broken into a two step procedure. Step one is to estimate the risk premia, while step two is to multiply the risk premia by the betas. I propose using all assets when estimating the spot risk premia, but limiting the Γ estimation to assets with

¹⁷ The λ error can be broken up into estimation error for β^C and estimation error for β^J . The $\frac{1-\alpha}{128-2\alpha}$ term handles the former while the $\frac{\varpi'-5\tau}{48/5}, \frac{1-\varpi'}{20}$ handles the latter. More specifically, $\frac{1-\varpi'}{20}$ is needed for the variance estimator to converge while $\frac{\varpi'-5\tau}{48/5}$ is needed for the risk premia estimator to converge.

infinite lifespans. This ensures that the risk premia estimation not limited to too few assets, while the Γ terms retain their consistency.

In practice, dropping assets may seem problematic. Most stocks do not exist forever. At the same time, stocks are interesting test assets and their pricing errors may be important to study. Hence, when computing the Γ terms empirically, I only drop stocks that exist for relatively short periods of time (<15 years). This approach is a heuristic one but still motivated by theory – it is generally harder to estimate first moments for assets with less data, so dropping short-lived assets removes highly noisy estimates from the sample. Alternatively, the Γ estimation could be fully limited to portfolios to ensure that all assets have infinite lifespans. This is not too big of a limitation, since it is common in the literature to use anomaly portfolios and industry portfolios as test assets. Hence, in the main text, I report the decomposition for both stocks and portfolios.

7.4 Proofs

In this section, I report the proofs. I also introduce and work with a new variable,

$$\Xi_t \equiv \beta_t \eta_t, \quad (49)$$

which shares many of the same properties as η_t . This new variable Ξ_t can be used to estimate the “ $\beta\lambda$ ” term in the drift. For simplicity, all of the lemmas will be focused on this variable. The lemmas for Ξ_t can trivially be extended to work for the decomposition terms; that is, we can simply define similar variables $\Xi_t^C \equiv (\beta_t^C, 0)\eta_t$ and $\Xi_t^J \equiv (0, \beta_t^J)\eta_t$ and apply the same following lemmas. The results would then follow for $\kappa_t \eta_t$ defined earlier. I will also assume that $H \geq 1$, so there is at least one jump factor.

7.4.1 Supporting Lemmas

Lemma 1 (Jump Beta Estimation Error)

Let $\psi \in (0, 1/2)$ and $\varphi > 0$. If $T_n \Delta_n^{1-\psi-\varphi} \rightarrow 0$, we have

$$P \left(\sup_{i \in I_n} \|\hat{\beta}_{n,i}^J - \beta_{iq_n \Delta_n}^J\| \leq v_n^2 T_n^{3/2} \Delta_n^\psi \right) \rightarrow 1$$

Proof. This is Lemma 4 from AJX. □

The following lemma is similar to Lemma 5 from AJX. The only difference is that the proof now needs to be done for the variable $\Xi_{n,i}^* \equiv \beta_{n,i}^* \eta_{n,i}^*$. Here, we have

$$\begin{aligned} \beta_{n,i}^* &= (\hat{\beta}_{n,i}^C, \beta_{iq_n \Delta_n}^J) \\ b_{n,i}^* &= (\beta_{n,i}^*)^\top \beta_{n,i}^* \\ \widehat{b_{n,i}^*}^{-1} &= \begin{cases} \widehat{b_{n,i}^*}^{-1} & \text{if } \zeta(b_{n,i}^*) > 1/v_n \\ 0 & \text{otherwise} \end{cases} \\ \eta_{n,i}^* &= \widehat{b_{n,i}^*}^{-1} (\beta_{n,i}^*)^\top. \end{aligned}$$

Additionally, as in AJX (A.7), I define

$$\mathcal{E}(Y)_{n,i}^* = \hat{Y}_{n,i} - Y_{n,i}^*, \quad \mathcal{E}(Y)_{n,i} = Y_{n,i}^* - Y_{iq_n \Delta_n} \quad (50)$$

for any of the above processes; this notation will be used to study the error of $\hat{\Xi}_{n,i}$. Lastly, define $\psi \in (0, 1/2)$ and pick $\varphi > 0$ so the following holds:

$$\psi = \frac{1}{2} - \frac{\varpi'}{10}, \quad 0 < \varphi < 1 - \psi - \tau.$$

Lemma 2 (Projection Error Convergence)

When restricted to the set $\Omega_n = \{\sup_{i \in I_n} \|\hat{\beta}_{n,i}^J - \beta_{iq_n \Delta_n}^J\| \leq v_n^2 T_n^{3/2} \Delta_n^\psi\}$, we have

$$\|\mathcal{E}(\Xi)_{n,i}^*\| \leq C z_{n,4}, \quad \text{with} \quad z_{n,4} = C v_n^7 T_n^{3/2} u_n^6 \Delta_n^{\psi-3} \quad (51)$$

Proof. Firstly, note the following,

$$\begin{aligned} z_{n,i} &\equiv v_n^2 T_n^{3/2} \Delta_n^\psi \\ \|\mathcal{E}(\beta)_{n,i}^*\| &\leq z_{n,1} = v_n^2 T_n^{3/2} \Delta_n^\psi \\ \|\mathcal{E}(b^{-1})_{n,i}^*\| &\leq z_{n,3} \leq C v_n^3 u_n^2 z_{n,1} / \Delta_n \\ \|\mathcal{E}(\eta)_{n,i}^*\| &\leq C v_n^4 u_n^4 z_{n,1} / \Delta_n^2 \\ \|\beta_{n,i}^*\| &\leq C v_n u_n^2 / \Delta_n \\ \|(b^{-1})_{n,i}^*\| &\leq v_n \\ \|\eta_{n,i}^*\| &\leq C v_n^2 u_n^2 / \Delta_n \end{aligned}$$

These results are based on Lemma 5 of AJX. Then, decomposing the error for Ξ , I have

$$\begin{aligned} \mathcal{E}(\Xi)_{n,i}^* &= \hat{\Xi}_{n,i} - \Xi_{n,i}^* \\ &= \hat{\beta}_{n,i} \hat{\eta}_{n,i} - \beta_{n,i}^* \eta_{n,i}^* \\ &= (\hat{\beta}_{n,i} - \beta_{n,i}^*) \hat{\eta}_{n,i} + \beta_{n,i}^* (\hat{\eta}_{n,i} - \eta_{n,i}^*) \\ &= (\hat{\beta}_{n,i} - \beta_{n,i}^*) (\hat{\eta}_{n,i} - \eta_{n,i}^*) + (\hat{\beta}_{n,i} - \beta_{n,i}^*) \eta_{n,i}^* + \beta_{n,i}^* (\hat{\eta}_{n,i} - \eta_{n,i}^*) \\ \implies \|\mathcal{E}(\Xi)_{n,i}^*\| &= \|(\hat{\beta}_{n,i} - \beta_{n,i}^*) (\hat{\eta}_{n,i} - \eta_{n,i}^*) + (\hat{\beta}_{n,i} - \beta_{n,i}^*) \eta_{n,i}^* + \beta_{n,i}^* (\hat{\eta}_{n,i} - \eta_{n,i}^*)\| \\ &\leq \|\mathcal{E}(\beta)_{n,i}^*\| \|\mathcal{E}(\eta)_{n,i}^*\| + \|\mathcal{E}(\beta)_{n,i}^*\| \|\eta_{n,i}^*\| + \|\beta_{n,i}^*\| \|\mathcal{E}(\eta)_{n,i}^*\| \\ &\leq C v_n^6 u_n^4 T_n^{3/2} \Delta_n^{\psi-2} z_{n,1} + C v_n^4 u_n^2 T_n^{3/2} \Delta_n^{\psi-1} + C v_n^5 u_n^6 z_{n,1} / \Delta_n^3 \\ &\leq C v_n^6 u_n^4 T_n^{3/2} \Delta_n^{\psi-2} z_{n,1} + C v_n^2 u_n^2 \Delta_n^{-1} z_{n,1} + C v_n^5 u_n^6 z_{n,1} \Delta_n^{-3} \\ &\leq C z_{n,1} v_n^2 u_n^2 \Delta_n^{-1} \left(v_n^4 u_n^2 T_n^{3/2} \Delta_n^{\psi-1} + 1 + v_n^3 u_n^4 \Delta_n^{-2} \right). \end{aligned}$$

Only one of the terms in the parentheses will dominate. Note that

$$v_n^4 u_n^2 T_n^{3/2} \Delta_n^{\psi-1} + v_n^3 u_n^4 \Delta_n^{-2} = v_n^3 u_n^2 \Delta_n^{-1} \left(v_n T_n^{3/2} \Delta_n^\psi + u_n^2 \Delta_n^{-1} \right).$$

From (A.8) in AJX, we know that $v^p T_n^{3/2} \Delta_n^\psi \rightarrow 0$ for all $p > 0$, while $u_n^2 \Delta_n^{-1} \geq 1$ by construction. Thus, the term associated with $\|\beta_{n,i}^*\| \|\mathcal{E}(\eta)_{n,i}^*\|$ dominates. This then gives

us

$$\|\mathcal{E}(\Xi)_{n,i}^*\| \leq C z_{n,1} v_n^5 u_n^6 \Delta_n^{-3}. \quad (52)$$

The right-hand-side is precisely $z_{n,4}$.

□

Lemma 3

Suppose that

$$\varpi < \min \left(\frac{\varpi' - 5\tau}{48/5}, \frac{1 - \varpi'}{20} \right).$$

Then, I have

$$\frac{z_{n,4}\sqrt{T_n}}{\sqrt{q_n\Delta_n}} + z_{n,4}\sqrt{T_n} + z_{n,4}^2\sqrt{T_n} + z_{n,4}v_n^3u_n^4\Delta_n^{-2}\sqrt{T_n} \rightarrow 0.$$

Proof. I prove convergence for each term.

- I begin by handling the first two terms. Since $\sqrt{q_n\Delta_n} \rightarrow 0$, we simply have to prove that $\frac{z_{n,4}\sqrt{T_n}}{\sqrt{q_n\Delta_n}} \rightarrow 0$. Squaring this term and expanding $z_{n,4}$, we can see that it suffices to show that

$$v_n^p T_n^4 \left(\frac{u_n^{12} \Delta_n^{2\psi-7}}{q_n} \right) \rightarrow 0.$$

Again, the $v_n \asymp \log(1/(T\Delta_n))$ term is negligible in size compared to the other terms, so we merely need to show that the remaining terms will converge to zero. Recall that $T_n \asymp \Delta_n^\tau$, $u_n \asymp \Delta_n^\varpi$, and $q_n \asymp \Delta_n^{-\varpi'}$. Thus, we simply have to prove that $\Delta_n^{-4\tau+12\varpi+2\psi-7+\varpi'} \rightarrow 0$. We can see that

$$\begin{aligned} -4\tau + 12\varpi + 2\psi - 7 + \varpi' &= -4\tau + 12\varpi + 1 - \varpi'/5 - 7 + \varpi' \\ &= -4\tau + (4/5)\varpi' + (-12)(1/2 - \varpi). \end{aligned}$$

Assuming that $\varpi' > 5\tau$ and $(1/2 - \varpi) < \frac{\varpi' - 5\tau}{12} \cdot (4/5)$ is sufficient to ensure that the above expression is positive. Consequently, we have $\frac{z_{n,4}\sqrt{T_n}}{\sqrt{q_n\Delta_n}} + z_{n,4}\sqrt{T_n} \rightarrow 0$.

- Next, is convergence for $z_{n,4}\sqrt{T_n}$. Note that $z_{n,4}\sqrt{T_n} \gg z_{n,4}$, while the former converges to zero. Hence, $z_{n,4} \rightarrow 0$. Therefore, we must have $z_{n,4}^2\sqrt{T_n} \rightarrow 0$.
- Last is $z_{n,4}v_n^3u_n^4\Delta_n^{-2}\sqrt{T_n}$. This is bounded by $Cv_n^{10}T_n^2u_n^{10}\Delta_n^{\psi-5}$. Then it suffices to show that

$$v_n^p T_n^4 u_n^{20} \Delta_n^{2\psi-10} \rightarrow 0.$$

Using the same argument as before, we can ignore v_n when proving convergence, since it increases merely at a log rate. Then, using the same definitions as before, we can see that the left-hand-side term is proportional to $\Delta_n^{-4\tau+20\varpi+2\psi-8}$. Next,

expanding the exponent, we get

$$\begin{aligned} -4\tau + 20\varpi + 2\psi - 10 &= -4\tau + 20\varpi + 1 - \varpi'/5 - 8 \\ &= -4\tau + 1 - \varpi'/5 - 20(1/2 - \varpi) \end{aligned}$$

Assume again that $\varpi' > 5\tau$. Additionally, assume that $(1/2 - \varpi) < \frac{1-\varpi}{20}$. Then, the expression above will be positive. As a result, we will have $z_{n,4}v_n^3u_n^4\Delta_n^{-2}\sqrt{T_n} \rightarrow 0$.

In total, all four processes converge to zero. □

Additional Processes

Now, I define the following processes involving $\Xi_{n,i}^* \equiv \beta_{n,i}^* \eta_{n,i}^*$.

$$U_{\Xi,n}^* = \frac{1}{T_n} \sum_{i=0}^{[n/q_n]-2} \Xi_{n,i}^* (P_{(i+2)q_n\Delta_n} - P_{(i+1)q_n\Delta_n}) \quad (53)$$

$$U'_{\Xi,n} = \frac{q_n\Delta_n}{T_n} \sum_{i=0}^{[n/q_n]-2} \Xi_{n,i}^* \bar{r}_{iq_n\Delta_n} \quad (54)$$

$$V_{\Xi,n}^* = \sum_{i=0}^{[n/q_n]-2} \Xi_{n,i}^* (P_{(i+2)q_n\Delta_n} - P_{(i+1)q_n\Delta_n}) (P_{(i+2)q_n\Delta_n} - P_{(i+1)q_n\Delta_n})^\top (\Xi_{n,i}^*)^\top \quad (55)$$

Note that these variables are constructed as in (A.9) of AJX. I begin by showing that $\hat{U}, \hat{U}'$, and $\hat{V}$ converge to $U^*, U^{*'},$ and V^* in the following lemma ([Lemma 4](#)). Afterwards, I will show that $U^*, U^{*'},$ and V^* converge in probability to the truth in [Lemma 6](#), [Lemma 7](#), and [Lemma 8](#).

Lemma 4 (Intermediate Convergence Proof)

We have

$$\begin{aligned} \sqrt{T_n} \|\hat{U}_{\Xi,n} - U_{\Xi,n}^*\| &\xrightarrow{p} 0 \\ \sqrt{T_n} \|\hat{U}'_{\Xi,n} - U'_{\Xi,n}^{*'}\| &\xrightarrow{p} 0 \\ \sqrt{T_n} \|\hat{V}_{\Xi,n} - V_{\Xi,n}^*\| &\xrightarrow{p} 0 \end{aligned}$$

Proof. The statement and proof are similar to Lemma 6 from AJX. We begin by restricting to the set Ω_n .

- First, for $U_{\Xi,n}$, we have:

$$\begin{aligned}
\|\sqrt{T_n}(\hat{U}_{\Xi,n} - U_{\Xi,n}^*)\| &\leq \left\| \frac{1}{\sqrt{T_n}} \sum_{i=0}^{[n/q_n]-2} \mathcal{E}(\Xi)_{n,i}^* \cdot (P_{(i+2)q_n\Delta_n} - P_{(i+1)q_n\Delta_n}) \right\| \\
&\leq C \frac{z_{n,4}}{\sqrt{T_n}} \sum_{i=0}^{[n/q_n]-2} \|P_{(i+2)q_n\Delta_n} - P_{(i+1)q_n\Delta_n}\| \\
\implies E\left(\|\sqrt{T_n}(\hat{U}_{\Xi,n} - U_{\Xi,n}^*)\| \cdot \mathbb{1}_{\Omega_n}\right) &\leq C \frac{z_{n,4}}{\sqrt{T_n}} \frac{n}{q_n} \sqrt{q_n\Delta_n} \\
&\leq C \frac{z_{n,4}\sqrt{T_n}}{\sqrt{q_n\Delta_n}}.
\end{aligned}$$

- Second is the proof for $U'_{\Xi,n}$. This is essentially the same proof as above and as AJX. Specifically, with $E(\|\bar{r}_{iq_n\Delta_n}\| \cdot \mathbb{1}_{\Omega_n}) \leq q_n\Delta_n$ and the fact that $\bar{r}_t$ is bounded, we have:

$$E\left(\|\sqrt{T_n}(\hat{U}'_{\Xi,n} - U'_{\Xi,n})\| \cdot \mathbb{1}_{\Omega_n}\right) \leq Cz_{4,n}\sqrt{T_n}.$$

- Third is $V_{\Xi,n}$, the variance process. Define

$$\begin{aligned}
\rho_i^n &= P_{(i+2)q_n\Delta_n} - P_{(i+1)q_n\Delta_n} \\
\zeta_i^n &= \mathcal{E}(\Xi)_{n,i}^* \rho_i^n.
\end{aligned}$$

So, we have the same decomposition from AJX,

$$\frac{1}{\sqrt{T_n}}(\hat{V}_{\Xi,n} - V_{\Xi,n}^*) = \frac{1}{\sqrt{T_n}} \sum_{i=0}^{[n/q_n]-2} \left(\zeta_i^n (\zeta_i^n)^\top + \Xi_{n,i}^* \rho_i^n (\zeta_i^n)^\top + \zeta_i^n (\rho_i^n)^\top (\Xi_{n,i}^*)^\top \right).$$

From [Lemma 2](#) and restriction to the set Ω_n , we have $\|\Xi_{n,i}^*\| \leq \|\beta_{n,i}^*\| \|\eta_{n,i}^*\| \leq Cv_n^3 u_n^4 / \Delta_n^2$. Consequently, we have,

$$\begin{aligned}
\left\| \frac{1}{\sqrt{T_n}}(\hat{V}_{\Xi,n} - V_{\Xi,n}^*) \right\| &\leq C \frac{z_{n,4}^2 + z_{n,4}v_n^3 u_n^4 \Delta_n^{-2}}{\sqrt{T_n}} \sum_{i=0}^{[n/q_n]-2} \|\rho_i^n\|^2 \\
E\left(\left\| \frac{1}{\sqrt{T_n}}(\hat{V}_{\Xi,n} - V_{\Xi,n}^*) \right\| \cdot \mathbb{1}_{\Omega_n}\right) &\leq C \frac{z_{n,4}^2 + z_{n,4}v_n^3 u_n^4 \Delta_n^{-2}}{\sqrt{T_n}} n\Delta_n \\
&\leq C(z_{n,4}^2 + z_{n,4}v_n^3 u_n^4 \Delta_n^{-2})\sqrt{T_n}.
\end{aligned}$$

Finally, we need to show that the errors converge to zero. That is, we need

$$\frac{z_{n,4}\sqrt{T_n}}{\sqrt{q_n\Delta_n}} + z_{n,4}\sqrt{T_n} + z_{n,4}^2\sqrt{T_n} + z_{n,4}v_n^3 u_n^4 \Delta_n^{-2}\sqrt{T_n} \rightarrow 0.$$

This result is proven in [Lemma 3](#). Thus, $\hat{U}, \hat{U}'$, and $\hat{V}$ converge to $U^*, U^{*'},$ and V^* in probability. □

Additional Properties

At this point, note that the properties described in section A.3 of AJX also hold for $U_{\Xi,n}^*$ and $U_{\Xi,n}'^*$. First, we have

$$\rho_n = q_n \Delta_n + 1/q_n \quad (56)$$

$$v_n^p \left(\rho_n^2 T_n + \frac{1}{q_n \Delta_n T_n} \right) \rightarrow 0, \quad (57)$$

which is based on (A.10) in AJX. Second, we have (A.12) from AJX, which also holds for Ξ_t and $\Xi_t \bar{r}_t$; formally, this is

$$\Xi_t, \Xi_t \bar{r}_t \text{ satisfy Equation 25 on each } \mathcal{L}'^{(j)} = (\tau_{j-1}, \tau_j + 2q_n \Delta_n]. \quad (58)$$

And, lastly, we have the following sets:

$$B_n = \bigcup_{j=1}^J B_n^j, \quad B_n^j = \{i : 1 \leq i \leq [n/q_n] - 2, iq_n \Delta_n \in \mathcal{L}^{(j)}\}. \quad (59)$$

These sets will be useful for tracking the estimation errors.

Lemma 5 (Expected Projection Error Bounds)

Suppose that

$$1 - 2\varpi \leq \frac{1 - \alpha}{64 - \alpha}.$$

Then, we have

$$i \in B_n \implies \|E_{iq_n}^n(\mathcal{E}(\Xi)_{n,i})\| \leq C\rho_n v_n^5, \quad E_{iq_n}^n(\|\mathcal{E}(\Xi)_{n,i}\|^2) \leq C\rho_n v_n^{10} \quad (60)$$

$$E_{iq_n}^n(\|\Xi_{n,i}^*\|^6) \leq C v_n^{18}. \quad (61)$$

Proof. The proof follows Lemma 7 of AJX. Begin by defining,

$$\hat{c}_{n,i}^{l,l'} = \frac{1}{q_n \Delta_n} \sum_{j=1}^{q_n} \Delta_{iq_n+j}^n Z^l \Delta_{iq_n+j}^n Z^{l'} \cdot \mathbb{1}_{\{|\Delta_{iq_n+j}^n Z^l| \leq u_n, |\Delta_{iq_n+j}^n Z^{l'}| \leq u_n\}}$$

$$\hat{c}_{n,i}'^{l,l'} = \frac{1}{q_n \Delta_n} \sum_{j=1}^{q_n} \Delta_{iq_n+j}^n X^l \Delta_{iq_n+j}^n X^{l'}.$$

From Lemma B.5 of Aït-Sahalia and Jacod (2014), we have

$$E_{iq_n}^n(|\hat{c}_{i,n}^{l,l'} - \hat{c}_{n,i}'^{l,l'}|^p) \leq C_p \Delta_n^{(2p-\alpha)\varpi-p+1} \leq C_p \sqrt{\Delta_n} \quad (62)$$

for $p \in [1, 32]$ and any l, l' . The latter inequality requires the restriction $1 - 2\varpi \leq \frac{1-\alpha}{64-\alpha}$, which is slightly stronger than the inequality from the original paper. Then, from Lemma 7 of AJX, we have,

$$|E_{iq_n}^n(\hat{c}_{i,n}^{l,l'} - \hat{c}_{n,i}'^{l,l'})| \leq C\rho_n, \quad E_{iq_n}^n(|\hat{c}_{i,n}^{l,l'} - \hat{c}_{n,i}'^{l,l'}|^p) \leq C_p \rho_n, \quad (63)$$

for $l' \leq K$ and $p \in [2, 32]$ on the set $\{i \in B_n\}$.

- First, I prove [Equation 60](#). Recall the following property from AJX, for some process Y , sequences $z_n, \bar{z}_n \geq 1$, and $p \geq 2$,

$$(\mathcal{P}_{z_n, \bar{z}_n, p}) : \quad \begin{cases} \text{on the set } \{i \in B_n\} \text{ and for all } p' \in [2, p], \text{ we have} \\ \|E_{iq_n}^n(\mathcal{E}(Y)_{n,i})\| \leq C\rho_n z_n, \quad E_{iq_n}^n(\|\mathcal{E}(Y)_{n,i}\|^{p'}) \leq C\rho_n \bar{z}_n^{p'}. \end{cases}$$

Based on [Equation 63](#), we have c_t^F and γ_t satisfying $(\mathcal{P}_{z_n, \bar{z}_n, p})$ with $p = 32$ and $z_n = \bar{z}_n = 1$. Then, like in AJX, we can successively bound more complex terms using the properties of $(\mathcal{P}_{z_n, \bar{z}_n, p})$. Specifically, we have $(c_t^F)^{-1}$ satisfying $(\mathcal{P}_{v_n, v_n, 32})$, β_t satisfying $(\mathcal{P}_{v_n, v_n, 16})$, b_t satisfying $(\mathcal{P}_{v_n^2, v_n^2, 8})$, $(b_t)^{-1}$ satisfying $(\mathcal{P}_{v_n^3, v_n^3, 8})$, η_t satisfying $(\mathcal{P}_{v_n^4, v_n^4, 4})$, and lastly Ξ_t satisfying $(\mathcal{P}_{v_n^5, v_n^5, 2})$. These results follow from a repeated application of Holders inequality. Furthermore, the last property is precisely [Equation 60](#).

- Second, I prove [Equation 61](#). As in AJX, using [Equation 62](#) with $p = 12$ gives

$$E_{iq_n}(\|\hat{c}_{n,i}^F\|^{12} + \|\hat{\gamma}_{n,i}\|^{12}) \leq C.$$

Then, we have

$$\begin{aligned} E_{iq_n}(\|\beta_{n,i}^*\|^{12}) &\leq C v_n^{12} \\ E_{iq_n}(\|\eta_{n,i}^*\|^{12}) &\leq C v_n^{24}, \end{aligned}$$

where these inequalities comes from the definition of $\beta_{n,i}^*$ and $\eta_{n,i}^*$; specifically, both involve truncations that produce the v_n terms. Finally, using Cauchy-Schwartz, it is easy to see that we must have

$$E_{iq_n}(\|\Xi_{n,i}^*\|^6) = E_{iq_n}(\|\beta_{n,i}^* \eta_{n,i}^*\|^6) \leq C v_n^{18}.$$

This is the third property in the statement of the lemma.

□

Lemma 6 (Convergence of U)

We have

$$\sqrt{T_n} \|U_{\Xi, n}^* - U_{\Xi, T_n}\| \xrightarrow{p} 0 \tag{64}$$

Proof. This proof follows Lemma 8 of AJX. To start, set $B'_n = \{0, 1, \dots, [n/q_n] - 2\} \setminus B_n$. Likewise, write $\sqrt{T_n}(U_{\Xi, n}^* - U_{\Xi, T_n}) = \sum_{l=1}^5 G_n^l$ where

$$\begin{aligned} \zeta_i^n &= \mathcal{E}(\Xi)_{n,i} (P_{(i+2)q_n \Delta_n} - P_{(i+1)q_n \Delta_n}), \quad G_n^1 = \frac{1}{\sqrt{T_n}} \sum_{i \in B_n} \zeta_i^n, \quad G_n^2 = \frac{1}{\sqrt{T_n}} \sum_{i \in B'_n} \zeta_i^n \\ \xi_i^n &= \int_{(i+1)q_n \Delta_n}^{(i+2)q_n \Delta_n} (\Xi_{iq_n \Delta_n} - \Xi_s) dP_s, \quad G_n^3 = \frac{1}{\sqrt{T_n}} \sum_{i \in B_n} \xi_i^n, \quad G_n^4 = \frac{1}{\sqrt{T_n}} \sum_{i \in B'_n} \xi_i^n \\ A_n &= ([n/q_n] - 2)q_n \Delta_n, \quad G_n^5 = -\frac{1}{\sqrt{T_n}} \int_{A_n} \Xi_s dP_s \end{aligned}$$

- Since $\{i \in B_n\} \in F_{iq_n}^n$ and $\mathcal{E}(\Xi)_{n,i} \cdot \mathbb{1}_{[i \in B_n]}$ is $F_{(i+1)q_n}^n$ -measurable, we have

$$\begin{aligned} \|E_{iq_n}^n(\zeta_i^n)\| &\leq E_{iq_n}^n\left(\|\mathcal{E}(\Xi)_{n,i}\| \cdot E_{(i+1)q_n}^n\left(\|P_{(i+2)q_n\Delta_n} - P_{(i+1)q_n\Delta_n}\|\right)\right) \\ &\leq C\rho_n v_n^5 q_n \Delta_n \end{aligned} \quad (65)$$

$$E_{iq_n}^n(\|\zeta_i^n\|^2) \leq C\rho_n v_n^{10} q_n \Delta_n \quad (66)$$

on the set $\{i \in B_n\}$. The bound on $\mathcal{E}(\Xi)_{n,i}$ comes from [Lemma 5](#) while the innovation in the price is clearly of order $q_n \Delta_n$. Now, we can bound the G_n^1 term using a martingale argument.

$$E(\|G_n^1\|) \leq \frac{C}{\sqrt{T_n}} \left(\frac{n}{q_n} \rho_n v_n^5 q_n \Delta_n + \sqrt{\frac{n}{q_n}} \sqrt{\rho_n v_n^{10} q_n \Delta_n} \right) \leq C v_n^5 \left(\rho_n \sqrt{T_n} + \sqrt{\rho_n} \right) \rightarrow 0 \quad (67)$$

The bound is slightly larger than in the original result but the convergence is unaffected.

- Next, using [Lemma 5](#) again with $\{i \in B'_n\} \in F_{iq_n}^n$ and $E_{(i+1)q_n}^n\left(\|P_{(i+2)q_n\Delta_n} - P_{(i+1)q_n\Delta_n}\|^2\right) \leq C q_n \Delta_n$, we obtain

$$E(\|\zeta_i^n\| \cdot \mathbb{1}_{[i \in B'_n]}) \leq C v_n^3 \sqrt{q_n \Delta_n} \cdot P(i \in B'_n). \quad (68)$$

Again, the bound converges slightly more slowly than in the original setting with η ; this is due to the higher power v_n term in [Equation 61](#). Then, using the argument from Lemma 8 of AJX, we have

$$E(\|G_n^2\|) \leq C v_n^3 \frac{\sqrt{q_n \Delta_n}}{\sqrt{T_n}} \cdot E\left(\sum_{i \geq 0} \mathbb{1}_{[i \in B'_n]}\right) \leq C \frac{v_n^3}{\sqrt{q_n \Delta_n T_n}} \rightarrow 0. \quad (69)$$

- Thirdly, we can use the fact that [Equation 25](#) applies to Ξ_t to make the same argument from AJX for G_n^3 . So, we have

$$\|E_{iq_n}^n(\xi_i^n)\| + E_{iq_n}^n(\|\xi_i^n\|^2) \leq C(q_n \Delta_n)^2 \quad (70)$$

on the set $\{i \in B_n\}$. To be precise, the bound above comes from the fact that the predictable quadratic variation $\bar{V}_t$ of the martingale component of the prices P is such that $at - \bar{V}_t$ is increasing. Consequently,

$$\begin{aligned} E(\|G_n^3\|) &\leq \frac{C}{\sqrt{T_n}} \left(\frac{n}{q_n} (q_n \Delta_n)^2 + \sqrt{\frac{n}{q_n}} q_n \Delta_n \right) \\ &\leq C(q_n \Delta_n \sqrt{T_n} + \sqrt{q_n \Delta_n}) \rightarrow 0 \end{aligned} \quad (71)$$

- Finally, note that Ξ_t is bounded due to β_t and η_t being bounded. From this, the

last two results of the proof follow,

$$E(\|G_n^4\|) \leq C \frac{1}{\sqrt{q_n \Delta_n T_n}} \rightarrow 0 \quad (72)$$

$$E(\|G_n^5\|) \leq C \frac{\sqrt{q_n \Delta_n}}{\sqrt{T_n}} \rightarrow 0. \quad (73)$$

The argument for G_n^4 is the same as the argument for G_n^2 . And, the argument for G_n^5 further uses the fact that the length of A_n smaller than $3q_n \Delta_n$.

In total, $\sqrt{T_n} \|U_{\Xi,n}^* - U_{\Xi,T_n}\| = \|\sum_{l=1}^5 G_n^l\| \xrightarrow{p} 0$.

□

Lemma 7 (Convergence of U')

We have

$$\sqrt{T_n} \|U_{\Xi,n}^* - U_{\Xi,T_n}'\| \xrightarrow{p} 0 \quad (74)$$

Proof. The proof follows trivially from Lemma 9 of AJX and Lemma 6 from above. Specifically, writing

$$\begin{aligned} \zeta_i^n &= q_n \Delta_n \mathcal{E}(\Xi)_{n,i} \bar{r}_{iq_n \Delta_n}, \quad G_n^1 = \frac{1}{\sqrt{T_n}} \sum_{i \in B_n} \zeta_i^n, \quad G_n^2 = \frac{1}{\sqrt{T_n}} \sum_{i \in B_n'} \zeta_i^n \\ \zeta_i^n &= \int_{(i+1)q_n \Delta_n}^{(i+2)q_n \Delta_n} (\Xi_{iq_n \Delta_n} \bar{r}_{iq_n \Delta_n} - \Xi_s \bar{r}_s) ds, \quad G_n^3 = \frac{1}{\sqrt{T_n}} \sum_{i \in B_n} \zeta_i^n, \quad G_n^4 = \frac{1}{\sqrt{T_n}} \sum_{i \in B_n'} \zeta_i^n \\ A_n &= ([n/q_n] - 2)q_n \Delta_n, T_n], \quad G_n^5 = -\frac{1}{\sqrt{T_n}} \int_{A_n} \Xi_s \bar{r}_s ds \end{aligned}$$

and repeating the previous proof will give the desired result.

□

Lemma 8 (Convergence of V)

We have

$$\sqrt{T_n} \|V_{\Xi,n}^* - V_{\Xi,T_n}\| \xrightarrow{p} 0 \quad (75)$$

Proof. This proof is similar to Lemma 10 in AJX. However, while η_t is $(K + H) \times M$, Ξ_t is a $M \times M$ matrix. To start, note that it is sufficient to prove that

$$\bar{V}_{\Xi,n} \equiv \frac{1}{\sqrt{T_n}} \sum_{i=0}^{[n/q_n]-2} \Xi_{n,i}^{*w,m} \Xi_{n,i}^{*w',m'} P_i^{n,m} P_i^{n,m'} - \frac{1}{\sqrt{T_n}} \int_0^{T_n} \Xi_t^{w,m} \Xi_t^{w',m'} d[P^m, P^{m'}]_t \xrightarrow{p} 0,$$

where $w, w', m, m' \in \{1, \dots, M\}$ are arbitrary and $P_i^{n,m} \equiv P_{(i+2)q_n \Delta_n}^m - P_{(i-1)q_n \Delta_n}^m$. Fur-

thermore, we have a decomposition similar to before. Write $\bar{V}_n = \sum_{l=1}^5 \bar{G}_n^l$ with

$$\begin{aligned}\bar{\zeta}_i^n &= \left(\mathcal{E}(\Xi)_{n,i}^{w,m} \mathcal{E}(\Xi)_{n,i}^{w',m'} + \mathcal{E}(\Xi)_{n,i}^{w,m} \Xi_{iq_n \Delta_n}^{w',m'} + \Xi_{iq_n \Delta_n}^{w,m} \mathcal{E}(\Xi)_{n,i}^{w',m'} \right) P_i^{n,m} P_i^{n,m'} \\ \bar{G}_n^1 &= \frac{1}{\sqrt{T_n}} \sum_{i \in B_n} \bar{\zeta}_i^n, \quad \bar{G}_n^2 = \frac{1}{\sqrt{T_n}} \sum_{i \in B'_n} \bar{\zeta}_i^n \\ \bar{\zeta}_i^n &= \int_{(i+1)q_n \Delta_n}^{(i+2)q_n \Delta_n} \left(\Xi_{iq_n \Delta_n}^{w,m} \Xi_{iq_n \Delta_n}^{w',m'} - \Xi_s^{w,m} \Xi_s^{w',m'} \right) d[P^m, P^{m'}]_s, \quad \bar{G}_n^3 = \frac{1}{\sqrt{T_n}} \sum_{i \in B_n} \bar{\zeta}_i^n \\ \bar{G}_n^4 &= \frac{1}{\sqrt{T_n}} \sum_{i \in B'_n} \bar{\zeta}_i^n, \quad \bar{G}_n^5 = -\frac{1}{T_n} \int_{A_n} \Xi_s^{w,m} \Xi_s^{w',m'} d[P^m, P^{m'}]_s.\end{aligned}$$

The proof now proceeds similar to [Lemma 6](#).

- Firstly, I handle the $\bar{\zeta}_i^n$ term. Note that we have

$$E_{iq_n}^n (\|\mathcal{E}(\Xi)_{n,i}\|^4) \leq \sqrt{E_{iq_n}^n (\|\mathcal{E}(\Xi)_{n,i}\|^2) E_{iq_n}^n (\|\mathcal{E}(\Xi)_{n,i}\|^6)} \leq C v_n^{14} \sqrt{\rho_n} \quad (76)$$

on $\{i \in B_n\}$ using [Equation 60](#) and [Equation 61](#). Next, using the fact that $E_{(i+1)q_n}^n (|P_i^{n,m}|^p) \leq C q_n \Delta_n$ for $p = 2, 4$, we now have,

$$\{i \in B_n\} \implies \begin{cases} |E_{iq_n}^n (\bar{\zeta}_i^n)| & \leq C v_n^{10} q_n \Delta_n \rho_n \\ E_{iq_n}^n (|\bar{\zeta}_i^n|^2) & \leq C v_n^{14} q_n \Delta_n \sqrt{\rho_n}. \end{cases}$$

These inequalities are obtained by noting that the $\mathcal{E}(\Xi)_{n,i}^{w,m} \mathcal{E}(\Xi)_{n,i}^{w',m'}$ term in $\bar{\zeta}_i^n$ dominates the other two terms; this is because Ξ_t is bounded due to β_t and η_t being bounded. Consequently, applying [Equation 60](#) gives the result for $|E_{iq_n}^n (\bar{\zeta}_i^n)|$, while applying [Equation 76](#) gives the result for $E_{iq_n}^n (|\bar{\zeta}_i^n|^2)$. Additionally, we have

$$E(\|\bar{\zeta}_i^n\| \cdot 1_{[i \in B'_n]}) \leq C v_n^6 q_n \Delta_n \cdot P(i \in B'_n)$$

for the same reason as in [Lemma 6](#); this result comes from iterated expectations, Cauchy-Schwartz, and [Lemma 5](#). Hence, repeating the proof from [Lemma 6](#), we have

$$\begin{aligned}E(|\bar{G}_n^1|) &\leq C v_n^{10} (\rho_n \sqrt{T_n} + \rho_n^{1/4}) \rightarrow 0 \\ E(|\bar{G}_n^2|) &\leq C \frac{v_n^4}{\sqrt{q_n \Delta_n T_n}} \rightarrow 0\end{aligned}$$

Convergence for both terms is obtained from [Equation 57](#).

- Secondly, I handle $\bar{G}_n^3, \bar{G}_n^4$, and $\bar{G}_n^5$. We can replace Ξ_t with $\Xi_t^{w,m} \Xi_t^{w',m'}$ and P by $Y = [P^m, P^{m'}]$ as in AJX Lemma 10. Consequently, based on the same lemma, I

have

$$E(\|\bar{G}_n^3\|) \leq C(q_n \Delta_n \sqrt{T_n} + \sqrt{q_n \Delta_n}) \rightarrow 0 \quad (77)$$

$$E(\|\bar{G}_n^4\|) \leq C \frac{1}{\sqrt{q_n \Delta_n T_n}} \rightarrow 0 \quad (78)$$

$$E(\|\bar{G}_n^5\|) \leq C \frac{\sqrt{q_n \Delta_n}}{\sqrt{T_n}} \rightarrow 0. \quad (79)$$

The last two equations follow from Y being a finite variation process and Ξ_t being bounded. The first equation follows from the same argument as [Lemma 6](#) but applied to Y .

□

7.4.2 Proof of Theorem 1

Proof. This proof follows that of Theorem 1 from AJX. Using [Lemma 6](#), [Lemma 7](#), and [Lemma 8](#), I have

$$\begin{aligned} \sqrt{T_n} \|\hat{U}_\Gamma - U_{\Gamma, T_n}\| &\xrightarrow{p} 0 \\ \sqrt{T_n} \|\hat{U}'_\Gamma - U'_{\Gamma, T_n}\| &\xrightarrow{p} 0 \\ \sqrt{T_n} (\hat{\Gamma}_n - \Gamma_{T_n}) - \frac{1}{\sqrt{T_N}} \bar{U}_{\Gamma, T_n} &\xrightarrow{p} 0. \end{aligned}$$

These equations are obtained by simply replacing Ξ_t in the earlier lemmas with $\varkappa_t$ defined earlier; since $\varkappa_t$ simply consists of $(\beta_t^C, \mathbf{0}_H)$ and $(\mathbf{0}_K, \beta_t^J)$, the results follow in a straightforward way. Next, I need to introduce the matrix $\mathcal{S}_2$, defined earlier in [Theorem 1](#), in order to limit estimation to assets with infinite lifespans. This narrows the convergence result to

$$\sqrt{T_n} (\mathcal{S}_2^\top \hat{\Gamma}_n - \mathcal{S}_2^\top \Gamma_{T_n}) - \frac{1}{\sqrt{T_N}} (\mathcal{S}_2^\top \bar{U}_{\Gamma, T_n} \mathcal{S}_2) \xrightarrow{p} 0, \quad (80)$$

where the idiosyncratic risk premia term is negligible due to Assumption [A.III](#). Additionally, note that $\mathcal{S}_2^\top \Gamma$ is simply $(S^\top \Gamma^C, S^\top \Gamma^J)$. As discussed earlier, the proof can be done without $\mathcal{S}$, but the result would be nonsensical. Specifically, since Ξ_t is set to zero for assets that are non-existent, the risk premia earned on such assets would be zero by construction; hence, estimates of Γ for assets with finite life-spans are not well-defined. So, estimation of Γ must be restricted using $\mathcal{S}$.

Now, for notational simplicity, I will assume that $\mathcal{S}$ is the identity matrix without loss of generality. Furthermore let m be an arbitrary asset with $\theta_m = \infty$ (infinite lifespan). Based on the earlier assumptions, the quadratic variation process V_{Γ, m, T_n} satisfies the following conditions:

$$\exists C_0 > 1 \quad \text{s.t.} \quad \frac{t}{C_0} \leq \zeta(V_{\Gamma, m, t}) \leq \|V_{\Gamma, m, t}\|, \quad E(\|V_{\Gamma, m, t}\|) \leq C_0 t. \quad (81)$$

Consequently, $E\left(\frac{1}{T_n} (\bar{U}_{\Gamma, m, T_n}^\perp)^2\right) \leq C_0$ and $\frac{1}{\sqrt{T_n}} \bar{U}_{\Gamma, m, T_n}$ is bounded in probability. Based on [Equation 80](#), this implies that $\sqrt{T_n} (\hat{\Gamma}_{m, n} - \Gamma_{m, T_n})$ is bounded in probability. Since

m was chosen arbitrarily in the set of all assets with finite lifespan, we must have $\sqrt{T_n}(\mathcal{S}^\top \hat{\Gamma}_n - \mathcal{S}^\top \Gamma_{T_n})$ bounded in probability. This gives the first part of [Theorem 1](#).

Now, I prove the second part of the theorem; this follows almost exactly from Theorem 1 in AJAX. To start, pick m arbitrarily from the set of assets with infinite lifespan. Assuming that $(1/t)V_{\Gamma,m,t}$ converges to a limit $V_{\Gamma,\infty}$ and using Corollary 2.3 of Crimaldi and Pratelli (2005), we can get the following CLT:

$$(V_{\Gamma,m,T_n})^{-1/2} \bar{U}_{\Gamma,m,T_n} \text{ converges in law to } \mathcal{N}(0, I_2). \quad (82)$$

To make this result useful, note that there exists a sequence of subsets $\Omega_n \in \Omega$ satisfying $P(\Omega_n) \rightarrow 1$ such that

- $(\hat{V}_{\Gamma,m,n})^{-1}$ exists on Ω_n
- $T_n(\hat{V}_{\Gamma,m,n})^{-1} \cdot \mathbb{1}_{\Omega_n}$ and $\frac{1}{T_n} \hat{V}_{\Gamma,m,n} \cdot \mathbb{1}_{\Omega_n}$ are bounded in probability
- $(\hat{V}_{\Gamma,m,n})^{-1/2}(V_{\Gamma,m,T_n})^{1/2}$ converges in probability to I_M on Ω_n .

To see why, first note that [Lemma 4](#) and [Lemma 8](#) show that $\hat{V}_{\Gamma,n}$ converges to V_{Γ,T_n} . But, the elements of these two matrix processes also make up $\hat{V}_{\Gamma,m,n}$ and V_{Γ,m,T_n} . Thus, we have $\sqrt{T_n} \|\hat{V}_{\Gamma,m,n} - V_{\Gamma,m,T_n}\| \xrightarrow{p} 0$ on Ω_n . The three statements above follow from this fact and Theorem 1 of AJAX. Moreover, the last statement gives

$$\sqrt{T_n}(\hat{V}_{\Gamma,m,n})^{-1/2} (\hat{\Gamma}_{m,n} - \Gamma_{m,T_n}) \xrightarrow{d} \mathcal{N}(0, I_2)$$

when combined with [Equation 82](#) on Ω_n . This proves the second part of [Theorem 1](#). □

8 Additional Tables and Figures

Table 28: FACTOR LITERATURE

Citation	Description	CZ Data Type	CZ Classification	JKP Classification
Abarbanell and Bushee, 1998	Change sales minus change Inventory	Accounting	Sales Growth	Profit Growth
Abarbanell and Bushee, 1998	Change sales minus change receivables			Profit Growth
Abarbanell and Bushee, 1998	Change gross margin minus change sales			Quality
Abarbanell and Bushee, 1998	Change sales minus change SG&A	Accounting	Sales Growth	Profit Growth
Abarbanell and Bushee, 1998	Labor force efficiency			Profit Growth
Abarbanell and Bushee, 1998	Change in capital inv (ind adj)	Accounting	Investment Growth	
Ali, Hwang, and Trombley, 2003	Idiosyncratic volatility from the CAPM (252 days)	Price	Volatility	Low Risk
Ali, Hwang, and Trombley, 2003	Idiosyncratic volatility from the CAPM (21 days)			Low Risk
Ali, Hwang, and Trombley, 2003	Idiosyncratic volatility from the q-factor model			Low Risk
Alwathainani, 2009	Earnings consistency	Accounting	Earnings Growth	
Amihud and Mendelson, 1986	Bid-ask spread	Trading	Liquidity	
Amihud, 2002	Amihud Measure	Trading	Liquidity	Size
Anderson and Garcia-Feijóo, 2006	CAPEX growth (2 years)	Accounting	Investment Growth	Investment
Anderson and Garcia-Feijóo, 2006	CAPEX growth (3 years)	Accounting	Investment Growth	Investment
Ang, Chen, and Xing, 2006	Downside beta	Price	Risk	Low Risk
Ang et al., 2006	Idiosyncratic volatility from the Fama-French ...	Price	Volatility	Low Risk
Ang et al., 2006	Return volatility	Price	Volatility	Low Risk
Ang et al., 2006	Idiosyncratic risk	Price	Volatility	
Asness, Frazzini, and Pedersen, 2019	Quality minus Junk: Composite			Quality
Asness, Frazzini, and Pedersen, 2019	Quality minus Junk: Profitability			Quality
Asness, Frazzini, and Pedersen, 2019	Quality minus Junk: Growth			Quality
Asness, Frazzini, and Pedersen, 2019	Quality minus Junk: Safety			Profit Growth
Asness et al., 2020	Highest 5 days of return scaled by volatility			Skewness
Asness et al., 2020	Market correlation			Seasonality
Bhandari, 1988	Debt-to-market	Price	Leverage	Value
Balakrishnan, Bartov, and Faurel, 2010	Quarterly return on assets	Accounting	Profitability	Quality
Balakrishnan, Bartov, and Faurel, 2010	Change in quarterly return on assets			Profit Growth
Bali, Cakici, and Whitelaw, 2011	Maximum daily return	Price	Volatility	Low Risk
Bali, Engle, and Murray, 2016	Idiosyncratic skewness from the q-factor model			Skewness
Bali, Engle, and Murray, 2016	Total skewness	Price	Risk	Skewness
Bali, Engle, and Murray, 2016	Idiosyncratic skewness from the CAPM			Skewness

continued...

Factor Literature (continued)

Citation	Description	CZ Data Type	CZ Classification	JKP Classification
Bali, Engle, and Murray, 2016	Idiosyncratic skewness from the Fama-French 3-...	Price	Risk	Skewness
Bali et al., 2017	Highest 5 days of return			Low Risk
Ball et al., 2016	Operating profits-to-lagged book assets	Accounting	Profitability	Quality
Ball et al., 2016	Cash-based operating profits-to-book assets	Accounting	Profitability	Quality
Ball et al., 2016	Cash-based operating profits-to-lagged book as...			Quality
Banz, 1981	Market Equity	Price	Size	Size
Barbee, Mukherji, and Raines, 1996	Sales-to-market	Accounting	Valuation	Value
Barber et al., 2001	Consensus Recommendation	Analyst	Recommendation	
Barber et al., 2001	Down forecast EPS	Analyst	Earnings Forecast	
Barber et al., 2001	Up Forecast	Analyst	Earnings Forecast	
Barry and Brown, 1984	Firm age based on CRSP	Other	Info Proxy	
Barth, Elliott, and Finn, 1999	Number of consecutive quarters with earnings i...			Quality
Basu, 1983	Earnings-to-price	Price	Valuation	Value
Belo and Lin, 2012	Inventory growth	Accounting	Profitability	Investment
Belo, Lin, and Vitorino, 2014	Brand capital investment	Accounting	Investment Alt	
Belo, Lin, and Bazdresch, 2014	Hiring rate	Other	Investment Alt	Investment
Blitz, Huij, and Martens, 2011	Residual momentum t-6 to t-1	Price	Momentum	Momentum
Blitz, Huij, and Martens, 2011	Residual momentum t-12 to t-1			Momentum
Blume and Husic, 1973	Price	Price	Other	
Bouchaud et al., 2019	Operating cash flow to assets			Profitability
Bouchaud et al., 2019	Change in operating cash flow to assets			Profit Growth
Boudoukh et al., 2007	Net payout yield	Price	Valuation	Value
Boudoukh et al., 2007	Payout yield	Price	Valuation	Value
Bradshaw, Richardson, and Sloan, 2006	Net external financing	Accounting	External Financing	
Bradshaw, Richardson, and Sloan, 2006	Net debt issuance	Accounting	External Financing	Seasonality
Bradshaw, Richardson, and Sloan, 2006	Net equity issuance	Accounting	External Financing	Low Risk
Bradshaw, Richardson, and Sloan, 2006	Net total issuance			Low Risk
Brennan, Chordia, and Subrahmanyam, 1998	Dollar trading volume	Trading	Volume	Size
Cohen and Frazzini, 2008	Customer momentum	Other	Lead Lag	
Cen, Wei, and Zhang, 2006	Analyst earnings per share	Analyst	Profitability	
Chan, Jegadeesh, and Lakonishok, 1996	Earnings announcement return	Price	Earnings Event	
Chan, Jegadeesh, and Lakonishok, 1996	Earnings forecast revisions	Analyst	Earnings Forecast	

continued...

Factor Literature (continued)

Citation	Description	CZ Data Type	CZ Classification	JKP Classification
Chan, Lakonishok, and Sougiannis, 2001	R&D-to-market	Accounting	R&D	Size
Chan, Lakonishok, and Sougiannis, 2001	R&D-to-sales			Leverage
Chan, Lakonishok, and Sougiannis, 2001	Advertising Expense	Accounting	R&D	
Chandrashekar and Rao, 2011	Cash Productivity	Accounting	Profitability Alt	
Chen, Hong, and Stein, 2002	Breadth of ownership	13F	Ownership	
Chordia, Subrahmanyam, and Anshuman, 2001	Coefficient of variation for share turnover	Trading	Liquidity	Profitability
Chordia, Subrahmanyam, and Anshuman, 2001	Coefficient of variation for dollar trading vo...	Trading	Liquidity	Profitability
Cohen and Lou, 2012	Conglomerate return	Price	Lead Lag	
Cohen, Diether, and Malloy, 2013	R&D ability	Accounting	Other	
Cooper, Gulen, and Schill, 2008	Asset Growth	Accounting	Investment	Investment
Corwin and Schultz, 2012	The high-low bid-ask spread			Leverage
Daniel and Titman, 2006	Equity net payout	Accounting	External Financing	Value
Daniel and Titman, 2006	Intangible return using BM	Accounting	Long Term Reversal	
Daniel and Titman, 2006	Intangible return using CFtoP	Accounting	Long Term Reversal	
Daniel and Titman, 2006	Intangible return using EP	Accounting	Long Term Reversal	
Daniel and Titman, 2006	Intangible return using Sale2P	Accounting	Long Term Reversal	
Daniel and Titman, 2006	Share issuance (5 year)	Accounting	External Financing	
Datar, Naik, and Radcliffe, 1998	Share turnover	Trading	Volume	Low Risk
De Bondt and Thaler, 1985	Long-term reversal	Price	Long Term Reversal	Investment
De Bondt and Thaler, 1985	Medium-run reversal	Price	Long Term Reversal	
Dechow et al., 2001	Short Interest	Other	Short Sale Constraints	
Dechow, Sloan, and Soliman, 2004	Equity duration	Price	Valuation	Value
Desai, Rajgopal, and Venkatachalam, 2004	Operating cash flow-to-market	Accounting	Valuation	Value
Dharan and Ikenberry, 1995	Exchange Switch	Event	Other	
Dichev, 1998	Ohlson O-score	Accounting	Default Risk	Profitability
Dichev, 1998	Altman Z-score			Leverage
Dichev and Piotroski, 2001	Credit Rating Downgrade	Event	Other	
Diether, Malloy, and Scherbina, 2002	EPS Forecast Dispersion	Analyst	Volatility	
Dimson, 1979	Dimson beta			Low Risk
Fama and French, 1992	Assets-to-market	Accounting	Valuation	Value
Fama and French, 1992	Book leverage	Accounting	Leverage	Leverage
Fama and French, 1992	Sales growth (1 quarter)			Investment

continued...

Factor Literature (continued)

Citation	Description	CZ Data Type	CZ Classification	JKP Classification
Fama and French, 1992	Book to market using December ME	Accounting	Valuation	
Fairfield, Whisenant, and Yohn, 2003	Change in long-term net operating assets	Accounting	Investment	Investment
Fama and MacBeth, 1973	Market Beta	Price	Risk	Low Risk
Fama and French, 1996	Price momentum t-12 to t-1			Momentum
Fama and French, 2015	Operating profits-to-lagged book equity			Profitability
Fama and French, 2015	Operating profits-to-book assets			Quality
Fama and French, 2015	Operating profits-to-book equity	Accounting	Profitability	Profitability
Foster, Olsen, and Shevlin, 1984	Standardized earnings surprise	Analyst	Earnings Growth	Profit Growth
Francis et al., 2004	Earnings persistence			Seasonality
Francis et al., 2004	Earnings volatility			Leverage
Francis et al., 2004	Earnings variability			Low Risk
Frankel and Lee, 1998	Intrinsic value-to-market	Analyst	Valuation	Profitability
Frankel and Lee, 1998	Analyst Optimism	Analyst	Other	
Frankel and Lee, 1998	Predicted Analyst forecast error	Accounting	Earnings Forecast	
Franzoni and Marin, 2006	Pension Funding Status	Accounting	Composite Accounting	
Frazzini and Pedersen, 2014	Frazzini-Pedersen market beta	Price	Other	Low Risk
George and Hwang, 2004	Current price to high price over last year	Price	Momentum	Momentum
Gompers, Ishii, and Metrick, 2003	Governance Index	Other	Other	
Guo, Lev, and Shi, 2006	IPO and no R&D spending	Event	R&D	
Hafzalla, Lundholm, and Van Winkle, 2011	Percent operating accruals	Accounting	Accruals	Accruals
Hafzalla, Lundholm, and Van Winkle, 2011	Percent total accruals	Accounting	Accruals	Accruals
Hahn and Lee, 2009	Asset tangibility	Accounting	Asset Composition	Leverage
Hartzmark and Solomon, 2013	Dividend seasonality	Event	Payout Indicator	
Harvey and Siddique, 2000	Coskewness	Price	Risk	Seasonality
Haugen and Baker, 1996	Capital turnover	Accounting	Cash Flow Risk	Quality
Haugen and Baker, 1996	Return on equity	Accounting	Profitability	Profitability
Haugen and Baker, 1996	Volume to market equity	Trading	Volume	
Haugen and Baker, 1996	Volume Trend	Trading	Volume	
Hawkins, Chamberlin, and Daniel, 1984	EPS forecast revision	Analyst	Earnings Forecast	
Heston and Sadka, 2008	Years 16-20 lagged returns, nonannual	Price	Other	Seasonality
Heston and Sadka, 2008	Years 16-20 lagged returns, annual	Price	Other	Seasonality
Heston and Sadka, 2008	Years 11-15 lagged returns, nonannual	Price	Other	Seasonality

continued...

Factor Literature (continued)

Citation	Description	CZ Data Type	CZ Classification	JKP Classification
Heston and Sadka, 2008	Years 11-15 lagged returns, annual	Price	Other	Seasonality
Heston and Sadka, 2008	Years 6-10 lagged returns, nonannual	Price	Other	Seasonality
Heston and Sadka, 2008	Years 2-5 lagged returns, annual	Price	Other	Seasonality
Heston and Sadka, 2008	Years 2-5 lagged returns, nonannual	Price	Other	Investment
Heston and Sadka, 2008	Year 1-lagged return, nonannual	Price	Other	Momentum
Heston and Sadka, 2008	Year 1-lagged return, annual	Price	Other	Profit Growth
Heston and Sadka, 2008	Years 6-10 lagged returns, annual	Price	Other	Seasonality
Hirshleifer et al., 2004	Net operating assets	Accounting	Asset Composition	Debt Issuance
Hirshleifer et al., 2004	Change in net operating assets	Accounting	Investment	Investment
Hirshleifer, Hsu, and Li, 2013	Citations to RD expenses	Other	Profitability Alt	
Hirshleifer, Hsu, and Li, 2013	Patents to RD expenses	Other	Profitability Alt	
Hong and Kacperczyk, 2009	Sin Stock (selection criteria)	Other	Other	
Hou and Moskowitz, 2005	Price delay SE adjusted	Price	Lead Lag	
Hou and Moskowitz, 2005	Price delay coeff	Price	Lead Lag	
Hou and Moskowitz, 2005	Price delay r square	Price	Lead Lag	
Hou and Robinson, 2006	Industry concentration (sales)	Other	Other	
Hou and Robinson, 2006	Industry concentration (assets)	Other	Other	
Hou and Robinson, 2006	Industry concentration (equity)	Other	Other	
Hou, 2007	Industry return of big firms	Price	Lead Lag	
Hou, 2007	Earnings surprise of big firms	Accounting	Lead Lag	
Hou, Xue, and Zhang, 2015	Quarterly return on equity			Profitability
Hou, Xue, and Zhang, 2015	Change in quarterly return on equity			Profit Growth
Huang, 2009	Cash flow volatility			Low Risk
Jegadeesh, 1990	Short-term reversal	Price	Short-Term Reversal	Skewness
Jegadeesh and Titman, 1993	Momentum (12 month)	Price	Momentum	
Jegadeesh and Titman, 1993	Price momentum t-3 to t-1			Momentum
Jegadeesh and Titman, 1993	Price momentum t-9 to t-1			Momentum
Jegadeesh and Titman, 1993	Price momentum t-6 to t-1	Price	Momentum	Momentum
Jegadeesh et al., 2004	Change in recommendation	Analyst	Recommendation	
Jegadeesh and Livnat, 2006	Standardized Revenue surprise	Accounting	Sales Growth	Profit Growth
Jiang, Lee, and Zhang, 2005	Firm age			Leverage
Lakonishok, Shleifer, and Vishny, 1994	Free cash flow-to-price	Accounting	Valuation	Low Risk

continued...

Factor Literature (continued)

Citation	Description	CZ Data Type	CZ Classification	JKP Classification
Lakonishok, Shleifer, and Vishny, 1994	Sales Growth (3 years)			Investment
Lakonishok, Shleifer, and Vishny, 1994	Sales Growth (1 year)	Accounting	Sales Growth	Investment
Lamont, Polk, and Saaá-Requejo, 2001	Kaplan-Zingales index			Seasonality
Lev and Nissim, 2004	Taxable income-to-book income	Accounting	Other	Seasonality
Li, 2011	R&D capital-to-book assets	Accounting	Asset Composition	Leverage
Litzenberger and Ramaswamy, 1979	Dividend yield	Accounting	Valuation	Value
Liu, 2006	Number of zero trades with turnover as tiebrea...			Low Risk
Liu, 2006	Number of zero trades with turnover as tiebrea...	Trading	Liquidity	Low Risk
Liu, 2006	Number of zero trades with turnover as tiebrea...			Low Risk
Lockwood and Prombutr, 2010	Growth in book equity	Accounting	Investment	
Loh and Warachka, 2012	Earnings streak length	Accounting	Earnings Growth	
Loh and Warachka, 2012	Earnings surprise streak	Accounting	Earnings Growth	
Lou, 2014	Growth in advertising expenses	Accounting	Investment Alt	
Loughran and Wellman, 2011	Ebitda-to-market enterprise value	Accounting	Valuation	Value
Lyandres, Sun, and Zhang, 2008	Change PPE and Inventory	Accounting	Investment	Investment
Lyandres, Sun, and Zhang, 2008	Growth in book debt (3 years)	Accounting	External Financing	Debt Issuance
Menzly and Ozbas, 2010	Customers momentum	Other	Lead Lag	
Menzly and Ozbas, 2010	Suppliers momentum	Other	Lead Lag	
Michaely, Thaler, and Womack, 1995	Dividend Initiation	Event	Payout Indicator	
Michaely, Thaler, and Womack, 1995	Dividend Omission	Event	Payout Indicator	
Miller and Scholes, 1982	Price per share			Size
Novy-Marx, 2011	Operating leverage	Accounting	Other	Quality
Novy-Marx, 2012	Price momentum t-12 to t-7	Price	Momentum	Profit Growth
Novy-Marx, 2013	Gross profits-to-assets	Accounting	Profitability	Quality
Novy-Marx, 2013	Gross profits-to-lagged assets			Quality
Ortiz-Molina and Phillips, 2014	Liquidity of book assets			Investment
Ortiz-Molina and Phillips, 2014	Liquidity of market assets			Leverage
Palazzo, 2012	Cash-to-assets	Accounting	Asset Composition	Leverage
Pástor and Stambaugh, 2003	Pastor-Stambaugh liquidity beta	Price	Liquidity	
Penman, Richardson, and Tuna, 2007	Book-to-market enterprise value	Accounting	Valuation	Value
Penman, Richardson, and Tuna, 2007	Net debt-to-price	Accounting	Leverage	Leverage
Penman, Richardson, and Tuna, 2007	Leverage component of BM	Accounting	Leverage	

continued...

Factor Literature (continued)

Citation	Description	CZ Data Type	CZ Classification	JKP Classification
Piotroski, 2000	Piotroski F-score	Accounting	Composite Accounting	Profitability
Pontiff and Woodgate, 2008	Net stock issues	Accounting	External Financing	Low Risk
Prakash and Sinha, 2013	Deferred Revenue	Accounting	Investment Alt	
Ritter, 1991	Initial Public Offerings	Event	External Financing	
Ritter, 1991	IPO and age	Event	Other	
Rajgopal, Shevlin, and Venkatachalam, 2003	Order backlog	Accounting	Sales Growth	
Richardson et al., 2005	Change in common equity	Accounting	Investment	Investment
Richardson et al., 2005	Change in financial liabilities	Accounting	External Financing	Debt Issuance
Richardson et al., 2005	Change in long-term investments	Accounting	Investment	Seasonality
Richardson et al., 2005	Change in short-term investments			Profit Growth
Richardson et al., 2005	Change in net financial assets	Accounting	Investment Alt	Debt Issuance
Richardson et al., 2005	Change in current operating liabilities	Accounting	External Financing	Investment
Richardson et al., 2005	Change in noncurrent operating assets			Investment
Richardson et al., 2005	Change in net noncurrent operating assets			Investment
Richardson et al., 2005	Change in current operating assets	Accounting	Investment Alt	Investment
Richardson et al., 2005	Change in current operating working capital			Accruals
Richardson et al., 2005	Total accruals	Accounting	Investment Alt	Accruals
Richardson et al., 2005	Change in noncurrent operating liabilities			Debt Issuance
Rosenberg, Reid, and Lanstein, 1985	Book-to-market equity	Accounting	Valuation	Value
Scherbina, 2008	Decline in Analyst Coverage	Analyst	Earnings Event	
Sloan, 1996	Operating accruals	Accounting	Accruals	Accruals
Soliman, 2008	Change in Net Working Capital	Accounting	Investment Alt	
Soliman, 2008	Assets turnover	Accounting	Sales Growth	Quality
Soliman, 2008	Return on net operating assets	Accounting	Investment Alt	Profitability
Soliman, 2008	Profit margin			Profitability
Spiess and Affleck-Graves, 1999	Debt Issuance	Event	External Financing	
Stambaugh and Yuan, 2017	Mispricing factor: Performance			Quality
Stambaugh and Yuan, 2017	Mispricing factor: Management			Investment
Thomas and Zhang, 2002	Inventory change	Accounting	Investment Alt	Investment
Thomas and Zhang, 2011	Tax expense surprise	Accounting	Other	Quality
Titman, Wei, and Xie, 2004	Abnormal corporate investment	Accounting	Investment	Debt Issuance
Tuzel, 2010	Real estate holdings	Accounting	Asset Composition	

continued...

Factor Literature (continued)

Citation	Description	CZ Data Type	CZ Classification	JKP Classification
Valta, 2016	Convertible debt indicator	Event	External Financing	
Xie, 2001	CAPEX growth (1 year)	Accounting	Accruals	Investment
Xing, Zhang, and Zhao, 2010	Volatility smirk near the money	Options	Option Risk	
Yan, 2011	Put volatility minus call volatility	Options	Option Risk	
Zhang, 2006	Firm Age - Momentum	Price	Momentum	

This table is based on factors compiled by Chen and Zimmermann (2020) and Jensen, Kelly, and Pedersen (2021). The description column is based on the descriptions from Jensen, Kelly, and Pedersen (2021); for factors that do not appear in that paper, I use the description from Chen and Zimmermann (2020). The “CZ” data type and classifications columns are based on Chen and Zimmermann (2020). The “JKP” classification column is based on Jensen, Kelly, and Pedersen (2021). Since the CZ and JKP factor datasets do not perfectly overlap, there may be factors (and thus classifications) from one paper but not the other. Factors that do overlap are given a classification from both papers.

9 References

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